

LakeCat

covers lakecat (0.1.1)

An Engine-Close Catalog Foundation for QueryGraph

Alexy Khrabrov

chiefscientist.org

&

Codex with ChatGPT 5.5

Contents

1 LakeCat	2
1.1 Preface	2
1.2 Catalogs, Iceberg, and What LakeCat Adds	2
1.2.1 What a catalog is	2
1.2.2 What Iceberg makes the catalog responsible for	3
1.2.3 What LakeCat adds without changing Iceberg	4
1.3 The Vocabulary	4
1.4 The Boundary Model	7
1.5 Where LakeCat Stands Today	9
1.6 Why the Catalog Stays Thin	10
1.7 The Architecture	12
1.7.1 The CatalogStore seam	12
1.7.2 The read path	13
1.7.3 The commit path	14
1.7.4 The durable spine	16
1.8 The Siblings and the Engine Path	16
1.8.1 Grust: the graph boundary	16
1.8.2 TypeSec: the authorization boundary	17
1.8.3 Sail and the v3→v4 path	17
1.8.4 The semantic handoff: Croissant, CDIF, OSI, OpenLineage	18
1.8.5 The full stack	18
1.8.6 Implementation shape	19
1.9 Workflow Examples	20
1.9.1 Starting The Catalog	20
1.9.2 Registering The Warehouse Shape	22
1.9.3 Storage Profiles And Credential Roots	26
1.9.4 A PySpark User Reads Iceberg	31
1.9.5 A Notebook Requests Credentials	32
1.9.6 A View Becomes Part Of The Catalog Story	33
1.9.7 QueryGraph Bootstrap	38
1.9.8 Draining The Outbox	52
1.9.9 An Agentic QGLake Flow	66
1.10 Operating The Book's Example System	67
1.11 Standards And Engine Boundary Decision Record	68
1.11.1 Catalog Concepts In Plain Terms	68
1.11.2 Decision Matrix	72
1.11.3 The Proper-Noun Test	74
1.11.4 Why The Engine Owns The Truth	74
1.11.5 Workflow Consequences	75
1.12 First Release Readiness	75
1.13 Future Work	81

1 LakeCat

1.1 Preface

LakeCat is a Rust-native, Iceberg-compatible catalog foundation for QueryGraph. It begins from one conservative claim and one ambitious one. The conservative claim: an ordinary Iceberg REST catalog must keep working for ordinary engines — Spark, Flink, Trino, DuckDB, PyIceberg, Sail — with no new protocol to learn. The ambitious claim: the *next* catalog also has to be a governed control plane for Rust-first planning, semantic graph handoff, lineage, and agent access.

LakeCat holds both at once by drawing a sharp line. It keeps Iceberg compatibility at the boundary and moves the work that needs deep table-format knowledge to the engine that can reason about it. In this repository the catalog is **LakeCat** and the engine-side lakehouse implementation is **Sail**. Graph behavior belongs to **Grust**; governance and capability proof belong to **TypeSec**; the end-to-end integration target is **QueryGraph**.

This book builds from first principles. Chapter 2 explains what a catalog is, what Iceberg makes the catalog responsible for, and what LakeCat adds without changing the table format. Chapter 3 fixes the vocabulary — the single hardest thing in a layered system — so every later term has an owner. Chapter 4 draws the boundary model: which repository owns which concept, and which ideas might someday become neutral, standardizable profiles. The middle chapters walk the live architecture: the service spine, the read path, the commit path, and the durable store. The final chapters cover the engine boundary and the v3→v4 path, the graph/security/lineage handoffs, the QueryGraph/QGLake acceptance flow, worked examples, and the first-release scope.

Each concept is defined once and then used. Where a later chapter needs a term, it links back rather than restating it.

1.2 Catalogs, Iceberg, and What LakeCat Adds

1.2.1 What a catalog is

A data catalog is often described as a place that lists datasets. That is true but too small. A real catalog is the control plane between names, storage, metadata, identity, and intent. At minimum it answers four questions:

1. What table does this name mean?
2. Where is its current metadata?
3. Who may read, write, plan, or administer it?
4. What changed, when, and under whose authority?

In a traditional database the catalog is embedded in the engine: one system owns table definitions, statistics, permissions, and the transaction log. A lakehouse pulls that apart. Data files live in object storage, metadata files live beside them, and several engines read and write the same tables. The catalog becomes the agreement point — it maps a logical table name to the current metadata pointer and arbitrates updates to that pointer.

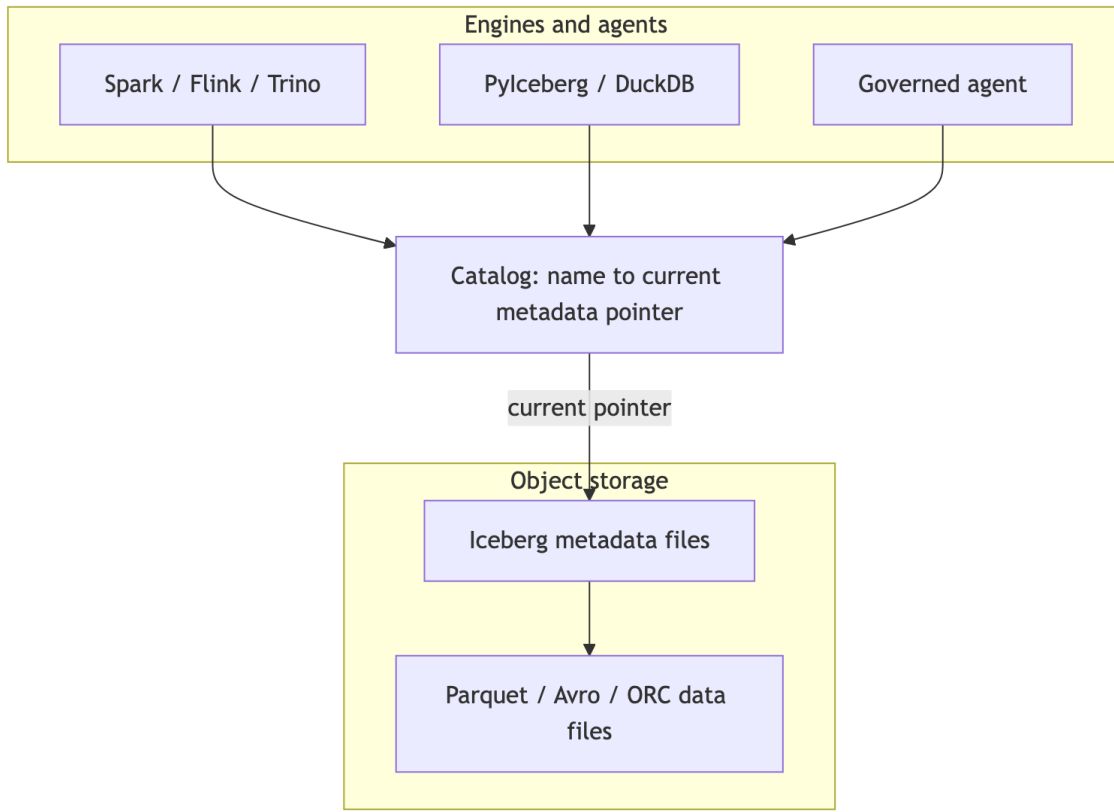


Figure 1: Diagram 1

That pointer is deceptively important. If the catalog points at metadata version 17, the table *is* version 17. If a writer prepares version 18 and wins the compare-and-swap, the table becomes 18; if it loses, nothing partially changes. The catalog is not the table format, but it is where table history becomes visible and durable. For human analytics this sounds like bookkeeping. For agentic systems it is a trust boundary: a catalog can know the principal, warehouse, namespace, table, snapshot, requested columns, row restriction, storage profile, and policy receipt — and if it captures that before planning and commits it after state changes, it becomes a governed control plane rather than a passive address book.

1.2.2 What Iceberg makes the catalog responsible for

Apache Iceberg is a table format for large analytic tables. Its core idea is to put the table’s truth in explicit metadata files so engines can plan reads and validate writes without directory listing or fragile storage conventions. A current metadata file names schemas, partition specs, sort orders, snapshots, and properties; snapshots point to manifest lists; manifest lists point to manifests; manifests describe data and delete files.

The catalog’s role in Iceberg is intentionally narrow. Standard clients must be able to load table metadata, create namespaces and tables, commit changes, and sometimes receive credentials or scan tasks — over a documented REST shape:

```

GET /v1/config
GET /v1/{prefix}/namespaces
POST /v1/{prefix}/namespaces/{ns}/tables
GET /v1/{prefix}/namespaces/{ns}/tables/{table}
POST /v1/{prefix}/namespaces/{ns}/tables/{table} # commit: requirements +
updates

```

The rule that governs everything else in this book: **a normal client must never have to call a non-standard endpoint to read an ordinary table.** If it does, compatibility is already broken.

1.2.3 What LakeCat adds without changing Iceberg

LakeCat’s promise is *compatibility first, evidence second, semantics above the catalog*. A Spark or PyIceberg client sees an Iceberg REST catalog. A QueryGraph or governed-agent client may ask for richer proof. Both share the same table because the portable truth stays in Iceberg metadata and the extra evidence sits beside it — never inside the table format.

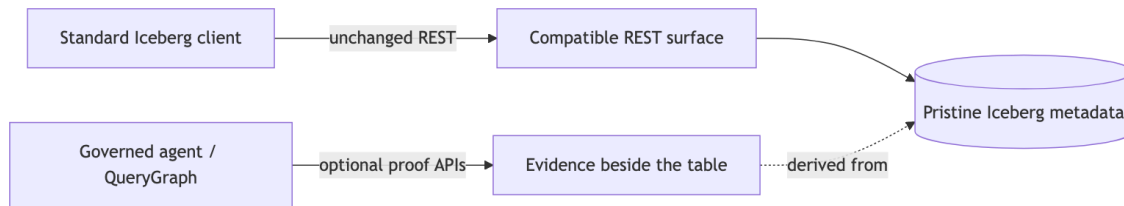


Figure 2: Diagram 2

Iceberg metadata stays pristine. Policy, graph, lineage, and agent state are *derived* control-plane or graph data. The table remains an Iceberg table.

1.3 The Vocabulary

A layered system fails the moment its words blur. The most common mistake is to call every useful LakeCat feature an “Iceberg extension,” which makes the standard boundary too large. The test is simple: **ask what breaks if a client knows nothing about LakeCat.** If a PySpark job cannot load, commit, or drop a table without the concept, it belongs to standard compatibility. If PySpark keeps working but operators, governed agents, or QueryGraph gain stronger evidence, the concept is an additive surface.

Every term in this book falls into one of six categories.

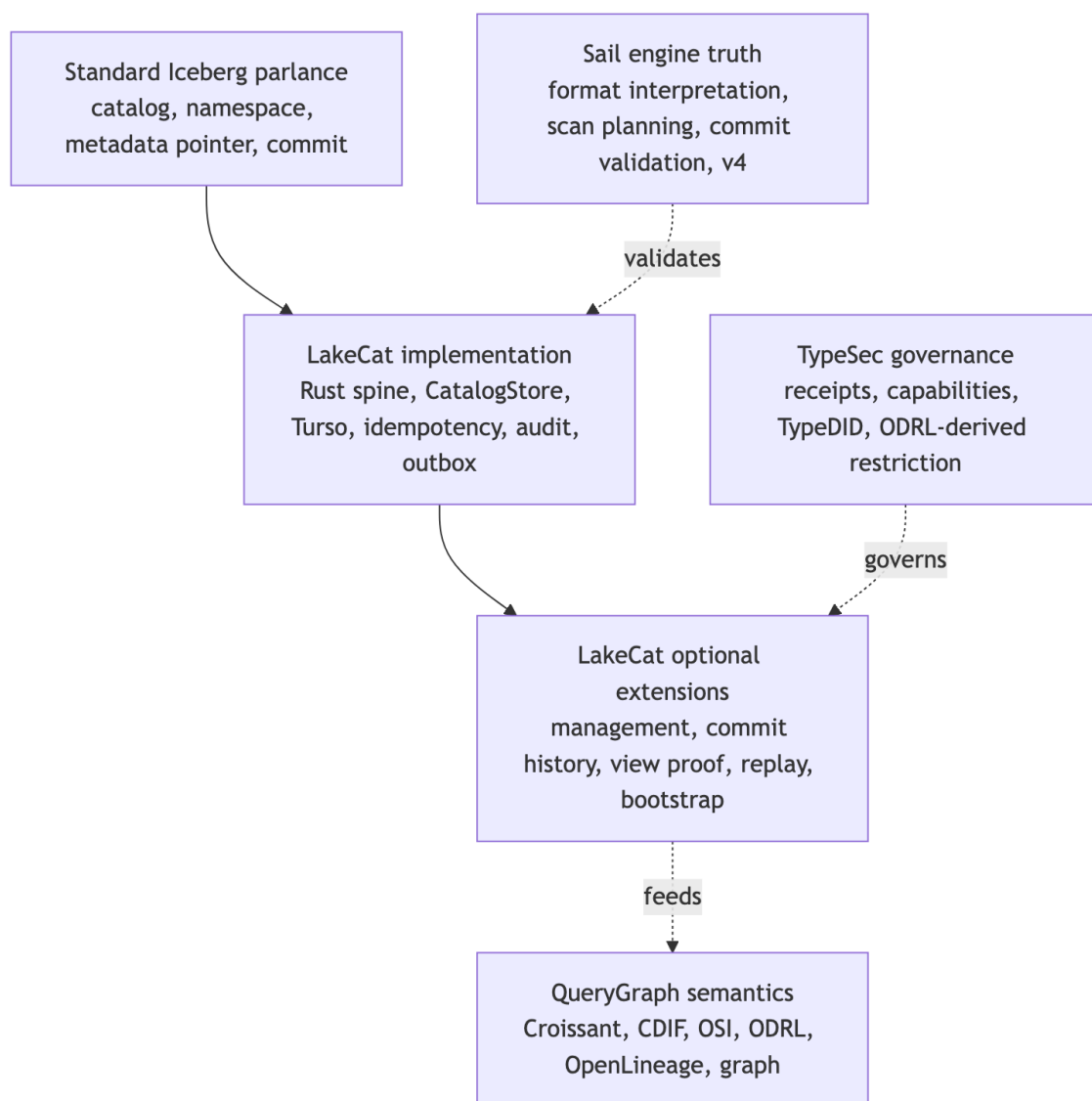


Figure 3: Diagram 3

Standard Iceberg parlance — the words Iceberg already owns: catalog, namespace, table identifier, current metadata location, snapshot, manifest list, manifest, data file, delete file, schema and partition evolution, optimistic commit, REST compatibility. LakeCat must implement these faithfully. Changing their meaning means losing compatibility.

LakeCat implementation — how this Rust catalog satisfies the contract reliably: the service spine, the CatalogStore trait, the Turso-backed durable store, normalized idempotency rows, pointer logs, audit rows, outbox rows, redaction rules, and replay validators. These make ordinary Iceberg behavior atomic, inspectable, and replayable. They are not extensions; they are good engineering behind a standard surface.

LakeCat optional extensions — additive APIs beside the standard path: management inventory, commit-history inspection, view proof, credential-root posture, replay verification, OpenLineage projection, and QueryGraph/QGLake bootstrap bundles. They help operators, agents, and QueryGraph without becoming hidden requirements for ordinary table access.

TypeSec governance — authority and receipt semantics: who may act, capability proof, TypeDID envelopes, ODRL-derived restrictions, and agent posture. TypeSec *decides*; LakeCat *carries and records* the decision.

QueryGraph semantics — composition above the catalog: Croissant, CDIF, OSI, ODRL, OpenLineage, and Grust graph projections built from the governed source of truth.

Sail engine truth — table-format interpretation, metadata-as-data, scan planning, commit validation, and typed v4 behavior. The catalog binds its proof to engine truth rather than reimplementing the format.

The single reference below replaces the per-claim restatements that earlier drafts repeated; read each LakeCat claim through its category.

Claim	Standard reading	Category	Portable idea (if any)
Rust service / catalog spine	Iceberg needs a catalog authority, not a language	LakeCat implementation	“A catalog can prove what it committed, planned, vended, and emitted”
Turso-backed durable store	Iceberg needs durable state + atomic pointer movement, not a named DB	LakeCat implementation	CAS, exact-retry, row/content binding, pointer-history proof
REST namespace/table routes + commit CAS	Standard compatibility	Standard Iceberg	— (must stay ordinary)
Idempotency, pointer logs, audit/outbox, replay validation	Mostly outside the table contract	LakeCat implementation / extension	Stable catalog-event identity; scoped replay admission
Governed scan with policy receipt	Iceberg gives scan inputs, not receipts	TypeSec governance	Engine-proved effective projection + predicate
Credential vending posture	Catalog-adjacent, not ordinary table semantics	TypeSec governance	Bounded, redacted, engine-neutral credential profile
QueryGraph / QGLake / OpenLineage handoff	Not required for table access	QueryGraph semantics	Redacted, replayable lineage/view/commit evidence
Typed Iceberg v4 behavior	Belongs to the format as it evolves	Sail engine truth	Engine-owned v4 interpretation, not catalog JSON parsing

1.4 The Boundary Model

The vocabulary tells you which category a word lives in. The boundary model says which *repository* owns the work and why. LakeCat is deliberately thin: it keeps identity, tenancy, metadata-pointer state, policy gates, idempotent commits, and integration events — and delegates everything reusable.

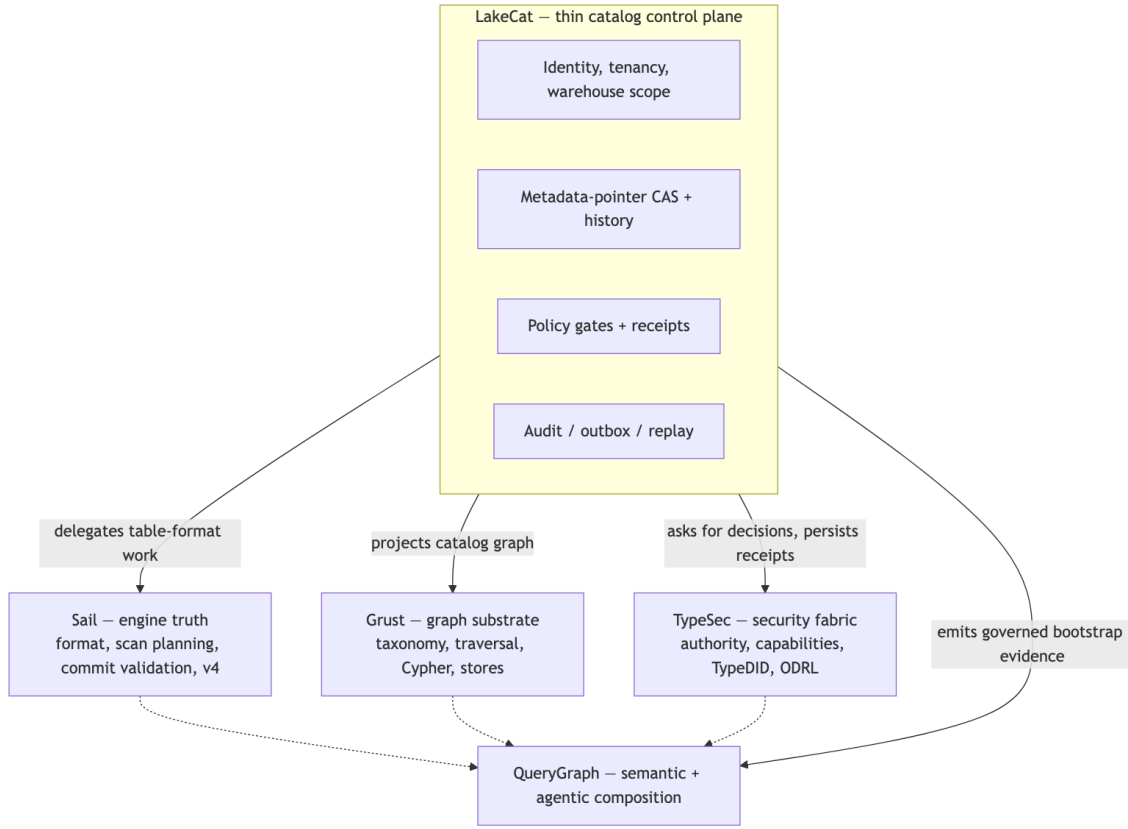


Figure 4: Diagram 4

The ownership rule, stated once:

Concern	Owner	LakeCat keeps only
Iceberg format, manifests, scan planning, pruning, delete handling, v4	Sail	The call into Sail and the proof binding its result
Graph schema, taxonomy, traversal, stores, Cypher	Grust	The catalog-facing sink/projection boundary
Authorization, policy composition, capabilities, TypeDID, credential decisions	TypeSec	The request for a decision and the persisted receipt
Croissant/CDIF/OSI/ODRL/OpenLineage composition, agent workflows	QueryGraph	The governed bootstrap bundle it emits
Identity, tenancy, pointer CAS, idempotency, audit, outbox, replay	LakeCat	All of it — this is the thin catalog

This is also the answer to the recurring “is it an extension or a standard?” question. LakeCat is opinionated in code and modest in standardization. “Use Rust,” “use Turso,” “use TypeSec,” “import QGLake,” “project into Grust” are project choices, not proposals. The portable ideas are narrower and stated without product names: *reject idempotency drift; record redacted pointer history; emit transactional catalog-event identity; admit only scoped replay evidence; prove a governed scan was narrowed by an engine*. A proposal that forced every Iceberg catalog to understand QueryGraph, TypeSec, Grust, or Turso would narrow the ecosystem; a proposal that says “a catalog may publish redacted, replayable commit evidence” leaves room for many implementations. Prove the stronger shape locally, then extract only the database-neutral, policy-neutral, engine-neutral part.

1.5 Where LakeCat Stands Today

LakeCat is not only a design. The implementation already has a Rust service/catalog spine, a Turso-backed durable store, Iceberg REST-compatible namespace and table paths, hardened commit and replay evidence, governed scan and credential proof, and QueryGraph/QGLake handoff surfaces. Read each of those through its category (Chapter 3) rather than flattening them into “Iceberg extensions”: some are ordinary catalog implementation, some are optional LakeCat APIs, some are TypeSec governance proof, and only a few are plausible future standardization candidates.

The most important word is **beside**. New capability sits beside the Iceberg REST path, never in front of it. A standard client never has to present a TypeDID, parse a QueryGraph bundle, read a Grust edge, or inspect an OpenLineage receipt to load a normal table. So the same catalog shows a different face to each caller:

Caller	What they see
PySpark / PyIceberg user	Ordinary Iceberg REST. Configure a catalog, create a namespace, write and load tables. Pointer-log, audit, and outbox rows exist but are invisible to table semantics.
Platform operator	A hardened catalog transaction log: idempotency outcomes, pointer movement, redacted conflict proof, storage-profile and credential posture, pending outbox, replay-validation failures.
Governed agent	A narrowed access path: TypeSec decides the capability, LakeCat binds the receipt to the catalog action, Sail plans the effective scan, the agent gets bounded work instead of broad credentials.
QueryGraph importer	A proof-bearing bootstrap: table, view, management, credential, scan, commit-history, OpenLineage, and graph-import anchors that must line up before the semantic layer is accepted.

That separation also keeps standardization honest. The interesting portable shapes are never “LakeCat uses Turso” or “QueryGraph imports a bundle”; they are smaller behaviors — idempotent commit replay, catalog pointer history, governed credential vending, proof-carrying scan planning, redacted conflict evidence, lineage receipt binding. The posture, in five rules:

1. Keep base Iceberg REST behavior strict and boring.
2. Keep LakeCat proof surfaces optional for the clients that need them.
3. Keep TypeSec and QueryGraph semantics out of Iceberg metadata.
4. Use Sail for reusable table-format and planning semantics.
5. Promote only small, interoperable proof profiles once real workflows prove multiple engines and catalogs need them.

1.6 Why the Catalog Stays Thin

The most dangerous failure mode for a “smart” catalog is becoming a partial engine. It starts innocently — validate a schema here, expand a manifest list there, peek at a format-version field, check a delete file. Each small parser looks cheaper than an engine call. Over time the catalog grows a second Iceberg implementation with weaker tests, fewer real execution users, and quiet drift from how the planner actually behaves.

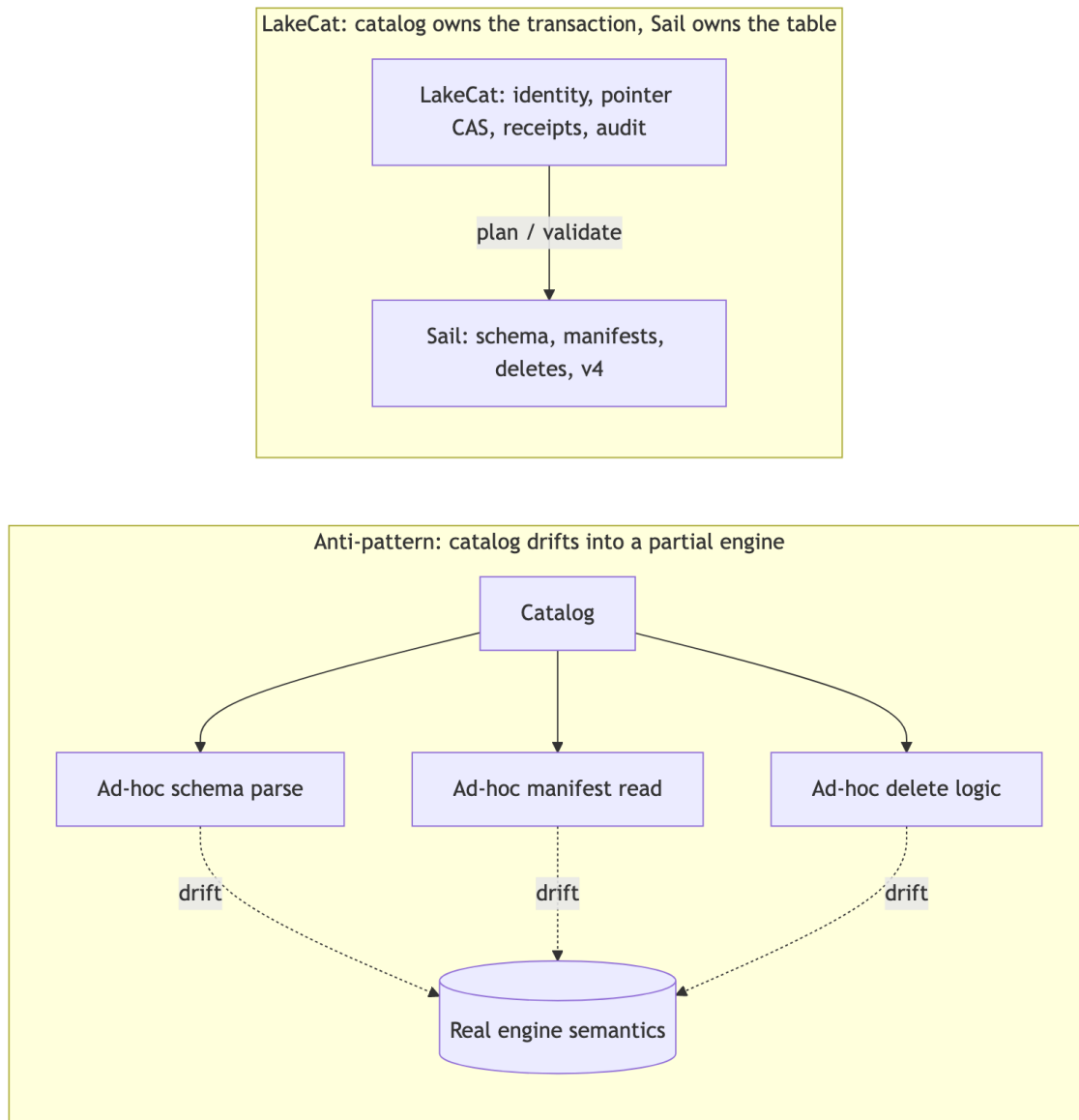


Figure 5: Diagram 5

LakeCat takes the opposite path: **the catalog owns the transaction; Sail owns the table semantics**. Sail is the right home because it is Rust-native and already holds the structures this needs — generated Iceberg REST models, table providers, manifest pruning, metadata-as-data paths, commit plumbing, and format-version handling. Anything that needs field-id binding, schema/partition evolution, manifest metrics, delete association, row lineage, v4 metadata trees, or snapshot/branch selection moves toward Sail.

This matters because intermediation, left passive, *loses* information. A pass-through catalog sees only the table name, the pointer, the caller, and maybe a credential request; the engine sees

schema, manifests, statistics, deletes, and filters; governance sees policy; lineage sees an after-the-fact event. Each system gets a shard of the truth, and the operational cost is concrete:

- policy can be checked before access but not carried *into* scan planning;
- credentials can be vended without proving why a raw-credential exception was allowed;
- lineage can say something happened but cannot bind to the exact governed plan, snapshot, policy, and metadata.

The next generation of callers — agents, notebooks, services, model pipelines — needs stronger guarantees that live precisely between catalog state and engine planning: a policy enforced before a scan is planned; column restrictions that narrow the projection before file tasks exist; policy-derived row predicates that become mandatory filters; a stateless `fetchScanTasks` that cannot widen a prior governed plan; short-lived, scoped, audited credentials; idempotent commit retries; and graph/lineage side effects that reflect committed state rather than best-effort handler work. A catalog too far from the engine can only check a policy and hand back a pointer, hoping the client preserves the restriction. LakeCat keeps the catalog boundary thin and binds its proof to Sail’s plan so the restriction is real.

1.7 The Architecture

Thin does not mean trivial. LakeCat owns the durable catalog state that must be correct even when external sinks are down, and nothing more. Its responsibilities are exactly:

- serve the Iceberg REST Catalog API for standard clients;
- model projects, warehouses, namespaces, tables, views, and storage profiles;
- persist metadata pointers and compare-and-swap commit history;
- validate idempotency keys and replay only matching commit bodies;
- resolve request identity from headers, bearer tokens, agents, and TypeDID envelopes;
- ask TypeSec for authorization decisions and persist the receipts;
- route scan planning and commit preparation through Sail;
- record audit and outbox events inside the catalog transaction;
- drain committed events to Grust and OpenLineage sinks;
- publish a QueryGraph bootstrap bundle.

Everything else is deliberately excluded — LakeCat does not invent a table format, fork manifest pruning, own graph traversal, decide security semantics, or author QueryGraph’s business model. Those belong to Sail, Grust, TypeSec, and QueryGraph respectively (Chapter 4).

1.7.1 The CatalogStore seam

All durable state sits behind one trait, so the memory store (for tests and embedded use) and the Turso store (for durable local deployments) are interchangeable. The interesting methods are the table lifecycle and the idempotent replay hook:

```
#[async_trait]
pub trait CatalogStore: Send + Sync + 'static {
    async fn create_namespace(&self, warehouse: &WarehouseName, namespace:
        Namespace)
        -> LakeCatResult<()>;
```

```

    async fn list_tables(&self, warehouse: &WarehouseName) ->
LakeCatResult<Vec<TableRecord>>;
    async fn create_table(&self, table: TableRecord) ->
LakeCatResult<TableRecord>;
    async fn load_table(&self, ident: &TableIdent) -> LakeCatResult<TableRecord>;

    // Optimistic pointer advancement: the catalog transaction.
    async fn commit_table(&self, ident: &TableIdent, commit: TableCommit)
        -> LakeCatResult<TableRecord>;

    // Exact idempotent replay: same key + same request hash -> stored response.
    async fn replay_table_commit(
        &self,
        ident: &TableIdent,
        idempotency_key: &str,
        idempotency_request_hash: &str,
    ) -> LakeCatResult<Option<TableRecord>>;

    // Compact pointer-log history for operators and QueryGraph.
    async fn table_commit_records(&self, ident: &TableIdent, start: u64, end:
Option<u64>)
        -> LakeCatResult<Vec<TableCommitRecord>>;
}

```

1.7.2 The read path

A read begins like a standard catalog request and ends with a *governed* Sail plan. The policy restriction becomes part of planning, not a note beside it.

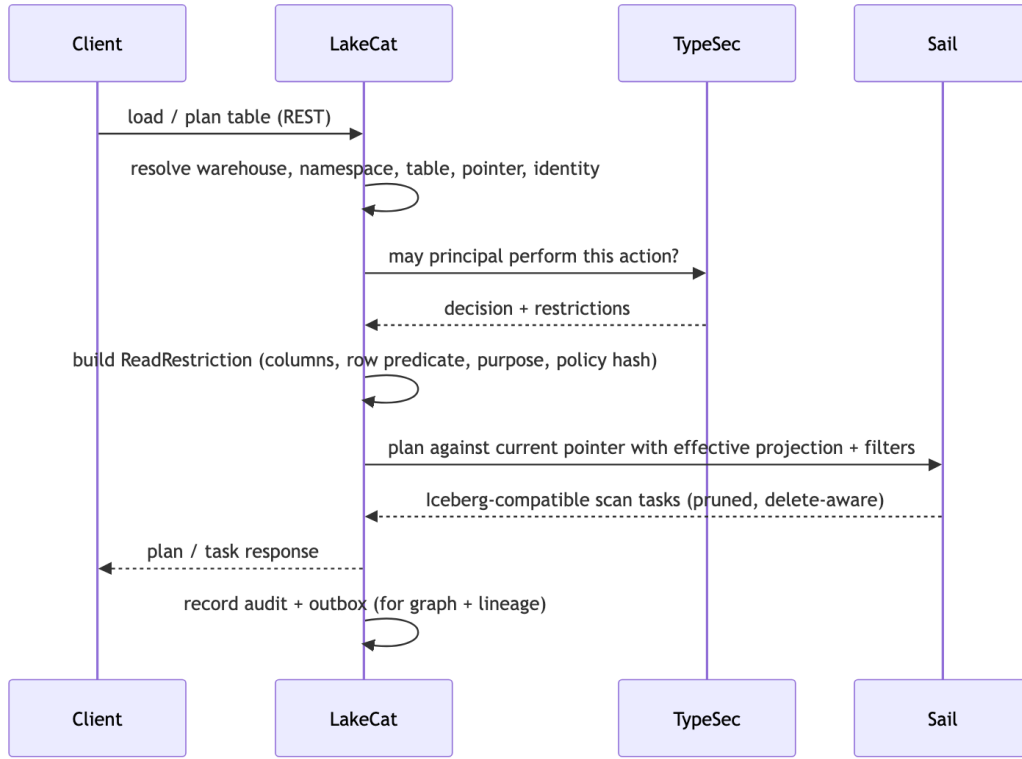


Figure 6: Diagram 6

The governing rule is **narrow, never widen**. An empty client projection under a column restriction means *the allowed columns*; a client projection may narrow further but cannot widen. LakeCat records both the requested and the effective projection (and stats fields) as replay evidence, and recomputes the restriction on every stateless `fetchScanTasks` so a stale token cannot expand back to all columns. Outbox admission enforces that the governed scan replay carries the same read-restriction at the top level and inside `authorization-receipt.context`, a nonblank purpose, and a positive `max-credential-ttl-seconds`, and rejects unknown fields — so graph, lineage, and QGLake evidence can never inherit a claim the receipt did not capture.

1.7.3 The commit path

The write path follows the same principle: LakeCat owns the transaction, Sail owns Iceberg validation and metadata assembly.

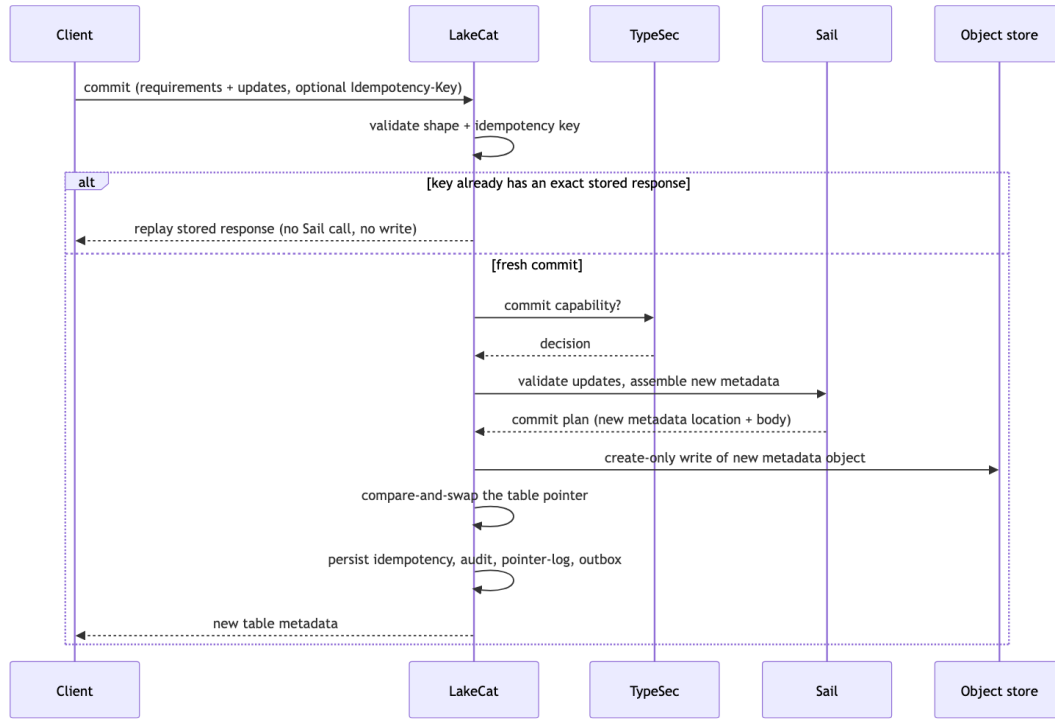


Figure 7: Diagram 7

Three invariant groups make this safe, each enforced once:

- **Metadata-write safety.** The new object must be a concrete child of the table's matched storage-profile prefix — never the current pointer, never the profile root, never a path with literal or percent-encoded dot segments, query strings, fragments, userinfo, or credential markers. Writes are create-only; if the store CAS loses after a write, LakeCat makes a bounded retry to delete the orphaned object and otherwise preserves the original conflict.
- **Idempotency.** Keys are 1–128 ASCII chars (Idempotency-Key / x-lakecat-idempotency-key, matching if both present). The same key + same body replays the stored response before any Sail call or write; the same key + a different body conflicts. Only audit-safe hashes are stored — never raw keys or secrets. Ordinary clients may commit without a key; they simply get no replay proof.
- **Redaction.** Every commit error — pointer conflict, overwrite, prefix mismatch, cleanup failure, backend error — reports sha256: hashes of the metadata location and failure detail, never raw object paths, storage roots, profile ids, or backend text. A conflict still looks like an ordinary Iceberg conflict to the client.

Commit records carry compact evidence — Iceberg format version, current snapshot id, request/response/idempotency/policy hashes, principal — so operators and QueryGraph can answer audit questions from the pointer log without parsing full metadata. Operators read it through a governed endpoint:

```
curl -s -H 'x-lakecat-principal: operator@example.com' \
  http://127.0.0.1:3000/management/v1/warehouses/local/namespaces/default/tables/
events/commits
```

That read itself enters the outbox as `table.commits-listed` and drains as lineage plus Commit graph anchors keyed by table and sequence — an auditable trail without giving anyone direct database access or turning LakeCat into a graph query engine.

1.7.4 The durable spine

The durable local store uses the Rust `turso` crate behind the `turso-local` feature. Object storage remains the source of Iceberg metadata files; Turso holds the atomic pointer and the control-plane memory: projects and warehouses, storage profiles, namespaces and tables, the metadata pointer log, idempotency records, soft deletes, policy bindings, audit events, and outbox events. This mirrors the Iceberg contract — metadata files describe the table; catalog state decides which file is current.

Two disciplines keep the spine trustworthy:

- **Decoded-row binding.** Every read reconciles the decoded JSON against the row’s own key columns before returning it — a warehouse against its project and storage root, a namespace against its path, a policy binding against its table and enforced flag, a storage profile against its prefix/provider/mode, a table body against its identity. A corrupted row index can therefore never point a valid-looking record at the wrong tenant, path, or policy.
- **Strict outbox draining.** LakeCat projects a batch to graph and lineage *first*, then acknowledges the whole batch; if projection fails, nothing is acknowledged, and an under-count acknowledgement is a hard error rather than a quiet partial success. The drain refuses unknown event types (they stay pending rather than vanish), rejects event-id/payload-hash drift, and validates governed-read and commit evidence — policy-hash arrays must be non-empty full digests, top-level and receipt read-restriction must match, purpose and TTL must be present — before any sink sees the event. Malformed source evidence stays available for retry instead of being promoted into a QGLake handoff.

1.8 The Siblings and the Engine Path

Chapter 4 named the four sibling repositories. This chapter shows how LakeCat talks to each — what it hands off, and what it deliberately refuses to own.

1.8.1 Grust: the graph boundary

Catalog events naturally form a graph: a server contains projects, a project contains warehouses, a warehouse contains namespaces and credential-rooted storage profiles, a namespace contains tables, and a table has columns, snapshots, commits, policies, scan plans, principals, and lineage runs. QueryGraph wants that graph — but LakeCat must not become a graph database.

LakeCat emits a bounded envelope of stable catalog facts through a catalog-facing sink; Grust owns the taxonomy, stores, traversal, and Cypher:

Server CONTAINS Project	Warehouse HAS_STORAGE_PROFILE StorageProfile
Project CONTAINS Warehouse	Table GOVERNED_BY Policy

Warehouse CONTAINS Namespace	Principal CAN_PLAN ScanPlan
Namespace CONTAINS Table	Commit DERIVED_FROM Snapshot
Table HAS_COLUMN Column	LineageRun USED_BY QueryGraphModel

High-cardinality file and manifest facts stay queryable through Sail’s metadata-as-data rather than being smuggled into the event sink. Storage-profile and credential-vend events replay with redacted evidence only — `secret-ref-present`, the provider, never the secret URI — so Query-Graph can see *that* a principal attempted credential-root access without seeing the credential. The `grust-turso-local` feature proves the boundary end to end: LakeCat writes catalog events into a Grust-owned Turso graph store and Grust Cypher reads them back, with LakeCat never parsing Cypher or executing traversal.

1.8.2 TypeSec: the authorization boundary

LakeCat is a policy *enforcement* point, not the author of security semantics. Every externally meaningful action — config reads; namespace and table lifecycle; scan planning; commit; credential vending; policy management; graph and lineage reads — passes through TypeSec. LakeCat gathers context (principal and agent DID, bearer subject, warehouse/namespace/table, columns, snapshot, requested credential duration, purpose, active bindings), asks TypeSec for a decision, persists the receipt with audit-safe hashes, and applies the restriction *before* Sail plans.

This is where ODRL becomes operational. A policy may say a principal reads only certain columns, only for a purpose, or only under a row predicate. LakeCat parses the minimal enforceable subset — accepting camel-case, kebab-case, and JSON-LD operand forms — and **fails closed**: missing or deny-shaped operators, blank allowed-column lists, blank purposes, or disagreeing purpose sources are rejected rather than guessed. Composition and reasoning stay in TypeSec; LakeCat does not grow a parallel security language.

Credential vending is the audited exception, not the default. Governed Sail-planned reads are the norm. When credentials must issue, TypeSec checks the `credentials.issue` capability for the exact secret reference and LakeCat returns only scoped, short-lived configuration — capping TTL to the tightest `max-credential-ttl-seconds` across all policy locations, replacing any issuer-supplied `lakecat.*` evidence with catalog-derived values, and recording only hashed prefixes in audit. The replay-admission rules from Chapter 5 (matching top-level and receipt restrictions, full digests, closed field sets) apply identically here, so a credential or raw-exception event cannot drift before it reaches graph, lineage, or QGLake proof.

1.8.3 Sail and the v3→v4 path

Sail is a Rust-native engine — Arrow, DataFusion, generated Iceberg REST models, catalog-provider seams, manifest pruning, metadata-as-data — so LakeCat can *ask* for typed Iceberg behavior instead of parsing just enough JSON to survive. The questions LakeCat sends to Sail are the ones that require table-format knowledge:

- which field ids satisfy this projection?
- which filters are enforceable at planning time?
- which manifests and files survive pruning?
- which delete files must accompany the selected data files?

- which scan tasks are children of a governed parent plan?
- which v4 fields are known, preserved as passthrough, or not yet safe to read?

LakeCat evolves under three rules: conform to Iceberg v3 for ordinary clients; preserve unknown/emerging v4 metadata without claiming settled semantics; prefer typed Sail support the moment it exists, using JSON passthrough only as a bridge. Today the v4 bridge is deliberately narrow — when LakeCat sees `format-version: 4`, `lakecat-sail` extracts the stable envelope (table UUID, location, schema id, snapshot id, sequence number, manifest-list path, default spec, field names), can plan a governed manifest-list scan task from it, and re-validates the signed plan task on `fetchScanTasks` so a stateless fetch can neither drift to a different manifest list nor widen the governed projection. Pruning and typed metadata-tree semantics wait for Sail-owned v4 support.

1.8.4 The semantic handoff: Croissant, CDIF, OSI, OpenLineage

QueryGraph needs a semantic picture, not just physical access. LakeCat publishes one without pretending to be QueryGraph. The bootstrap bundle carries Semantic Croissant and CDIF projections (dataset/field discovery), an OSI handoff of stable anchors, ODRL artifacts and TypeSec policy context, OpenLineage events for catalog changes and plans, a Grust-ready graph envelope, and a manifest that hashes every artifact. The boundary is careful: LakeCat publishes stable anchors and governed source metadata; QueryGraph authors the business model. Because OpenLineage events drain from the durable outbox in `created_at,event_id` order *after* the catalog transaction, lineage reflects committed state rather than a handler’s best-effort side effect.

1.8.5 The full stack

When LakeCat is done, a standard engine still loads and commits tables without knowing QueryGraph exists, while governed callers get the richer path:

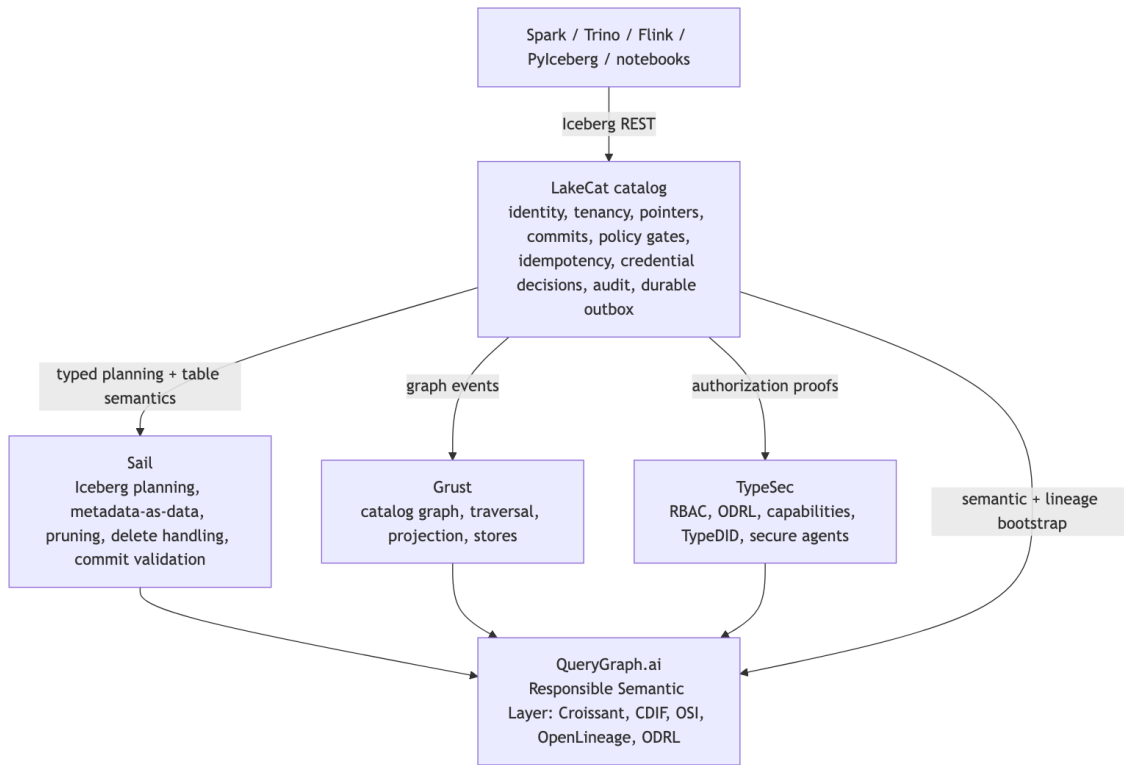


Figure 8: Diagram 8

1.8.6 Implementation shape

The workspace expresses the architecture directly:

```

crates/
lakecat-core      stable IDs, errors, time, config, content hashes
lakecat-api       Iceberg REST request/response adapters
lakecat-store     catalog state traits + Turso-backed implementation
lakecat-sail      Sail provider bridge and privileged planning client
lakecat-graph     catalog-facing Grust sink/adapters
lakecat-security  TypeSec integration and authorization receipts
lakecat-lineage   OpenLineage projection and event receipts
lakecat-querygraph Croissant/CDIF/OSI/ODRL/OpenLineage bootstrap projection
lakecat-service   axum service, middleware, auth, routing
lakecat-cli       admin, local demo, conformance, bootstrap export
  
```

Feature gates keep integrations honest, and embedded defaults stay safe for tests so a memory-store test never accidentally depends on a sibling repo:

```

sail-local        local Sail APIs for planning and provider integration
typesec-local     local TypeSec APIs for governance and TypeDID verification
grust-local       local Grust APIs for catalog graph projection
grust-turso-local Grust's Turso backend for durable catalog graph projection
turso-local       the Turso-backed durable store
  
```

The runtime honors the same line: without `sail-local`, the deferred Sail seam *rejects* scan planning rather than fabricating an empty plan, so any real read reflects the engine that interprets Iceberg metadata, never a catalog-shaped placeholder. Standard compatibility lives at `/catalog/v1`; management APIs, `/querygraph/v1/bootstrap`, and feature-gated Sail planning sit beside it, never in front of it. (The first-release gate and dependency contract that hold this together are covered in the release chapter.)

1.9 Workflow Examples

The catalog is easiest to understand by watching it participate in ordinary work. LakeCat should not ask users to think about graph, lineage, security, and Sail every time they read a table. Those systems should appear when they matter: at the boundary where a name is resolved, a policy is enforced, a plan is created, credentials are withheld or issued, and a durable event is replayed.

The examples below use one table, `local.default.events`, but the pattern is the same for larger warehouses. The important point is not the exact sample data. It is the catalog role in each workflow.

1.9.1 Starting The Catalog

A local operator starts LakeCat as an Iceberg REST catalog plus management surface:

```
cargo run -p lakecat-service --features sail-local,turso-local,typesec-local,grust-local
```

The standard catalog path is still `/catalog/v1`. The management and QueryGraph surfaces sit beside it:

```
/catalog/v1
/management/v1
/querygraph/v1/bootstrap
```

A simple health-oriented configuration read shows the split. Standard engines care about the Iceberg endpoints. Operators and QueryGraph care about the management and bootstrap endpoints.

```
curl -s http://127.0.0.1:3000/catalog/v1/config
```

The defaults intentionally separate compatibility from future capability:

```
{
  "defaults": [
    {"key": "lakecat.compatibility", "value": "iceberg-rest"},
    {"key": "lakecat.format.baseline", "value": "iceberg-v1-v3"},
    {"key": "lakecat.format.v4", "value": "extension-ready"},
    {"key": "lakecat.format.v4.bridge", "value": "json-passthrough"},
    {"key": "lakecat.format.v4.typed-sail", "value": "unavailable"}
  ]
}
```

That means LakeCat can preserve and replay emerging v4 metadata through the Sail JSON bridge, but it is not claiming typed Sail v4 semantics yet. The same defaults are stored in catalog config-read replay evidence, and malformed replay that omits the v4 bridge

posture is rejected before graph or OpenLineage projection. The replay defaults must also be ordinary string key/value entries with duplicate-free keys, so a saved outbox event cannot say both `lakecat.format.v4.typed-sail=unavailable` and `lakecat.format.v4.typed-sail=available`. Those key/value entries are closed over just key and value before replay is acknowledged. A saved config-read event cannot hide an extra compatibility, v4, integration, or application claim inside a default or override entry and have that claim travel into graph, OpenLineage, or QGLake config proof beside the checked catalog contract. LakeCat also rejects unsupported extra `lakecat.format.v4*` defaults, such as preview typed-Sail keys, because those would make the bridge posture sound more settled than the current Sail-owned typed v4 surface proves. Config overrides are held to the same honesty rule for v4 posture: until typed Sail v4 support is available, replay evidence cannot use an override to claim `lakecat.format.v4.typed-sail=available` or introduce another v4 bridge key. Catalog config replay now also preserves the advertised endpoint list. That is not a new protocol requirement for standard clients; it is proof that the configuration LakeCat projected to graph and OpenLineage still contained the ordinary Iceberg REST surface. Replay validation requires the config endpoint, namespace list/create endpoints, table create endpoint, table load endpoint, and table commit endpoint for both the default and warehouse-prefixed catalog routes before the config read can become compatibility evidence. Replay validation also requires LakeCat's governed access endpoints: plan, fetch-scan-tasks, and credentials. Those routes are not a new table format and not a QueryGraph dependency for ordinary reads. They are additive catalog APIs that let governed clients ask LakeCat, TypeSec, and Sail for proof-carrying plans, task fetches, or audited credential decisions over the same standard Iceberg tables. Replay validation also preserves the additive integration surfaces that make LakeCat useful as the QueryGraph foundation: `/querygraph/v1/bootstrap` and `/management/v1/lineage/drain`. These are not standard Iceberg REST table operations and they are not required for PySpark or another ordinary Iceberg client to load a table. They are LakeCat/QueryGraph/OpenLineage control-plane endpoints. Their presence in config evidence proves that a QGLake import, OpenLineage replay, or agentic management workflow saw the same integration contract that LakeCat later projects into graph and lineage systems.

That proof now survives into saved lineage-drain artifacts and compact QGLake hand-off summaries, not only service admission. A `catalog.config-read` drain summary carries three compact fields: the advertised config defaults, config overrides, and endpoint list. The compact handoff summary then promotes the same evidence into `lakecatReplayVerification.catalogConfigProof`, alongside the principal, authorization action, graph count, replay hashes, and OpenLineage hashes for the config-read event. QGLake verification checks those fields again when reading the saved drain, when reading the compact summary, and when comparing captured LakeCat replay sidecars. A handoff cannot keep the `catalog.config-read` event while dropping `lakecat.format.v4.typed-sail=unavailable`, adding a preview `lakecat.format.v4*` key, using an override to rewrite v4 posture, omitting the standard REST, governed access, bootstrap, or lineage-drain endpoints, or archiving a captured replay file whose config proof differs from the summary. Raw lineage-drain summary construction now applies the same fail-closed shape checks before that compact proof is returned: config defaults and overrides must remain `ConfigEntry` arrays with duplicate-free string keys and string

values, and endpoint evidence must remain a duplicate-free nonblank string array containing the required standard REST, governed plan/fetch, bootstrap, and lineage-drain routes LakeCat advertised at service replay. That makes the config proof replayable outside the service process. QueryGraph can trust that the compatibility and integration contract it imports is the same contract LakeCat admitted before graph and OpenLineage projection. The same rule now applies to raw `querygraph.bootstrap` summaries. Bootstrap itself is not standard Iceberg parlance: it is LakeCat/QueryGraph handoff evidence built beside the Iceberg REST path. Its source facts still describe standard catalog state such as warehouse identity, table count, view count, and accepted table/view stable ids, but its bundle hash, Grust graph hash, OpenLineage hash, Croissant/CDIF/OSI/ODRL artifact hashes, standards list, authorization receipt, and TypeSec-style request-identity hashes are additive LakeCat/QueryGraph/TypeSec proof. Raw summary construction now runs the same service replay validator over that bootstrap envelope before compact QGLake proof can inherit any of it. A saved drain cannot keep a `querygraph.bootstrap` event while shrinking the standards list, replacing artifact arrays with objects, drifting table/view counts away from verified manifests, inventing a malformed principal, or attaching short TypeDID/agent hashes to an otherwise valid-looking bootstrap proof. When config evidence carries optional tenant-root records, the same admission rule applies to sensitive roots: a raw `server-record.endpoint-url` must carry the matching full `endpoint-url-hash`, and a raw `warehouse-record.storage-root` must carry the matching full `storage-root-hash` before config discovery can be projected or archived as QGLake proof.

The bridge is intentionally conservative, but it should not reject Iceberg metadata that Sail has already decoded. Manifest expansion now emits null partition slots as JSON `null` and recursively encodes nested Sail partition literals into JSON objects, arrays, and explicit key/value map entries. That keeps standard Iceberg REST fetch responses usable for richer partition tuples without pretending LakeCat owns a full typed v4 implementation.

At this point the catalog is already doing more than route HTTP. It has a warehouse identity, a store, a governance engine, a Sail planning seam, a graph sink, and a lineage sink. Embedded defaults keep the local loop small, but the same trait boundaries can point to Turso, TypeSec, Grust, and Sail.

1.9.2 Registering The Warehouse Shape

An operator usually starts with management objects. A server groups projects. A project groups warehouses. A warehouse owns namespaces, tables, views, storage profiles, policy bindings, and the metadata pointer state that standard engines see through Iceberg REST.

```
curl -s -X PUT http://127.0.0.1:3000/management/v1/servers/prod \
  -H 'content-type: application/json' \
  -d '{
    "display-name": "Production LakeCat",
    "endpoint-url": "https://lakecat.example.com",
    "properties": {
      "owner": "platform"
    }
  }'
```

```
curl -s -X PUT http://127.0.0.1:3000/management/v1/projects/resilience \
-H 'content-type: application/json' \
-d '{
  "display-name": "Resilience Desk",
  "server-id": "prod",
  "properties": {
    "environment": "demo"
  }
}'
```

```
curl -s -X PUT http://127.0.0.1:3000/management/v1/projects/resilience/
warehouses/local \
-H 'content-type: application/json' \
-d '{
  "display-name": "Local QGLake Warehouse",
  "storage-root": "file:///tmp/lakecat/qglake",
  "properties": {
    "querygraph": "enabled"
  }
}'
```

These writes are not Iceberg table metadata. They are catalog control-plane state. LakeCat persists them durably, records authorization receipts, and writes outbox events. When the outbox drains, server, project, warehouse, and storage-profile and policy-binding changes become catalog graph events; the same management changes also become OpenLineage receipts. QueryGraph can later learn the management shape without requiring every Iceberg client to understand it. Project, server, and warehouse tenant-root replay is checked before projection: project evidence must carry a matching project id, optional valid server scope, and string-map public properties; server evidence must carry a valid server id, optional valid endpoint URL or full endpoint-url-hash, and string-map properties; warehouse evidence must carry a valid warehouse, project id, optional valid storage root or full storage-root-hash, and string-map properties. Service admission also closes those nested project, server, and warehouse record objects over their route-produced fields, so unexpected tenant-root, endpoint, or storage-root claims fail before acknowledgement, graph projection, OpenLineage projection, or QGLake proof can inherit them. LakeCat also closes the top-level management upsert payloads for `project.upserted`, `server.upserted`, and `warehouse.upserted`, so a replay sidecar cannot append unverified endpoint, storage-root, project-scope, lineage, graph, QueryGraph, or application claims beside checked route identity, nested record evidence, optional project scope, and authorization receipt evidence. The wrapped outbox envelopes for `project.upserted`, `server.upserted`, and `warehouse.upserted` are closed as well: only the audit event id, event type, and checked inner payload are accepted, which keeps tenant-root replay evidence from gaining extra management claims outside the schema LakeCat actually verifies. Policy-binding upsert replay is checked before projection too: the evidence must carry a valid policy id, warehouse, optional namespace/table scope, an enforcement flag, the captured ODRL material, and an `odr1`-hash that matches that material. The hash must be a full SHA-256-shaped digest before LakeCat compares it to the ODRL body. LakeCat does not reason

over that ODRL during replay, but malformed binding shape or drifted ODRL content proof fails closed before the policy anchor can be delivered to graph or lineage sinks. Service admission also closes the nested policy object over the route-produced fields, so unexpected ODRL, governance, scope, or enforcement claims fail before acknowledgement, graph projection, OpenLineage projection, or QGLake proof can inherit them. It also closes the top-level `policy-binding.upserted` payload, so a replay sidecar cannot append unverified ODRL, governance, scope, lineage, graph, QueryGraph, or application claims beside checked warehouse, policy object, ODRL content hash, enforcement state, and authorization evidence. The wrapped `policy-binding.upserted` envelope is closed too, so ODRL or governance claims cannot be smuggled beside an otherwise valid inner policy-binding replay payload. Raw lineage-drain summaries now use the same service replay validators for `policy-binding.upserted`, `project.upserted`, `server.upserted`, and `warehouse.upserted` before compact QGLake management proof inherits them. That keeps compact replay proof from becoming a looser path for malformed management ids, endpoint or storage-root hashes, ODRL hashes, wrapper fields, or authorization receipts. Those management upserts must also carry a valid authorization receipt principal, so the catalog graph and OpenLineage stream never accept actorless tenant-root, storage-profile, or policy mutations. Namespace lifecycle replay is checked before projection as well: create, load, and drop events must carry a valid warehouse and either a valid namespace path or non-empty namespace component array. A malformed namespace lifecycle event stays pending and reaches neither the Grust-facing graph sink nor OpenLineage. Service replay closes those namespace lifecycle payloads over event - type, authorization-receipt, warehouse, and namespace, so an archived create, load, or drop cannot attach unverified namespace, scope, replay, OpenLineage, or QueryGraph claims beside valid standard catalog evidence. It also closes the wrapped outbox envelope for namespace lifecycle events, so those claims cannot be placed beside an otherwise valid checked inner namespace payload. Raw lineage-drain summaries now reuse those same validators for `namespace.listed`, `namespace.created`, `namespace.loaded`, and `namespace.dropped`: namespace inventory counts must match the listed namespace paths and remain duplicate-free, lifecycle namespaces must be valid paths or component arrays, receipt actions must match the event type, and closed wrappers cannot carry unverified QueryGraph or lineage sidecars before compact QGLake standard catalog proof inherits the evidence. Catalog read replay has the same fail-closed shape: `catalog.config-read` events must carry a valid warehouse, and `namespace.listed` events must carry both a valid warehouse and an unsigned namespace count before the read evidence can be projected. These standard catalog reads and namespace lifecycle events must also carry a valid authorization receipt principal before delivery, so Iceberg-compatible control-plane replay remains attributable. Management-list replay is checked before delivery too: `policy-binding`, `project`, `server`, `storage-profile`, and `warehouse list` events must carry unsigned counts, warehouse-scoped lists must carry a valid warehouse, and optional project scope on warehouse-list replay must be a non-empty, syntactically valid project identifier before those reads can become replay evidence. Raw management-list summaries now use that same service replay validator before compact QGLake proof inherits the inventory. This keeps management proof as a LakeCat/QueryGraph control-plane extension around catalog state, not a loose JSON appendix: list events must still carry the event-matching management action, valid authorization receipt, closed wrapper schema, valid warehouse or project scope where applicable, required count-aligned ID arrays, and duplicate-

free valid identifiers before QueryGraph can treat server, project, warehouse, policy-binding, or storage-profile inventory as accepted proof. Malformed management identifiers are reported with hash evidence rather than raw tenant-root text. Raw summaries enforce the same closed payload schema, actor evidence, action match, and allowed decision before compact proof is built. They also require a nonblank receipt engine and RFC3339 checked_at timestamp, so a server.listed replay cannot be accepted under an unrelated table action, with unverified QueryGraph or OpenLineage claims, with missing or denied authorization, with an untraceable decision engine, with malformed decision time, or without a valid authorization principal. View replay is checked at the same boundary: view list events must carry valid warehouse, namespace, and count evidence, and the count must match the listed view names before graph or OpenLineage projection. View create/load/drop evidence must carry a valid warehouse, namespace, and non-empty view name before projection too. View list and lifecycle replay must also carry a valid authorization receipt principal before delivery, preserving actor evidence for QueryGraph view proofs. A view list is read-side catalog evidence, so the service requires its authorization receipt action to be view-load; a view-manage receipt is valid for mutations but not for replaying view.listed. Raw lineage-drain summaries enforce the same count binding and read-side action before compact QGLake proof is built, so an archived view inventory cannot inflate discovery counts or be accepted under mutation authority. View lifecycle replay is action-bound too: view.upserted requires view-manage, view.loaded requires view-load, and view.dropped requires view-drop before LakeCat emits graph or OpenLineage evidence. The nested view evidence is closed over the catalog route's view shape too: warehouse, namespace, name, store-assigned view-version, SQL, dialect, schema version, columns, and properties. If a sidecar also carries top-level warehouse or namespace evidence, it must match the nested view object before delivery. A replay sidecar cannot drift the view scope, add an extra QueryGraph, lineage, governance, or application claim inside that view object, and have it acknowledged as catalog evidence. Raw lineage-drain summaries now use the same service replay validators for view.listed, view.upserted, view.loaded, and view.dropped, so compact QGLake view proof cannot inherit a looser view list, action-drifted receipt, malformed version guard, or sidecar-modified lifecycle payload than full replay would accept. Table lifecycle replay now follows the same rule:

Active view state is protected before replay as well. A Turso row selected as warehouse local, namespace default, and view active_customers must decode to that same view before LakeCat returns it, lists it, updates it, or drops it. The memory store applies the same check to keyed active-view reads. This is not an Iceberg view extension; it is LakeCat's durable row/content guard around the control-plane view state that later produces view receipt chains and QGLake proof. create, load, delete, and restore events must carry a valid root table identity, and any payload warehouse, namespace, table-name, or soft-delete table evidence must agree with that identity before the event can be acknowledged. Their authorization receipts must also carry the matching lifecycle action: table-create, table-load, table-drop, or table-restore, along with an allow decision, engine, and checked-at timestamp. Delete replay preserves the table-format generation through nested soft-delete format-version evidence. Older sidecars may spell that field as format_version, but LakeCat rejects a soft-delete object that carries both aliases before acknowledgement, graph projection, or OpenLineage projection. That keeps archived delete

proof from hiding conflicting table-format evidence behind the spelling LakeCat happens to read first. Server endpoint URLs are operator-visible management metadata, so LakeCat keeps them deliberately plain: they must be absolute http or https URLs, and they cannot include query strings, fragments, or URI userinfo. Rejected submissions return `server-endpoint-url-hash=sha256:... evidence` rather than echoing the submitted endpoint. Warehouse replay does not forward the raw storage root to graph or lineage consumers. The drained payload replaces `storage-root` with `storage-root-hash`, so QueryGraph can bind tenant evidence to a configured root without receiving the local filesystem path or bucket URI.

1.9.3 Storage Profiles And Credential Roots

Storage profiles bind a warehouse to physical storage roots and credential issuance policy. A local profile can return scoped local file configuration. A remote profile should usually reference a secret store and require TypeSec to authorize issuance before any resolver sees the secret reference. Warehouse storage roots are validated before memory or Turso persistence: query strings, fragments, URI userinfo, and literal or percent-encoded dot path segments fail with `warehouse-storage-root-hash=sha256:... evidence` rather than echoing the submitted root. LakeCat rejects profiles whose declared provider conflicts with the URI scheme of the location prefix, so a credential root cannot claim to be local while pointing at an S3 prefix. Those provider/location mismatch errors follow the same redaction rule as replay: they name provider labels and a `storage-profile-prefix-hash=sha256:... evidence`, not the raw storage root. When multiple profiles in the same warehouse could match a table, LakeCat uses the longest matching location prefix. If two profiles tie on that longest prefix, LakeCat fails closed rather than guessing which credential root or metadata-object boundary should apply. The ambiguity error reports the competing profile ids and `location-prefix-hash=sha256:... evidence`, not the raw storage root. The location prefix itself must be plainly addressed: LakeCat rejects literal and percent-encoded dot path segments, query strings, fragments, and URI userinfo before the profile can reach memory or Turso persistence. Traversal-shaped or decorated storage-profile prefixes fail with `storage-profile-prefix-hash=sha256:... evidence` rather than echoing the raw prefix, token-like query value, or embedded userinfo. The management route pins the same operator-facing behavior, so a rejected storage-profile upsert does not leak the submitted decorated prefix. It also rejects unsafe issuance-mode combinations: `local-file-no-secret` is for file storage only, while `short-lived-secret-ref` is for configured remote providers such as S3, GCS, and Azure. Those mismatches fail with the same `storage-profile-prefix-hash=sha256:... evidence` anchor and without echoing the raw storage prefix or submitted `secret-ref`, so operators can correlate the credential-root error without turning the management API into a credential leak. The `public-config` map is only for non-secret routing hints such as region, endpoint labels, and operational purpose. LakeCat rejects secret-looking public keys and values, so raw tokens, passwords, access keys, and credential query parameters must move behind `secret-ref` and the TypeSec-authorized resolver path. That rule is enforced both when a profile is built from a management request and when a storage profile is revalidated before memory or Turso persistence, so deserialized control-plane records cannot bypass the public-config guard. Public-config validation failures also use `public-config-key-hash=sha256:... evidence` rather than echoing the submitted key or value, because even a rejected key name may contain a secret-looking identifier. LakeCat also reserves

credential-evidence keys such as `lakecat.storage-profile-id`, `lakecat.storage-provider`, `lakecat.credential-mode`, and `lakecat.max-credential-ttl-seconds`; operators may still publish non-secret hints such as `lakecat.endpoint`, but they cannot shadow catalog-owned proof in the eventual credential response. Replay admission re-checks the same public-config shape for `storage-profile.upserted` and `credentials.vend-attempted`, so archived storage-profile or credential sidecars cannot smuggle reserved LakeCat proof keys or secret-like public hints after the management/store guards have already run. The `secret-ref` field itself must remain a clean external secret-store locator: LakeCat rejects query strings, URI fragments, and userinfo before persisting a storage profile, so token-like material cannot hide inside a decorated secret URI. It also rejects literal and percent-encoded dot path segments, so a credential root cannot rely on traversal-like spelling before a resolver sees it. Unsupported credential-root schemes and malformed secret-root paths are rejected with `secret-ref-hash=sha256:... evidence` instead of echoing the submitted secret reference. The same hash-only rule applies to invalid secret-ref URI syntax, decorated URI forms, and embedded secret-like material such as password or token assignments. Management upsert and list responses follow the same redaction rule. They do not echo the raw `secret-ref`; they return `secret-ref-present`, `secret-ref-provider`, and `secret-ref-hash` so operators and QueryGraph can verify that a credential root exists and correlate it without learning the secret-store path. When LakeCat selects the storage profile for a table, the location prefix is also matched on a storage-root boundary. A profile for `s3://lakecat/events` applies to that exact root and to children such as `s3://lakecat/events/tenant-a/table`, but it does not apply to a sibling path such as `s3://lakecat/events-shadow/table`. That keeps credential roots from accidentally governing more storage than their configured prefix describes; if no stored profile matches, LakeCat falls back to an inferred governed-read profile for the table location. The same check runs after the credential issuer returns: `loadCredentials` rejects any returned prefix broader than the selected profile before LakeCat attaches canonical response evidence, so a custom issuer cannot widen catalog-owned storage scope. The rejection exposes only `credential-prefix-hash` and `storage-profile-prefix-hash` evidence, and LakeCat records no `credentials.vend-attempted` replay event for that failed issuer response. Public configuration on the selected storage profile remains non-secret evidence. Service replay and raw lineage-drain credential summaries both re-check that public-config keys are non-secret, values are strings, and LakeCat-reserved credential-evidence keys do not appear there. Rejections carry `public-config-key-hash=sha256:...` rather than the raw key or value, so compact credential proof cannot smuggle token-shaped hints or structured secret blobs beside an otherwise valid credential-vend event.

```
curl -s -X PUT \
  http://127.0.0.1:3000/management/v1/warehouses/local/storage-profiles/local-
events \
  -H 'content-type: application/json' \
  -d '{
    "location-prefix": "file:///tmp/lakecat/qglake/events",
    "provider": "file",
    "issuance-mode": "local-file-no-secret",
    "public-config": {
```

```

        "lakecat.purpose": "developer-loop"
    }
}'

curl -s -X PUT \
  http://127.0.0.1:3000/management/v1/warehouses/local/storage-profiles/s3-events \
  -H 'content-type: application/json' \
  -d '{
    "location-prefix": "s3://lakecat/events",
    "provider": "s3",
    "issuance-mode": "short-lived-secret-ref",
    "secret-ref": "vault://kv/lakecat/events",
    "public-config": {
      "lakecat.region": "us-west-2",
      "lakecat.purpose": "production-events"
    }
  }'
```

The catalog row stores the public profile and secret reference, not raw cloud keys. A later credential request is checked against TypeSec and against the effective read restriction for the target table. Agents with fine-grained table restrictions are steered to governed Sail-planned reads instead of raw credentials. Trusted humans can receive audited standard credentials only when policy allows the exception. For production secret managers, LakeCat keeps a provider-dispatch seam rather than hard-coding credentials into catalog state. `vault://` can resolve through the built-in Vault HTTP backend when Vault environment configuration is present. `aws-sm://`, `gcp-sm://`, and `azure-kv://` can dispatch to explicitly configured provider backends after TypeSec authorizes the exact `secret-ref` resource. They can also use LakeCat's built-in file-backed provider roots for local or single-node deployments: `LAKECAT_AWS_SECRETS_MANAGER_FILE_DIR`, `LAKECAT_GCP_SECRET_MANAGER_FILE_DIR`, and `LAKECAT_AZURE_KEY_VAULT_FILE_DIR`. Each directory contains JSON credential config files named as the full SHA-256 digest of the exact secret reference, without the `sha256:` prefix, plus `.json`. For example, `gcp-sm://lakecat/events` is authorized as that exact TypeSec resource and then resolved from a hash-named JSON file under the configured GCP root. If no backend is configured, those providers fail closed with an operator-readable not-configured error, and denied TypeSec decisions do not call the backend or read the file at all. Configured provider backends receive the same policy-derived `max-credential-ttl-seconds` cap that LakeCat records in the read restriction, and returned credentials must preserve that cap in `lakecat.max-credential-ttl-seconds`. LakeCat rewrites duplicate TTL config entries into one effective value before returning credentials, preserving a stricter issuer TTL when it is valid and otherwise falling back to the policy cap. It also rewrites LakeCat-owned profile, provider, mode, principal, and governed-read-required evidence after issuance. For `secret-ref`-backed profiles it also derives `lakecat.secret-ref-provider` and `lakecat.secret-ref-hash` from the selected storage profile, so a cloud secret backend cannot make the response look like a different catalog decision, secret-provider path, or secret-reference anchor. Replay admission treats that evidence as structural too: `secret-ref` providers and hashes must be nonblank when

secret-ref-present is true, and provider/hash fields must be absent when secret-ref-present is false, no matter how a corrupted pending event encodes them. The service tests for the REST credential endpoint prove this response shape directly, not just through helper functions. LakeCat also rejects any credential whose returned prefix is outside the storage profile's location-prefix, so a misconfigured cloud secret backend cannot widen a table's storage scope after TypeSec has authorized the secret reference. That failure remains hash-only and stops before credential-vend replay evidence is recorded. The audit event for the credential attempt records redacted credential-response-evidence: the response prefix is hashed, LakeCat-owned proof fields are kept as canonical values, and issuer-owned config is hashed rather than copied. That keeps OpenLineage and QueryGraph replay useful without turning lineage into a credential leak. For secret-ref-backed profiles the redacted response evidence includes the catalog-derived lakecat.secret-ref-provider and lakecat.secret-ref-hash, while the storage-profile replay evidence includes secret-ref-provider and a full secret-ref-hash; outbox admission rejects any credential response whose provider or hash proof drifts from the selected profile before graph or OpenLineage projection. The nested storage-profile proof is still checked even when no credentials are returned: provider and issuance mode must be compatible, and secret-reference presence must match the mode. That keeps blocked credential attempts from projecting a weaker credential-root proof than storage-profile management would accept. Raw lineage-drain summaries now enforce that same nested storage-profile posture before returning compact event proof. A summary cannot carry a raw secret-ref, a short location-prefix-hash, a short secret-ref-hash, a provider/hash field when secret-ref-present is false, a missing provider/hash when it is true, or a provider/issuance-mode combination that service replay would reject. That closes the gap between accepted replay and the compact lineage-drain artifact QueryGraph later imports. The storage-profile and credential-vend service tests pin that producer-side location-prefix-hash evidence is already a full SHA-256 digest before QGLake receives the compact locationPrefixHash proof. The trusted-human credential-vending route test pins that the committed outbox payload contains this redacted proof for the audited raw-credential exception path. The blocked-agent route pins the other side of the same contract: when Sail-planned reads are required and no raw credentials are returned, the outbox records an explicit empty credential-response-evidence array rather than leaving replay to infer why no credential proof exists. A not-configured resolver error reports the provider label and a secret-ref-hash=sha256:... value, not the raw secret URI, so the operator can correlate configuration without leaking the credential root. Resolver validation errors for malformed Vault and TypeSec environment references follow the same rule: wrong schemes, missing Vault mounts or paths, and invalid environment-variable names produce hash evidence instead of echoing the malformed secret reference. Generic provider detection and resolver URI parsing follow that rule too, including unsupported provider schemes, so malformed credential-root strings cannot leak through production resolver diagnostics. Once a configured resolver is authorized to run, backend lookup and secret payload parse failures still stay hash-only: LakeCat returns the secret-reference hash and an error-detail hash instead of the environment variable name, Vault path, token, namespace, backend exception text, cloud secret-manager ARN or account path, or malformed secret fields. That rule applies both to the built-in Vault and environment resolvers and to explicitly configured AWS Secrets Manager, GCP Secret Manager, and Azure Key Vault style backend seams, including the file-backed provider roots.

The file-backed roots are not a claim that LakeCat has cloud SDK support for those providers; they are a redacted built-in backend that lets the same production-shaped secret-ref dispatch run locally while SDK resolvers are added later. Secret payload parsing also rejects malformed credential configuration before issuance, including blank config keys in either object-shaped secrets, ConfigEntry-array secrets, or Vault's nested data object. That keeps secret-manager output from turning into ambiguous Iceberg client credential configuration. When storage-profile changes replay into lineage/OpenLineage evidence, LakeCat does not forward the full secret-store URI or raw storage root. The committed audit/outbox payload keeps secret-ref-present and secret-ref-provider so QueryGraph can verify that a production credential root exists without learning the Vault, cloud secret manager, or TypeSec environment path. It also records a full location-prefix-hash instead of raw location-prefix, so replayed evidence can bind a credential root to a storage scope without exposing the bucket, path, or local filesystem root to downstream consumers. Warehouse replay follows the same shape: storage-root becomes storage-root-hash before graph and lineage projection, keeping the tenant root replayable without exposing the raw root itself. The drain also rejects unsafe storage-profile upsert evidence before delivery: storage-profile.upserted must carry a full location-prefix-hash and must not carry raw location-prefix or raw secret-ref values. If secret-reference presence is true, the replay evidence must carry a provider and full secret-ref-hash; if presence is false, provider and hash evidence must be absent. The same admission check validates the credential-root identity before projection: profile id must be non-empty, the nested warehouse must be valid and match any top-level warehouse field, and provider plus issuance mode must use LakeCat's supported storage-profile vocabulary. Service admission also closes the nested storage-profile object over the redacted producer schema for both storage-profile.upserted and credentials.vend-attempted replay. Unexpected nested storage-profile fields fail before acknowledgement, graph projection, OpenLineage projection, or QGLake proof can inherit unverified credential-root or storage-scope claims. Provider and issuance-mode compatibility is replay-checked as well: local-file-no-secret is only valid for the file provider, and short-lived-secret-ref is only valid for cloud object providers. Secret-reference presence must also agree with issuance mode: short-lived secret-ref profiles must carry redacted secret-reference proof, while governed and no-secret profiles cannot carry secret-reference proof. The drain summary lifts the same proof into compact fields alongside the profile id and provider. QGLake replay verification now checks the provider and issuance-mode pair in both compact handoff summaries and raw lineage-drain artifacts, so a saved handoff cannot turn a local file credential root into an S3 root, or claim a secret-reference mode on a file provider, without being rejected before import. That means a saved handoff can prove the credential root was configured without handing the next system a secret path. Raw storage-profile summaries now use the same service replay validator before the compact proof is built. That matters because storage-profile.upserted is a LakeCat/TypeSec credential-root management event, not an Iceberg metadata extension. Standard Iceberg clients do not need it to load a table, but agents and QueryGraph need it to know which redacted storage root and secret-reference posture were admitted by the catalog. A raw lineage drain can no longer keep a storage-profile upsert while dropping the authorization receipt, drifting the top-level warehouse from the nested profile, using a malformed profile id, turning a file provider into a secret-ref

profile, adding a raw location-prefix, or sneaking LakeCat-reserved public-config keys beside an otherwise hash-shaped credential-root proof.

1.9.4 A PySpark User Reads Iceberg

A PySpark user should not need to know about QueryGraph. They configure Spark's Iceberg REST catalog and point it at LakeCat:

```
from pyspark.sql import SparkSession

spark = (
    SparkSession.builder
        .appName("lakecat-events")
        .config("spark.sql.catalog.lakecat", "org.apache.iceberg.spark.SparkCatalog")
        .config("spark.sql.catalog.lakecat.type", "rest")
        .config("spark.sql.catalog.lakecat.uri", "http://127.0.0.1:3000/catalog/v1")
        .config("spark.sql.defaultCatalog", "lakecat")
        .getOrCreate()
)

events = spark.table("default.events")
events.select("event_id", "severity").where("severity = 'critical']").show()
```

For an unrestricted principal, the flow looks like a normal Iceberg read:

1. Spark asks LakeCat to resolve default.events.
2. LakeCat checks the principal and table capability.
3. LakeCat returns an Iceberg-compatible table response.
4. Spark plans the read using Iceberg metadata.

For a governed principal, the more interesting path is server-side planning. The user request still looks ordinary, but LakeCat derives the mandatory restriction before Sail sees the plan. If policy allows only event_id and severity, then a wider client projection is narrowed:

```
client asks:      event_id, severity, raw_payload
policy allows:    event_id, severity
Sail receives:    event_id, severity
```

The catalog does not trust the client to remember that. The restriction is re-applied when scan tasks are fetched, and the audit payload records the policy hash, narrowed columns, row predicate, storage location, metadata location, principal, requested stats fields, and effective stats fields. The fetch response also carries LakeCat extension evidence for the exact required-projection and required-filters derived from the authorized capability. That makes a stateless fetchScanTasks replay prove the restriction was re-applied, not merely that the original policy object was echoed. The QGLake fixture verifier checks those fields directly when it fetches scan tasks, so a local acceptance run fails if the response drops either the narrowed projection or the mandatory row predicate proof. The same verifier now checks that the exported policy binding, scan planning extension, and fetch extension all preserve the server-derived purpose and policy-derived max-credential-ttl-seconds cap before compact replay proof can

be accepted. The scan-planned replay proof also carries `plannedRequestedStatsFields` and `plannedEffectiveStatsFields`, so QGLake can prove a broader stats request, the server-derived narrowing, and the final effective stats fields that match the allowed columns. Saved handoff summaries and captured replay output are rejected if those fields disappear, widen, or drift apart. Planned and fetched scan replay also reject non-LakeCat, decorated, or credential-bearing plan-task values before graph, OpenLineage, or QGLake proof can inherit them. It also cross-checks the embedded ODRL policy projection against the structured binding so QueryGraph cannot import a stale copy while LakeCat verifies a different one.

1.9.5 A Notebook Requests Credentials

Credential vending is deliberately different from scan planning. Returning storage credentials gives the client broader power than returning a governed task list, so LakeCat treats it as an exception path.

```
curl -s \
  -H 'x-lakecat-principal: agent:triage' \
  http://127.0.0.1:3000/catalog/v1/local/namespaces/default/tables/events/
credentials
```

For an agent bound by a fine-grained restriction, LakeCat should fail closed:

```
{
  "credentials": [],
  "lakecat:credential-block-reason": "fine-grained read restriction requires
Sail-planned reads"
}
```

That empty credential response is not a missing feature. It is the intended agentic posture. The agent should ask LakeCat to plan the read through Sail, not receive raw storage reach. The audit event records the decision and the lineage outbox can replay the credential-vend attempt with the same block reason.

For a trusted human principal, policy can allow an audited raw credential exception. LakeCat still records the same context:

```
principal: analyst:maya
principal-kind: human
table: local.default.events
decision: raw credential exception allowed
reason: trusted human principal
restriction: present in receipt
lineage: credential vend attempted
```

The contrast matters for operators. They can prove that agents were kept on the governed path while a human exception was explicit, policy-backed, and replayable.

1.9.6 A View Becomes Part Of The Catalog Story

Views are catalog objects too. LakeCat stores durable view records with SQL, dialect, schema version, typed columns, properties, creator, and warehouse scope. They can be managed through the management API or through warehouse-prefixed catalog routes.

```
curl -s -X PUT \
  http://127.0.0.1:3000/management/v1/warehouses/local/namespaces/default/views/
events_view \
  -H 'content-type: application/json' \
  -d '{
    "sql": "select event_id, severity from default.events where severity =
'\''critical'\''",
    "dialect": "spark-sql",
    "schema-version": 1,
    "columns": [
      {
        "name": "event_id",
        "data-type": {"type": "long"},
        "nullable": false,
        "comment": "Stable event identifier"
      },
      {
        "name": "severity",
        "data-type": {"type": "string"},
        "nullable": false,
        "comment": "Operational severity"
      }
    ],
    "properties": {
      "owner": "resilience-desk"
    }
  }'
```

This is not Iceberg business metadata glued onto a table. It is catalog state about a view object. LakeCat records `view.listed`, `view.upserted`, `view.loaded`, and `view.dropped` audit events. Outbox replay projects listing reads to namespace-scoped graph and OpenLineage evidence, and projects single-view changes and reads to catalog-facing View graph events plus LakeCat OpenLineage view dataset receipts. QueryGraph bootstrap can then include views with OSI hashes, store-assigned view versions, view-aware graph edges, and OpenLineage view counts. Before any view lifecycle event is acknowledged from replay, LakeCat closes the top-level lifecycle payload over the fields current producers emit and checks that the nested view object contains only the catalog's view fields: warehouse, namespace, name, view-version, SQL, dialect, schema version, columns, and properties. That same replay gate rejects top-level warehouse or namespace evidence that drifts from the nested view object. That keeps a replay bundle from smuggling unverified graph, lineage, policy, QueryGraph, or application facts beside or inside view lifecycle evidence, and it prevents a view event from being projected under a different catalog scope than the view record itself. LakeCat applies the same posture one layer lower in the Turso local store:

active-view load, namespace list, guarded mutation, and drop paths decode the view record and bind it back to the durable views row columns. A corrupted row cannot silently remap an active view into another namespace or name before that view becomes replay, graph, OpenLineage, or QueryGraph bootstrap evidence. The lineage-drain summary also carries compact view replay identity:

```
{
  "name": "events_view",
  "view-version": 1,
  "schema-version": 1,
  "dialect": "sql"
}
```

That view-version is assigned by the durable store on each upsert, not by the caller. It is the first compatibility bridge toward Iceberg view commit history: QueryGraph can compare a bootstrap view artifact with the catalog's current view version today. A caller that wants optimistic commit behavior can include expected-view-version on the next upsert:

```
curl -s -X PUT \
  http://127.0.0.1:3000/management/v1/warehouses/local/namespaces/default/views/
events_view \
  -H 'content-type: application/json' \
  -d '{
    "sql": "select event_id from default.events where severity =
'\''critical'\''",
    "dialect": "spark-sql",
    "schema-version": 2,
    "expected-view-version": 1
  }'
```

If another writer has already advanced the view, LakeCat returns a conflict before it replaces the current view or appends a receipt. Omitting the field keeps the compatibility behavior: the store assigns the next version. The guard must be a positive store-assigned version. expected-view-version=0 is rejected as a bad request before LakeCat changes the active view or appends any view-version receipt. The same guard can protect deletion:

```
curl -s -X DELETE \
  'http://127.0.0.1:3000/management/v1/warehouses/local/namespaces/default/views/
events_view?expected-view-version=2'
```

If the current view is no longer version 2, LakeCat returns a conflict before it removes the view or appends a tombstone receipt. Accepted guarded mutations also carry their expected-view-version into the audit/outbox payload. During lineage drain, LakeCat turns that into compact replay evidence, so QueryGraph can distinguish “the replacement happened at version 2” from “the replacement was guarded by version 1 and then produced version 2.” Replay admission rejects view lifecycle evidence that omits the positive store-assigned view-version or carries a non-positive guarded expected-view-version, so graph and OpenLineage sinks never observe a versionless view lifecycle fact. Replay admission also checks the authorization receipt action:

upsert uses `view-manage`, load uses `view-load`, and drop uses `view-drop`, so a valid view artifact cannot be replayed under a weaker or unrelated catalog action.

LakeCat also writes a compact view-version receipt in the durable store. The receipt records the stable view id, assigned version, previous version, previous receipt hash, content hash, principal, operation, and timestamp. That makes the compact receipt list a hash chain: version 2 points at the version 1 receipt hash, and a later tombstone points at the last upsert receipt hash. Fuller version-log semantics remain a Sail-aligned implementation target. When a view is dropped, LakeCat appends a compact tombstone receipt instead of inventing a new view version: the receipt keeps `view-version` at the last durable version, sets operation to drop, links to the previous receipt, and preserves the last content hash so QueryGraph or an operator can prove which catalog view state was removed. If the same view name is later recreated, the new upsert continues after the latest receipt in that durable chain. A create, drop, and recreate sequence therefore looks like version 1 upsert, version 1 drop tombstone, version 2 upsert linked to the tombstone receipt, not two unrelated version-1 chains for the same stable view id.

```
{
  "event-type": "view.upserted",
  "view-warehouse": "local",
  "view-namespace": ["default"],
  "view-name": "events_view",
  "view-stable-id": "lakecat:view:local:default:events_view",
  "view-version": 2,
  "expected-view-version": 1,
  "graph-events": 2,
  "lineage-events": 1,
  "replay-event-hashes": ["sha256:..."],
  "replay-open-lineage-hashes": ["sha256:..."]
}
```

A `view.listed` replay is intentionally namespace-scoped. It records the warehouse, namespace, view count, graph and lineage projection counts, and receipt hashes for the list read without fabricating a single `view-stable-id`.

QGLake acceptance compares both that `view-stable-id` and `view-version` with the accepted QueryGraph bootstrap view artifacts. That closes a small but important gap: the bootstrap bundle may say a view was exported, and the drain evidence can now prove the corresponding view catalog event was replayed at the same durable catalog version with graph and lineage receipts. When the event was guarded, QGLake replay JSON also includes `expectedViewVersion`, preserving the optimistic version that LakeCat checked before accepting the mutation. The live QGLake fixture uses that path for deletion: after QueryGraph accepts the transient view in the bootstrap bundle, the fixture drops it with `expected-view-version` equal to the accepted durable view version.

When QueryGraph bootstrap is replayed through the outbox, LakeCat includes only compact receipt hashes:

```
{
  "event-type": "querygraph.bootstrap",
  "view-artifact-count": 1,
  "view-version-receipt-hashes": ["sha256:..."]
}
```

QGLake acceptance requires one non-empty receipt hash for each accepted view artifact. That keeps normal Iceberg view access standard, but gives the QueryGraph handoff a durable proof that the exported view version came from LakeCat's catalog spine. Compact handoff verification also binds those bootstrap receipt hashes to `viewReceiptChainProof.views[].acceptedReceiptHash` exactly, so a saved summary cannot splice a valid-looking QueryGraph bootstrap receipt array from another accepted view proof. Raw lineage-drain summary construction now also rejects malformed `table-artifacts` or `view-artifacts` fields instead of counting them as zero, so a corrupted bootstrap artifact claim cannot vanish from the handoff evidence before QueryGraph import. The fixture also exercises the deletion side of the same workflow: it creates a transient view, accepts a QueryGraph bootstrap that contains that view, drops the view, reads the receipt chain through the governed management endpoints by view name and namespace, and then requires lineage-drain replay to include `view.dropped`, `view.version-receipts-listed`, and `view.version-receipt-chains-listed` evidence with non-empty tombstone receipt hashes and namespace chain hashes. The service route tests pin those produced `receipt-hash`, `view-hash`, and `chain-hash` fields as full SHA-256 digest evidence before the QGLake verifier consumes them. LakeCat also validates the ordered `previous-receipt-hash` links before marking a namespace chain as `chain-verified`, so QueryGraph can reject a replay that contains hashes but not a coherent chain. The lower store layer now applies the same structural guard when view receipts are read: both memory and Turso-backed receipt reads reject forged `previous-receipt-hash` links before service replay, OpenLineage projection, or QGLake handoff can consume the chain. The mutation path uses that same guard before extending history. A guarded or unguarded view upsert/drop first validates the existing durable receipt chain, then computes the latest receipt hash, and only then appends the next receipt. If a stored receipt has a forged `previous-receipt-hash`, LakeCat rejects the new mutation before changing the active view record or writing another receipt. For the Turso-backed store, LakeCat also compares the decoded receipt JSON to the row/query scope. A receipt row selected for one warehouse, namespace, and view cannot carry JSON that claims another view and still be returned, grouped into a namespace chain, or used as the latest receipt for the next mutation. That comparison now includes the durable receipt row columns themselves: `view_version_receipts.warehouse`, `namespace_path`, and `view_name` must agree with the decoded receipt before LakeCat returns receipt history or appends the next receipt. A valid-looking receipt body cannot ride a corrupted row index into another namespace or view.

The verifier is fail-closed on version progression too. The first receipt must be a version-1 upsert with no previous version or receipt hash; zero-version chains, first-receipt tombstones, and first receipts with forged previous-link fields fail before the chain can be marked verified. Later upserts must point at the previous receipt hash and advance exactly one durable view version. Drop tombstones must point at the previous receipt hash, cite the previous view version, and keep

view-version equal to the accepted version that was deleted. Unsupported operations and forged previous-receipt-hash links fail the same check. That lets QueryGraph reject a chain that is cryptographically linked but lies about how the catalog view advanced. Replay admission also binds the chain header back to the receipt body. A verified chain cannot claim a larger receipt-count, a different latest-view-version, a different latest-operation, or a tombstone flag that does not agree with the last receipt. That prevents namespace-level receipt-chain evidence from shedding receipts or presenting a forged head while keeping individually plausible receipt links. The top-level hash arrays are bound to the same structure: chain-hashes, receipt-hashes, and drop-receipt-hashes must exactly cover the nested verified chains, receipt bodies, and tombstone receipts. A replay event cannot declare a convenient hash set that is merely SHA-256-shaped while the embedded chain tells a different story.

QueryGraph and operators can also read the compact receipt chain directly from the governed management surface:

```
curl -s \
  http://127.0.0.1:3000/management/v1/warehouses/local/namespaces/default/views/
events_view/version-receipts \
  -H 'x-lakecat-agent-did: did:example:resilience-agent'
{
  "receipts": [
    {
      "stable-id": "lakecat:view:local:default:events_view",
      "view-version": 1,
      "previous-view-version": null,
      "operation": "upsert",
      "view-hash": "sha256:...",
      "receipt-hash": "sha256:..."
    },
    {
      "stable-id": "lakecat:view:local:default:events_view",
      "view-version": 1,
      "previous-view-version": 1,
      "previous-receipt-hash": "sha256:...",
      "operation": "drop",
      "view-hash": "sha256:...",
      "receipt-hash": "sha256:..."
    },
    {
      "stable-id": "lakecat:view:local:default:events_view",
      "view-version": 2,
      "previous-view-version": 1,
      "previous-receipt-hash": "sha256:...",
      "operation": "upsert",
      "view-hash": "sha256:...",
      "receipt-hash": "sha256:..."
    }
  ]
}
```

```

    ]
}

```

The response is catalog evidence, not Iceberg table metadata. It lets QueryGraph verify the version chain, including tombstones after the current view row is gone, while keeping the richer view history model available for a future Sail-owned implementation.

When the caller needs discovery rather than a known view name, the management surface can return all view receipt chains in a namespace, including chains for views that no longer appear in the active view list:

```

curl -s \
  http://127.0.0.1:3000/management/v1/warehouses/local/namespaces/default/view-
  version-receipt-chains \
  -H 'x-lakecat-agent-did: did:example:resilience-agent'
{
  "chains": [
    {
      "stable-id": "lakecat:view:local:default:events_view",
      "chain-hash": "sha256:...",
      "chain-verified": true,
      "latest-view-version": 1,
      "latest-operation": "drop",
      "tombstoned": true,
      "receipt-count": 2,
      "receipts": ["..."]
    }
  ]
}

```

That tombstone read is replayable too. LakeCat projects `view.version-receipts-listed` and `view.version-receipt-chains-listed` as lineage evidence, not as graph topology. The graph taxonomy stays in Grust; LakeCat only proves that the governed read saw the tombstone receipt needed to explain why a previously accepted view is now deleted. The namespace response also carries a deterministic chain-hash over the chain identity and ordered receipt hashes, and lineage-drain summaries replay that value as `view-version-receipt-chain-hashes` together with a verified-chain count. The QGLake fixture now fails if the namespace-level receipt-chain read, its chain hash, or its verified-chain evidence is absent from lineage-drain replay, so QueryGraph acceptance can depend on compact chain evidence without scraping store internals.

1.9.7 QueryGraph Bootstrap

QueryGraph should import LakeCat facts through a verified handoff, not by scraping service internals. The bootstrap endpoint publishes a bundle with artifact hashes:

The exported graph includes a tenant spine:

```

Catalog HAS_SERVER Server
Server HAS_PROJECT Project

```

Project HAS_WAREHOUSE Warehouse
Warehouse HAS_NAMESPACE Namespace

When LakeCat has durable management rows, those graph nodes come from the stored `ServerRecord`, `ProjectRecord`, and `WarehouseRecord`. That means a `QueryGraph` import can see the real server id, project id, warehouse id, display names, and tenant relationships that operators manage through LakeCat's management API. Replay evidence deliberately redacts storage roots and server endpoint URLs into hashes, so `QueryGraph` can prove which management state was observed without inheriting local paths, bucket roots, query tokens, URI fragments, or userinfo. In the bootstrap graph, `Server` nodes carry `endpointUrlHash` and `Warehouse` nodes carry `storageRootHash`; they do not carry the raw endpoint URL or storage root. The projection code and service route tests both pin those emitted fields as full SHA-256 digest evidence, so the producer and verifier agree on the shape before `QueryGraph` import. Authorized management responses can still show the configured endpoint URL or storage root to an operator; graph and lineage replay receive hash-only evidence. When those records do not exist yet, LakeCat falls back to the old deterministic default anchors so bootstrap remains compatible with minimal embedded tests and older import flows.

LakeCat also keeps the older `Catalog HAS_NAMESPACE Namespace` edge in the bundle so existing `QueryGraph` importers can keep working while newer flows read the tenant path. The tenant anchors and warehouse-to-namespace edges are part of the manifest-covered graph hash, so an importer or handoff verifier can reject a bundle whose namespace is silently detached from the warehouse or rebound to a different durable tenant chain. The local `QGLake` verifier enforces that shape: an accepted bootstrap must prove `Catalog -> Server -> Project -> Warehouse -> Namespace -> Table`, not merely that a table node exists somewhere in the graph. It also rejects bundles whose tenant graph nodes expose raw `endpointUrl` or `storageRoot` properties, even if the graph hash and bundle hash have been recomputed around those raw values. Accepted handoffs must use hash-only `endpointUrlHash` and `storageRootHash` evidence for those roots. Those fields are not merely labels with a `sha256:` prefix: the `QGLake` verifier requires a full 64-hex SHA-256 digest, so an importer cannot accept a rewritten bundle that replaces raw roots with placeholder hash text after the fact. The saved handoff verifier repeats that check against the archived `lakecat-bootstrap.json` artifact, so a later replay cannot pass with a compact summary whose bundle file has lost the tenant path or drifted from the summary hashes.

```
curl -s \
  -H 'x-lakecat-principal: agent:querygraph-importer' \
  http://127.0.0.1:3000/querygraph/v1/bootstrap \
  -o target/qglake/lakecat-bootstrap.json
```

The bundle contains catalog tables, views, policy bindings, graph artifacts, OpenLineage artifacts, Croissant/CDIF/OSI/ODRL projections, and a manifest that hashes what was emitted. The manifest is the import contract. `QueryGraph` can refuse a bundle whose graph hash, OpenLineage hash, table artifact hash, view artifact hash, or `QueryGraph` import-compatibility hash does not match. For view-bearing bundles, the import contract also carries compact receipt evidence for each exported view version:

```
{
  "querygraph-import": {
    "schema-version": "lakecat.querygraph.import-compat.v1",
    "view-count": 1,
    "view-receipt-evidence": [
      {
        "stable-id": "lakecat:view:local:default:events_view",
        "view-version": 1,
        "receipt-hash": "sha256:...",
        "receipt-chain-hash": "sha256:..."
      }
    ],
    "view-receipt-evidence-hash": "sha256:..."
  }
}
```

That gives QueryGraph a manifest-covered way to reject a view bootstrap that lost the accepted catalog receipt or detached the ordered receipt chain before the richer graph import begins.

LakeCat also enforces the same binding when a `querygraph.bootstrap` outbox event is replayed. The authorization receipt must carry a valid principal and the `graph-read` action, table artifact stable IDs must match the `verified-tables` manifest exactly, view artifact stable IDs must match `verified-views`, and view-version receipt stable IDs must match `verified-views`. A saved replay event that keeps valid-looking hashes while dropping actor proof, drifting to a lineage-read action, swapping in another table artifact, or borrowing another view's receipt evidence fails before graph or OpenLineage projection.

The QueryGraph side should verify the same bundle before importing it:

```
cd /Users/alexy/src/querygraph/qg-rust
```

```
cargo run -- lakecat-verify \
  --bundle /Users/alexy/src/lakecat/target/qglake/lakecat-bootstrap.json
```

```
cargo run -- lakecat-import \
  --bundle /Users/alexy/src/lakecat/target/qglake/lakecat-bootstrap.json \
  --output .querygraph/lakecat/import-plan.json
```

The importer checks the outer bundle hash, the manifest hashes, the QueryGraph-import compatibility hash, the graph hash, and view receipt evidence, including the accepted receipt hash and receipt-chain hash for each exported view. The graph envelope must be valid as a graph, not just valid JSON: for example, a table and a view in the same namespace must share one namespace node, not emit duplicate vertex ids. That validation belongs on the QueryGraph/Grust side, while LakeCat is responsible for producing a clean catalog-facing graph projection.

For the full local handoff, LakeCat carries a script that runs both sides without writing generated artifacts into the QueryGraph checkout:

```
scripts/qglake-handoff-local.sh
```

The script starts LakeCat on 127.0.0.1:18181, uses a Turso-backed local store under `target/qglake-handoff/`, generates the paired QGLake bootstrap bundle and lineage-drain response, runs `qglake-verify-replay`, then runs QueryGraph's `lakecat-verify` and `lakecat-import` against the same bundle. The resulting import plan is written to `target/qglake-handoff/querygraph-import-plan.json`. The same run also writes `target/qglake-handoff/handoff-summary.json`, a compact machine-readable record under the `lakecat.qglake.handoff-summary.v1` schema. It records the catalog URL, principal, table scope, LakeCat replay status from `lakecat.qglake.replay-verification.v1`, QueryGraph-verified table/view counts, and semantic bundle/graph/OpenLineage/import hashes plus standards accepted only after LakeCat replay, `lakecat-verify`, and `lakecat-import` agree. Before writing that compact summary, the harness also requires the governed scan replay evidence to include planned requested/effective projection, planned requested/effective stats fields, fetched required/effective projection, and fetched filters, with effective projection matching the server-derived read restriction. The compact verifier requires the catalog URL to be an absolute HTTP(S) endpoint and requires warehouse, namespace, and table scope to be present before accepting the summary. It rejects captured QueryGraph verify/import output whose warehouse no longer matches the summary. The compact handoff also carries `catalogConfigProof`, so the compatibility contract advertised by `/catalog/v1/config` remains present beside bootstrap, scan, management, credential, view, and commit-history proof. That object must preserve the same v4 bridge defaults, endpoint list, catalog-config authorization action, graph count, replay hashes, and OpenLineage hashes as the captured LakeCat replay output. A saved handoff therefore cannot pass raw drain verification and then silently drop or rewrite the config-read contract in the summary or sidecar. Both the compact `catalogConfigProof` object and the captured LakeCat replay `catalogConfig` object are closed over those compared fields, so neither can append an extra unverified v4 bridge, endpoint, authorization, graph, replay, or OpenLineage claim beside the accepted config-read proof. The saved LakeCat handoff verifier output repeats that same object under `lineageDrainArtifactSemantics.catalogConfigProof`, binding the verifier's statement about the raw drain artifact to the config proof actually carried by that artifact. Both the compact summary verifier and the saved lineage-drain verifier reject a config proof that drops required routes, including the standard Iceberg REST plan endpoint and the QueryGraph bootstrap endpoint. The same check covers the warehouse-prefixed plan route, so a saved handoff cannot keep default catalog discovery while silently weakening warehouse-scoped Iceberg planning. It also covers default and warehouse-prefixed `fetch-scan-tasks`, preserving the governed task-fetch surface that follows a Sail-planned scan, and default and warehouse-prefixed credential endpoints, preserving the audited credential-decision surface. The lineage-drain replay summaries are bound back to the drain-level `eventTypes` manifest as well. A saved handoff cannot add a compact replay summary for `storage-profile.upserted`, `querygraph.bootstrap`, or any other catalog event type unless the drain itself declared that event type as delivered. LakeCat checks this as a multiset rather than a simple set: repeated event types such as credential vending or scan-task fetching must appear in the same multiplicity in `eventTypes` and in the replay summary array. It also checks order: `eventTypes[i]` must name the same event type as replay summary `events[i]`. That makes the manifest a compact replay sequence proof instead of a loose inventory that could be reordered after the fact. The service applies the same discipline to

replay summary identities before returning the drain response: every summary event id must be nonblank and duplicate-free. That keeps later QGLake proof from inheriting an ambiguous event sequence even when counts and event types still add up. The standalone QGLake lineage-drain verifier checks the same identity rule for saved artifacts, so archived handoff bundles cannot bypass the live service manifest guard by editing only the replay summary event ids. It also embeds `querygraphVerification.verifiedTables` and `verifiedViews` directly in the compact summary. `verifiedTables` must include the stable LakeCat table id derived from that scope, such as `lakecat:table:local:default:events`; `verifiedViews` must include every accepted stable view id from LakeCat replay, such as `lakecat:view:local:default:active_customers_view`; and both arrays must match the QueryGraph table/view counts. LakeCat rejects duplicate entries in those manifests at outbox admission before graph or OpenLineage projection, and the compact handoff verifier repeats the check for archived summaries. Captured QueryGraph verify/import output must match those compact arrays exactly, which keeps a verified artifact set from being replayed against the wrong catalog tenant, table, or view. The `querygraphImportVerification` object is also a compact proof, not only a boolean: it repeats the QueryGraph import table/view ids, counts, bundle hash, graph hash, OpenLineage hash, QueryGraph import hash, and standards, and LakeCat rejects a summary unless those fields are SHA-256-shaped and match both `querygraphVerification` and the captured `lakecat-import` output. It also records structured request-identity, scan, management, credential, table-commit, and view replay evidence, plus compact `requestIdentityProof`, `queryGraphBootstrapProof`, `governedScanProof`, `tableCommitHistoryProof`, `viewReceiptChainProof`, `managementProof`, `storageProfileUpsertProof`, and `credentialVendingProof` objects that lift the replay principal proof, QueryGraph bootstrap/import proof, governed scan counts, pointer-log read proof, view version and receipt-chain proof, management-list counts, redacted credential-root proof, and credential-vending decision out of the full replay tree. The identity proof shows the principal subject and kind used for the replay, the request-identity source and state, the authorization receipt hash, and sanitized TypeDID envelope/proof hashes when a TypeDID envelope is present. The local QGLake fixture currently records the agent-header source with null TypeDID hash slots; a future TypeDID-envelope run can fill those slots without changing the handoff schema. The QueryGraph bootstrap proof shows the accepted bundle, graph, OpenLineage, and QueryGraph import hashes, the table and view artifact counts, standards, policy-binding count, agent delegation and summary signature hashes, view-version receipt hashes, and replay/OpenLineage sink hashes. Those core bootstrap hash anchors must also be SHA-256-shaped before the verifier compares them with QueryGraph's verify/import proof. In the compact handoff verifier, the QueryGraph bundle, graph, OpenLineage, import, bootstrap replay, and bootstrap OpenLineage anchors must be full `sha256:-prefixed` 64-hex digests, so saved summaries cannot use prefix-shaped placeholders as QueryGraph acceptance evidence. The scan proof shows LakeCat planned and fetched scan tasks through the governed path, including file, delete-file, and child plan-task counts with replay and OpenLineage hashes. Planned and fetched scan replay admission validates any carried plan-task as LakeCat-issued evidence, not as arbitrary client text. The commit-history proof shows the catalog pointer log was read back with commit count, sequence numbers, commit hashes, positive graph event evidence, and replay/OpenLineage hashes. The view receipt-chain proof shows QueryGraph's accepted view versions together with accepted receipt hashes,

accepted expectedViewVersion guard evidence when a mutation was guarded, tombstone receipt hashes, namespace chain hashes, verified-chain counts, positive graph event evidence for accepted view replay, and replay/OpenLineage hashes. The credential proof shows the restricted agent was blocked onto Sail-planned reads while the trusted human path used the audited raw-credential exception. The summary also records artifact paths, raw file hashes, captured LakeCat replay output, captured LakeCat handoff-summary verification output, captured QueryGraph verification output, captured QueryGraph import output, captured-output hashes for the LakeCat replay and QueryGraph verify/import JSON files, and service log path. The handoff verifier does not stop at byte hashes: it parses the saved LakeCat replay JSON and QueryGraph verify/import JSON captures and checks their replay schema/status, table and view counts, warehouse, verified table ids, bundle hash, graph hash, OpenLineage hash, QueryGraph import hash, and standards against the compact summary. It compares those standards across sections and independently requires the full QGLake standards set, so a compact handoff cannot omit ODRL or another required contract simply by making every section omit it consistently. It compares the captured LakeCat replay-evidence.requestIdentity and replay-evidence.queryGraphBootstrap objects with the compact request-identity and bootstrap proofs, including the principal, authorization hash, TypeDID hash slots, delegation and summary signature hashes, artifact counts, standards, replay hashes, and the accepted bundle, graph, OpenLineage, and QueryGraph import hashes. The compact verifier depends on the bootstrap manifest verifier having already rejected duplicate stable IDs across table projections, table artifact manifests, view projections, and view artifact manifests. That duplicate-free rule is not an Iceberg REST requirement for ordinary table reads; it is LakeCat/QGLake import proof. It prevents a semantic bundle from satisfying table or view counts by repeating the same stable ID, then letting QueryGraph believe it verified more catalog objects than the manifest uniquely proved. The compact verifier also requires the bootstrap proof to carry the same request-identity source and verification state as requestIdentityProof. The authorization receipt hashes are intentionally distinct proof slots: requestIdentityProof records the lineage-drain read receipt, while queryGraphBootstrapProof records the original bootstrap event receipt. The verifier requires both hashes to be full sha256:-prefixed 64-hex digests and requires their actions to keep the correct meaning: compact requestIdentityProof must be lineage-read, and compact queryGraphBootstrapProof must be graph-read. Those values are then bound back to their captured replay sections rather than forcing the receipt hashes to be equal. The same full-digest rule applies to the required agent delegation and agent summary-signature hashes in the bootstrap proof, so a saved handoff cannot replace those proof anchors with short readable placeholders. The compact requestIdentityProof object and the captured LakeCat replay requestIdentity object are also closed over their compared fields. A saved summary or replay sidecar cannot append an extra unverified actor, identity-source, TypeDID, authorization, or drain-read action claim beside the accepted request-identity proof. The compact queryGraphBootstrapProof object and captured LakeCat replay queryGraphBootstrap object are closed the same way. A saved summary or replay sidecar cannot append extra unverified bundle/import, artifact-count, standards, identity, TypeDID, authorization, delegation, view-receipt, replay, or OpenLineage claims beside the accepted bootstrap proof. The compact verifier also validates the TypeDID hash-slot shape directly: envelope and proof slots must be null or full sha256:-prefixed 64-hex digests, and a TypeDID proof hash cannot appear without the paired

envelope hash. As with authorization receipts, the request and bootstrap TypeDID hash slots are independently shaped replay evidence because they may come from different requests in the captured workflow. That keeps the compact handoff self-describing without moving TypeDID trust semantics out of TypeSec. Live request parsing now enforces the same boundary earlier: a caller that sends `x-lakecat-typedid-proof` without `x-lakecat-typedid-envelope` receives a hash-only `typedid-proof-hash` rejection, and a caller that sends agent delegation or agent summary proof headers without an agent-shaped identity receives only the matching proof hash. Those failures happen before governance or capability receipt creation, so raw proof material cannot become either policy context or operator-facing diagnostics. TypeDID verifier failures follow the same rule: malformed or rejected envelopes report only the envelope hash and error-detail hash, and a verified-subject mismatch reports only hashes of the verified and supplied principals before governance dispatch. LakeCat applies that redaction at the verifier trait boundary, so a custom TypeDID verifier can choose the error class without leaking raw envelope, DID, gateway, or payload text into the catalog response. The JSON output from `lakecat-cli qglake-verify-handoff` also carries the accepted lineage-drain identity source, identity state, and TypeDID envelope/proof hash slots in `lineageDrainArtifactSemantics`, so QueryGraph can index the verified drain boundary without reparsing the raw drain artifact. Before that boundary is accepted, the verifier reconciles the drain artifact's top-level delivered count, eventTypes list, graph event count, and lineage event count against the replay summary array. If a saved `lakecatHandoffVerifyOutput` artifact is named, LakeCat first requires its full `lakecatHandoffVerifyOutputHash`, then binds those saved drain identity semantics back to the compact `requestIdentityProof`, so a rehash cannot disguise drift in principal, authorization receipt, source/state, or TypeDID hash-slot evidence. The same self-verification pass compares the saved verifier output's delivered count, eventTypes, graph event count, and lineage event count with the archived lineage-drain artifact, so a rehashed verifier output cannot rewrite the drain manifest while keeping the artifact hash set intact. It compares captured `replay-evidence.scan` with `governedScanProof`, requiring positive plan task, scan-plan graph event, file task, delete file, and child plan task counts plus the planned and fetched read-restriction objects and the fetch-side required projection/filter evidence. The verifier rejects a summary if the fetched restriction drifts from the planned restriction, so the compact handoff proves the narrowed allowed columns, row predicate, and policy hashes alongside the planned/fetched replay and `OpenLineage` hashes that prove the Sail-planned read path. The compact Rust verifier requires both planned and fetched replay/`OpenLineage` arrays to contain full sha256:-prefixed 64-hex digests, so automation can reject incomplete or placeholder scan lineage without falling back to the shell harness. The compact `governedScanProof` object and the captured LakeCat replay `replay-evidence.scan` object are also closed over the fields LakeCat compares: counts, principal and authorization receipt evidence, read restrictions, requested and effective projection, stats-field evidence, fetch filters, replay hashes, and `OpenLineage` hashes. A handoff therefore cannot attach an extra unverified scan claim beside the checked proof and ask QueryGraph to index it as if it were part of the Sail-planned read. The planned/fetched read-restriction objects are closed too, as are their row-predicate children, so an archived proof cannot smuggle an unverified purpose, policy, predicate, projection, or credential-scope claim inside a restriction object whose core fields happen to match. The service outbox admission path enforces the same closed schema

before graph or OpenLineage projection, so the handoff verifier is confirming evidence that was already constrained at the catalog boundary rather than cleaning it up after delivery. Captured scan replay-line recomputation also reuses the governed read-restriction guard, so an archived replay artifact cannot make empty planned or fetched allowed-columns look like a readable operator summary. It also compares the captured replay-evidence.tableCommitHistory object with tableCommitHistoryProof, including the commit count, sequence numbers, commit hashes, replay principal subject/kind, authorization receipt hash/action, graph event count, replay hashes, and OpenLineage hashes that prove the pointer-log commit history was not rewritten between replay and summary and that the commit-history replay projected catalog graph evidence for the accepted actor. The compact verifier also requires the commit-history principal subject and kind to match the accepted QGLake handoff principal, requires the authorization receipt hash to be a full SHA-256 digest, requires the authorization action to be the read-side table-load action for table.commits-listed, requires the commit count to match the sequence-number and commit-hash arrays, requires every sequence number to be positive and strictly increasing, requires commit hashes to be duplicate-free, and requires positive graph event evidence plus replay and OpenLineage receipt hashes. Captured raw lineage-drain regressions cover both missing and drifted commit-history principal subject, principal kind, and authorization action, and raw summaries also re-check receipt principal, action, allowed decision, engine, and RFC3339 checked_at evidence when receipt evidence is present, so actor, action, and decision provenance must survive before the compact handoff proof exists. The service admission layer now rejects table.commits-listed source replay whose authorization receipt principal is missing or malformed, whose top-level principal-subject or principal-kind is missing, or whose top-level actor summary drifts from the receipt before acknowledgement, catalog graph projection, or OpenLineage projection, so the raw lineage-drain summary is never built from an actorless pointer-log read. It also binds the replayed warehouse, namespace, and table evidence to the durable outbox table identity before projection, so a source replay cannot project one table's pointer log as another table's history. Captured LakeCat replay-line recomputation enforces the same sequence invariant even when the captured replay JSON and compact summary agree on malformed sequence evidence, so operator-readable table-commit-history-replay text cannot launder zero, duplicated, or reordered pointer-log proof. It compares the captured replay-evidence.views object with viewReceiptChainProof, including accepted view receipts, accepted-view graph event counts, expected-version guard evidence, tombstone receipts, namespace receipt-chain hashes, and their replay/OpenLineage hashes, so durable view history stays tied to the saved LakeCat replay artifact. It also compares the tombstone branch's expectedViewVersion with the accepted view version, so a handoff cannot claim a governed deletion unless the saved LakeCat replay proves that deletion used the catalog's optimistic version guard; the standalone qglake-verify-handoff command enforces this match even when a summary is checked outside the local shell harness. The same standalone verifier now requires the compact view proof to keep viewCount consistent with the accepted view list, carry stable warehouse/namespace/name identity, prove viewVersion == acceptedViewVersion, and carry accepted receipt hashes, accepted receipt-chain hashes, positive accepted-view graph event evidence, tombstone receipt hashes, positive verified-chain counts, receipt-chain warehouse/namespace identity, namespace chain hashes, and replay/OpenLineage hashes. It also derives the expected lakecat:view:<warehouse>:<namespace>:<name>

stable ID from each accepted view's warehouse, namespace, and view name, so a saved handoff cannot keep a verified stable ID while drifting the visible component fields. Tombstone receipt entries use the same component binding before their `expectedViewVersion` guard evidence is accepted, so deletion proof cannot drift from the stable view identity either. The verifier also checks that each namespace receipt-chain summary's `verifiedChainCount` equals the number of structural chains[] entries and that every chain entry is covered by `chainHashes`. The compact proof and captured LakeCat replay sidecar are also closed over their verified view schema: top-level `viewReceiptChainProof`, each accepted-view entry, each tombstone entry, each receipt-chain group, each structural chain, and each receipt may carry only the fields LakeCat compares or uses in the structural digests. A saved handoff therefore cannot append unverified view lifecycle, tombstone, receipt-chain, principal, replay, or OpenLineage claims beside the checked view proof. The compact proof carries chains[] entries inside each namespace receipt-chain summary. Each chain entry keeps only catalog-facing evidence: stable view identity, the chain hash, the verified flag, latest view version, latest operation, tombstone state, receipt count, and per-receipt version, operation, view hash, principal subject, principal kind, recorded timestamp, receipt hash, and previous-link fields. The chain warehouse and namespace must match the enclosing namespace receipt-chain summary, and every receipt's stable ID, warehouse, namespace, and view name must match the chain identity. Each structural chain stable ID is also checked against its own warehouse, namespace, and view-name components, so compact evidence cannot splice receipts across views or namespaces while preserving hash-shaped fields. LakeCat now applies the same scope rule at raw service replay time: a `view.version-receipt-chains-listed` outbox event is rejected if any nested chain or receipt warehouse/namespace drifts from the top-level payload, if a chain stable ID does not match its warehouse, namespace, and view name, or if a receipt stable ID or view name does not match the chain identity before acknowledgement, graph projection, or OpenLineage projection. Structural chain bodies cannot repeat a `chainHash`. The enclosing namespace `chainHashes` and `receiptHashes` arrays are duplicate-free summaries of those same structural chains, and `receiptHashes` must match the structural per-receipt hashes exactly, so a compact proof cannot hide extra receipt hashes or omit receipts from the ordered chain bodies. Each structural `chainHash` is also recomputed from the same content-derived digest LakeCat service uses for view receipt chains: stable view identity, latest version, latest operation, tombstone state, and the ordered receipt hashes. A compact proof therefore cannot pair an accepted receipt-chain hash with a different ordered receipt body. LakeCat enforces the duplicate-free part before compact proof generation as well: outbox replay rejects duplicate receipt-hashes, drop-receipt-hashes, or chain-hashes in view receipt-list and receipt-chain events before graph or OpenLineage projection. For verified receipt chains, raw service replay also recomputes each structural chain-hash from the chain identity, latest state, tombstone posture, and ordered receipt hashes before projection, so an archived outbox event cannot simply invent a full SHA and repeat it in chain-hashes. Each structural `receiptHash` is recomputed too, using the same content-derived view-version receipt digest LakeCat service emits over stable view identity, version, previous-link fields, operation, view hash, principal, and recorded timestamp. That closes the gap between a valid-looking chain over opaque receipt hashes and a chain whose individual receipt bodies are themselves durable catalog facts. `qglake-verify-handoff` rejects a chain whose first receipt is not version 1 upsert, whose previous links do not point to the prior receipt, whose

upsert skips a version, whose drop advances the durable version, whose operation is unsupported, or whose latest receipt does not match the chain head. In the compact verifier, accepted-view receipt hashes, accepted receipt-chain hashes, tombstone receipts, namespace receipt/chain hashes, per-receipt hashes, and view replay/OpenLineage hashes must all be full sha256:-prefixed 64-hex digests, so a saved handoff cannot use readable placeholder strings as view acceptance evidence. It also requires `queryGraphBootstrapProof.viewVersionReceiptHashes` to match the accepted view receipt hashes exactly, so the compact summary cannot combine bootstrap view receipt evidence from one run with accepted-view proof from another. Accepted receipt-chain hashes and tombstone receipt hashes must be covered by structural chains[] evidence for the same stable view, not merely by another chain in the same namespace receipt-chain summary. A tombstoned view therefore carries both the accepted pre-drop chain and a structural drop chain whose final receipt is the tombstone. Tombstoned views must also include tombstone receipt evidence whose `expectedViewVersion` preserves the accepted view version. A consumer can reject a handoff whose view history claim lacks identity, accepted-version, count-aligned hash-chain evidence, duplicate-free exact structural receipt-hash coverage, content-derived chain-hash agreement, same-view accepted-chain coverage, tombstone guard evidence, same-view tombstone receipt coverage, or replay evidence before parsing the full replay tree. It compares captured LakeCat replay `replay-evidence.management` list counts with compact `lakecatReplayVerification.managementProof`, requiring positive server, project, warehouse, and storage-profile counts, positive graph event counts for server, project, warehouse, policy-binding, and storage-profile list replay, and a policy-binding count that matches the `QueryGraph` bootstrap proof. Those management-list counts are receipt-backed: the compact proof and captured replay must also agree on replay and OpenLineage hash arrays for `server.listed`, `project.listed`, `warehouse.listed`, `policy-binding.listed`, and `storage-profile.listed`. The same admission rule applies to the outbox event envelope itself. The store requires the top-level pending outbox payload to name the event type, and the service projection path now also requires a hash-bound durable row's inner replay payload to carry the same event-type before Grust, OpenLineage, QGLake, or `QueryGraph` sees it. A forged row cannot say "this is a table event" in the outer envelope while projecting an unbound inner payload into the graph or lineage stream. It also compares the captured LakeCat replay `replay-evidence.management.storageProfileUpsert` object with the compact `lakecatReplayVerification.storageProfileUpsertProof`, including the profile id, provider, issuance mode, location-prefix hash, secret-reference presence/provider/hash, replay hashes, and OpenLineage hashes. The compact verifier also requires those replay and OpenLineage arrays to contain full sha256:-prefixed 64-hex digests. It requires that location-prefix value to be a full sha256:-prefixed 64-hex digest and requires a redacted secret-reference provider plus full-digest `secretRefHash` whenever the proof says a secret reference is present. If the proof says no secret reference is present, the provider and hash may be omitted or null, but any other provider/hash value is rejected. The saved LakeCat replay must normalize the same way as the compact summary: omitted absent secret-reference provider/hash fields are accepted only when both the compact proof and captured replay omit them under `secretRefPresent=false`. The local QGLake handoff script applies the same nonblank provider rule to compact credential storage-profile proof, so credential proof cannot claim secret-reference presence while omitting the provider an-

chor. Source replay enforces the same full-digest secret-reference rule before compact proof generation, so the saved summary cannot launder short placeholder credential-root hashes through the lineage-drain artifact. The positive QGLake acceptance fixture covers the production-shaped case too: when the storage profile uses a secret reference, the compact handoff accepts the proof only when the storage-profile branch and both credential branches agree on the redacted provider and full `secretRefHash`, and the operator replay line carries `secret_ref_hash=sha256:...` rather than a raw secret URI. Those operator-facing management and credential replay lines also fail closed when secret-backed evidence has only a prefix-shaped or placeholder hash, so the human-readable proof cannot be weaker than the structured verifier. The storage-profile proof objects themselves are closed over the compared schema: `storageProfileUpsertProof` and captured `replay-evidence.management.storageProfileUpsert` may carry only the profile identity, provider, issuance mode, redacted location and secret-reference posture, principal and authorization receipt evidence, graph count, replay hashes, and OpenLineage hashes that LakeCat verifies. `QueryGraph` should treat any extra credential-root claim as untrusted until it is promoted into that closed proof contract and backed by replay evidence. This is a LakeCat/TypeSec governance extension around Iceberg access, not an Iceberg table-metadata extension. Raw lineage-drain summaries also apply the same storage-profile redaction and secret-reference posture before compact proof generation: nested storage-profile evidence must use the closed producer schema, full location and secret-reference hashes, coherent secret-ref presence/provider/hash fields, and a valid provider/issuance-mode pairing. A saved lineage-drain artifact therefore cannot hide a raw secret reference or launder a malformed credential root by silently dropping the bad fields. Raw lineage-drain management summaries now apply the same inventory posture to control-plane ID arrays: project ids, server ids, warehouse names, policy ids, and storage-profile ids must be string-shaped, nonblank, and duplicate-free before QGLake proof can inherit them. It also compares the captured `replay-evidence.credentials` restricted-agent and trusted-human branches with the compact `credentialVendingProof`, so a saved handoff cannot claim that agents were blocked onto Sail-planned reads or that humans used an audited exception unless the captured LakeCat replay proves the same decision. That equality includes `credentialPrefixHashes`, `authorizationReceiptHash`, and `authorizationReceiptAction`, which closes the gap where a captured replay artifact could report a different redacted returned-credential set or authorization action while the compact summary still looked valid. The credential proof envelope is closed at three levels: the top-level `credentialVendingProof` object may only contain the restricted-agent and trusted-human branches; each branch may only contain the principal, credential count, redacted prefix hashes, raw-exception decision, TTL cap, storage-profile anchor, authorization receipt, replay hashes, and OpenLineage hashes LakeCat checks; and each nested storage-profile anchor may only contain the redacted profile identity, provider, issuance mode, storage-scope hash, secret-reference posture, and graph count. Extra raw credential or storage-scope claims are not Iceberg metadata, and `QueryGraph` should not treat them as catalog truth unless LakeCat promotes them into this checked proof contract. The raw-exception decision itself is also closed in raw replay before compact handoff proof can inherit it: both the top-level and authorization receipt context `lakecat:raw-credential-exception` objects may carry only the requested posture, allowed/blocked posture, and reason that LakeCat compares. A captured replay artifact cannot smuggle an additional raw-credential entitlement beside an otherwise valid

blocked-agent or trusted-human branch. Both credential branches must carry a full authorization receipt hash, the `credentials-vend` authorization action, and `replay/OpenLineage` arrays whose entries are full `sha256:-`prefixed 64-hex digests, so the compact proof cannot replace credential receipt evidence with prefix-shaped placeholders or a different catalog action. They also carry `credentialPrefixHashes`: the restricted-agent branch must prove the array is empty when zero credentials were returned, while the trusted-human branch must prove the array length matches `credentialCount`, every entry is a full SHA-256 digest, and no prefix hash is repeated. The verifier also binds the operator-facing replay text back to the same proof fields. The captured top-level scan-replay line is recomputed from `governedScanProof`, including `plan/fetch` task counts, the policy-derived purpose, and the TTL cap. The captured top-level credential-replay line is recomputed from `credentialVendingProof`, including the restricted-agent block, trusted-human exception, TTL caps, and redacted storage-profile anchors. The captured management-replay line is also recomputed from `managementProof` and `storageProfileUpsertProof`, while `table-commit-history-replay` is recomputed from `tableCommitHistoryProof`. A saved artifact therefore cannot keep valid structured proof while presenting a different terminal transcript or principal-attribution story to an operator. Each credential branch carries the same redacted storage-scope anchor as the storage-profile upsert proof: `locationPrefixHash` binds the credential-vend attempt to the configured storage root without replaying the raw prefix. That anchor must be a full `sha256:-`prefixed 64-hex digest in both the storage-profile proof and each credential branch. LakeCat checks the storage-scope hash at lineage-drain replay time before the compact handoff summary is accepted, and the operator-readable credential replay line prints the same hash so captured terminal output cannot look complete while omitting the credential-root boundary. Source replay validates secret-reference shape on the credential branches themselves: if a credential-root proof says a secret reference is present, it must carry a non-empty provider and full `sha256:-`prefixed 64-hex `secretRefHash`; if it says no secret reference is present, provider and hash evidence must be absent. LakeCat now applies the same discipline before the outbox event is delivered: `credentials.vend-attempted` must carry a `credential-count` that matches its credential-response evidence, full SHA-256 prefix and issuer-config hashes for each returned credential, a full storage-profile `location-prefix-hash`, and non-contradictory secret-reference state. The event must also carry a valid authorization receipt principal before delivery, including blocked zero-credential attempts where no returned credential entry exists to repeat actor evidence. The top-level `storage-profile-id` must match the nested `storage-profile.profile-id`, even when no raw credentials were returned, and the nested `storage-profile.warehouse` must match the event table warehouse. The replay payload's table hint must also match the durable outbox table identity before acknowledgement, graph projection, or `OpenLineage` projection, so a credential decision for one table cannot be replayed as another table's credential-root evidence. If the top-level `secret-ref-present` field is missing, non-boolean, or different from `storage-profile.secret-ref-present`, the replay event is rejected before delivery. Raw lineage-drain summaries enforce the same binding before compact QGLake proof is built. That duplicate field is small, but it keeps compact credential proof from omitting whether the selected credential root depends on an external secret reference. Each returned credential entry must also agree with the catalog-derived storage-profile id, catalog profile id, storage provider, credential mode, authorization principal, receipt principal, governed-read marker, and any policy-derived TTL cap.

Returned credential entries must be duplicate-free by `prefix-hash`, and service replay now rejects unexpected fields inside each `credential-response` entry, so a replay event cannot count the same redacted credential twice or attach an unverified `credential-scope` claim beside the catalog-derived proof. The issuer config stays redacted, but its proof is still internally checked: when `issuer-config-entry-count` is zero, `issuer-config-hash` must match LakeCat's canonical hash of the empty issuer config array. Non-zero issuer config remains represented by count plus full digest evidence, without storing or replaying raw credential config. LakeCat carries those redacted prefix hashes into the raw lineage-drain summary as `credentialPrefixHashes`, and the QGLake verifier rejects the drain before compact proof generation if the prefix hashes are missing, count-drifted, short, or duplicated. A malformed credential replay event therefore remains pending instead of becoming graph or OpenLineage evidence. The raw service summary builder now applies the same fail-closed posture before the verifier runs: non-array credential response evidence, missing prefix hashes, malformed prefix hashes, duplicate prefix hashes, malformed issuer-config hashes, unsigned issuer-config entry counts, or a zero issuer-config entry count paired with a non-canonical empty issuer hash reject the raw lineage-drain summary instead of silently omitting redacted credential proof. It also reuses the service credential-response binding, so a raw summary cannot drift the returned credential away from the catalog storage-profile id, storage provider, issuance mode, receipt principal, governed-read posture, secret-ref posture, or authorized TTL while preserving a valid-looking redacted prefix hash. Credential replay also rejects a governed read-restriction that is missing from, or different from, the authorization receipt context, so credential TTL and blocked-agent evidence cannot drift away from the receipt that authorized the decision. The credential replay boundary now also treats a governed restriction without a nonblank purpose or a positive `max-credential-ttl-seconds` cap as malformed before acknowledgement, graph projection, or OpenLineage projection. That keeps a blocked zero-credential agent decision from becoming QGLake evidence unless it still proves the policy purpose and TTL cap that made raw credential vending inappropriate. Source replay and compact handoff verification both reserve `rawCredentialExceptionReason` for the audited trusted-human path; a restricted agent proof must be blocked with `blockReason` and cannot carry a raw exception reason. The local handoff harness now preserves the lineage-drain artifact before replay verification, so if any of these proof checks fail the operator still has the exact failed drain JSON for diagnosis. The compact handoff verifier also validates that credential proof directly: the restricted branch must name the accepted agent principal, carry the Sail-planned-read block reason, prove zero credentials, carry the policy-derived `maxCredentialTtlSeconds` cap, explicitly set `rawCredentialExceptionAllowed` to false, reject any non-null `rawCredentialExceptionReason`, and include replay/OpenLineage hashes; the trusted-human branch must name a human principal, prove a positive credential count, carry the same policy-derived TTL cap, carry the exact audited raw-credential exception reason, prove `blockReason` is null, and include replay/OpenLineage hashes. Both branches must also carry count-aligned, duplicate-free `credentialPrefixHashes`, with an empty array for the blocked branch and full SHA-256 returned-prefix hashes for any issued credential. The compact verifier has direct negative coverage for the credential-branch secret-reference rules too: blank providers are rejected when a secret ref is present, and non-null provider/hash evidence is rejected when no secret ref is present. That way a handoff cannot hide malformed provider or hash evidence behind an otherwise matching storage-profile upsert proof. The local handoff harness now

preserves those replay fields while building the compact summary, rather than relying on the nested raw replay blob alone: scan graph events and fetch-side projection/filter requirements, management graph events, storage-profile graph events, credential block/exception fields, and table commit-history graph events all move into the verifier-facing proof sections before `qglake-verify-handoff` runs. That makes the handoff repeatable from the LakeCat repo while keeping QueryGraph responsible for graph validation and import semantics. The handoff script refuses to write the summary unless LakeCat replay JSON contains request-identity evidence for the expected agent principal, an explicit identity source/state, an authorization receipt hash, and explicit TypeDID envelope/proof hash slots that are either null or full sha256:-prefixed 64-hex digests. A proof hash is valid only when the matching envelope hash is present. It also refuses to write the summary unless LakeCat replay JSON contains redacted `storageProfileUpsert` evidence with replay and OpenLineage hashes, and the accepted summary repeats that evidence as `lakecatReplayVerification.storageProfileUpsertProof`. QueryGraph gets proof that the credential root was configured, including the provider, issuance mode, the configured location-prefix hash, and a full sha256:-prefixed 64-hex digest of the secret reference, without receiving the underlying secret-store URI or full storage prefix in the compact proof. That compact proof must also name the principal subject and kind, carry a full authorization receipt hash, and prove the receipt action was `storage-profile-manage`. Captured LakeCat replay and the compact QGLake summary are compared field by field, so a saved handoff cannot turn an authorized storage-profile management action into an actorless credential-root fact or replay it under a weaker catalog action. The operator-readable management replay line now prints the same storage-scope hash and redacted secret-reference state, so a captured transcript cannot describe the credential root only by provider while omitting its redacted storage scope or secret-reference boundary. The saved handoff verifier recomputes that management line from compact proof fields before accepting captured output. The script also refuses to write a summary unless LakeCat replay proves both sides of credential vending: untrusted agents get no raw credentials, trusted humans receive only the audited standard exception, and both branches preserve the `max-credential-ttl-seconds` restriction as `maxCredentialTtlSeconds` in compact evidence. For reads, the summary similarly refuses to omit proof that scan planning and scan-task fetch both replayed with full digest-shaped sink receipt hashes. The compact scan proof must preserve the server-derived read restriction as a full restriction, not only as columns and filters: allowed columns, row predicate, purpose, full sha256:-prefixed policy-hash evidence, and `max-credential-ttl-seconds` must be present, and the planned and fetched restrictions must agree. Short readable policy anchors such as `sha256:policy-name` are rejected before QueryGraph receives the compact handoff proof. The fetched required filters must also be exactly the mandatory row predicate evidence, not a prefix with extra unverified filters appended. For catalog state, it refuses to omit proof that the table commit-history read replayed with sequence-number and commit-hash evidence. For views, it refuses to omit proof that accepted view versions line up with their receipt hashes and that the namespace-level tombstone and receipt-chain reads replayed with chain hashes and verified-chain counts. The service-side lineage-drain summary preserves the full receipt hash set from receipt-list reads and nested namespace receipt-chain payloads, so the QGLake verifier can prove that both upsert and tombstone receipts are covered by the replayed namespace chain.

This gives the semantic layer a responsible starting point. LakeCat says:

Here are the governed catalog objects.
Here are the policies that shaped planning.
Here are stable dataset and field anchors.
Here is the graph envelope.
Here is the OpenLineage replay evidence.
Here are the hashes that bind the handoff.

QueryGraph then owns the richer semantic work: metrics, dimensions, joins, business names, multi-dataset reasoning, and agent-facing synthesis.

1.9.8 Draining The Outbox

LakeCat records side effects as durable outbox events. Draining the outbox is what turns committed catalog facts into graph and lineage receipts:

```
curl -s -X POST \
  -H 'x-lakecat-principal: agent:lineage-drainer' \
  http://127.0.0.1:3000/management/v1/lineage/drain \
  -o target/qglake/lineage-drain.json
```

A useful drain response includes delivered event types, graph projection counts, lineage projection counts, receipt hashes, and the authorization proof for the drain request itself. That last part is easy to overlook. Reading the replay stream is also privileged, so LakeCat records that the drainer was allowed to read lineage evidence. Before LakeCat returns that raw lineage-drain summary or acknowledges delivery, it also checks the projection receipt arrays produced by the graph/lineage sink boundary: replay hashes and OpenLineage hashes must be count-aligned with lineage events, full SHA-256-shaped, and duplicate-free. That keeps a malformed sink receipt from inflating the proof QGLake later archives. Standard catalog reads replay too: `catalog.config-read` records a warehouse-scoped graph/OpenLineage fact for the Iceberg REST configuration endpoint; `namespace.listed` records the namespace listing at the warehouse; and `namespace.loaded` records the specific namespace resolved through the standard catalog route. For view events, the response includes the warehouse, namespace, view name, and QueryGraph-compatible stable id, so downstream acceptance can check replay identity without parsing the full audit payload. Table restores replay as table graph evidence plus a restore OpenLineage receipt, so a soft-deleted table returning to service is visible to QueryGraph without forcing LakeCat to invent restore-specific graph taxonomy. Management list reads for policy bindings, projects, servers, storage profiles, and warehouses replay as OpenLineage receipts too. They intentionally do not create list-specific graph nodes in LakeCat; Grust owns the reusable hierarchy and traversal model. The durable audit/outbox payload carries only the redacted stable ID arrays beside the counts: policy ids, project ids, server ids, storage-profile ids, and warehouse names. Before projection, LakeCat rejects a management-list event when the required ID array is missing, malformed, contains an invalid identifier, repeats an identifier, or no longer matches the recorded count. LakeCat also requires the authorization receipt to carry a valid principal, the event-matching catalog action, an affirmative allow decision, a non-empty engine, and an RFC3339 checked-at timestamp before acknowledging the event, so QueryGraph never has to

accept actorless or action-drifted management inventory replay. Service replay also closes the top-level payload schema for `namespace.listed`, `view.listed`, and management list events, so an archived inventory read cannot attach unverified namespace, view, management, replay, OpenLineage, or QueryGraph claims beside the checked count and ID/name/path evidence. The drain response lifts their counts, ID arrays, and management scope into compact fields, so QueryGraph can verify the control-plane read evidence without opening the raw lineage payload. It also carries replay and OpenLineage hash arrays for those management-list reads, so a compact handoff cannot prove only that the right number of management records existed while losing the identities or receipt evidence for the reads. The lineage-drain verifier rejects those source replay events when the ID arrays are missing, empty, duplicate-inflated, count-drifted, or when the receipt arrays are empty or not full SHA-256-shaped, or when the event no longer preserves nonblank principal subject/kind evidence and a full authorization receipt hash, so the compact managementProof starts from verified replay evidence rather than untrusted text. This is deliberately LakeCat/QGLake/TypeSec control-plane proof: the underlying namespace, table, warehouse, policy, and storage-profile inventory remains standard catalog state, while the actor, authorization, replay, and OpenLineage receipts prove how that state was read and projected for QueryGraph. The compact handoff verifier repeats that check with the stricter full sha256:-prefixed 64-hex digest shape for every management replay and OpenLineage array, and it verifies that `serverIds`, `projectIds`, `warehouseNames`, `policyIds`, and `storageProfileIds` match their recorded counts and are duplicate-free. When `warehouseProjectId` is present, it must be a non-empty listed project id. Saved summaries therefore cannot preserve only prefix-shaped placeholders for control-plane read receipts, inflate a count with repeated valid identities, detach a project-filtered warehouse inventory from the project list, or normalize malformed management identities later. Captured replay agreement checks the same ID arrays against the saved compact managementProof, so a handoff cannot keep valid artifact hashes while swapping the server, project, warehouse, policy, or storage-profile identities between source replay and the summary. When an operator preserves the verifier output as `lakecat-handoff-verify.json`, LakeCat re-checks the saved `capturedOutputSemantics.lakecatReplay` proof against the compact summary for every replay section, including management IDs, governed scans, commit history, view receipt chains, storage-profile evidence, and credential-vending proof. That makes the verifier output a replayable audit artifact instead of a second place where compact proof drift can hide. It also rejects management-list source replay without catalog graph projection evidence, keeping the durable server/project/warehouse, policy, and storage-profile facts visible to QueryGraph through Grust-facing graph events. Compact managementProof carries those graph event counts too, and captured replay agreement checks them, so the graph evidence cannot disappear between source replay and handoff verification. Policy binding upserts add a content anchor to that management proof. A policy list can prove that `agent-columns` was listed; it cannot prove which ODRL document `agent-columns` meant. LakeCat now carries a compact `policyUpsertProof` with `policyId`, `odrlHash`, graph event count, replay hashes, OpenLineage hashes, principal subject/kind, a full authorization receipt hash, and the `policy-manage` action. The raw lineage-drain verifier requires a matching `policy-binding.upserted` replay event, requires the policy id to be present in the policy list, requires the ODRL hash to be a full SHA-256 digest, and rejects action drift away from `policy-manage`. Captured replay agreement compares the same object

against the saved summary. That keeps QueryGraph from accepting a management proof that preserved the policy name but lost or swapped the policy document anchor or the authority under which it was recorded. Tenant-root upserts get the same hash-binding treatment. When a server replay event carries an `endpoint-url`, LakeCat recomputes `endpoint-url-hash` from that value before projection. When a warehouse replay event carries a `storage-root`, LakeCat recomputes `storage-root-hash` from that value before projection. This is not an Iceberg table-access rule; it is LakeCat/QGLake management proof. It keeps a replay event from pairing one raw endpoint or storage root with a valid-looking hash for another value, then asking Grust, OpenLineage, or QueryGraph to trust the mismatched tenant-root evidence. Raw lineage-drain summaries apply the same redaction-hash check before compact handoff proof is built, so the archived summary path cannot become weaker than delivery replay. The write side also checks ancestry before extending the tenant tree. Project upserts validate their parent server record, and warehouse upserts validate their parent project record. In proof terms, the child record does not inherit authority from a mere key lookup; it inherits from a decoded, scoped, valid parent management record. The QGLake acceptance workflow now establishes its server/project/warehouse tenant spine, performs governed server, project, warehouse, policy-list, policy-upsert, storage-profile-list, scan-planning, scan-task-fetch, and table commit-history reads before bootstrap, and rejects a drain that does not replay matching `server.listed`, `project.listed`, `warehouse.listed`, `policy-binding.listed`, `policy-binding.upserted`, `storage-profile.listed`, `table.scan-planned`, `table.scan-tasks-fetched`, and `table.commits-listed` evidence. Request-identity and bootstrap replay are checked before any compact handoff proof is built: the drain authorization, bootstrap authorization, QueryGraph bundle/import hashes, agent delegation hash, agent summary signature hash, and TypeDID envelope/proof hashes must be full sha256:-prefixed 64-hex digests, and a TypeDID proof hash cannot appear without its paired envelope hash. For scan replay, the typed drain summary carries scan-plan task counts, scan-plan graph event evidence, fetched file-scan, delete-file, and child-plan task counts, along with planned and fetched OpenLineage receipt hashes. Source replay validation now also requires planned/fetched replay and OpenLineage receipt arrays to be full SHA-256 digests, and the compact handoff verifier repeats that full-digest check for the saved governedScanProof arrays. The scan read restriction itself is part of that proof: both source replay and compact plannedReadRestriction/fetchedReadRestriction evidence require policy-hashes to be non-empty full sha256:-prefixed 64-hex digests, so a self-consistent handoff cannot smuggle placeholder policy names or empty policy anchors through a field that later readers treat as integrity evidence. The outbox drain checks the same digest shape and non-empty requirement before acknowledging any pending event that carries `read-restriction.policy-hashes`, including the copy embedded in the authorization receipt context, so malformed source evidence is stopped before it becomes delivered replay material. Scan replay also requires the top-level read restriction to match the authorization receipt context exactly before delivery, so policy narrowing cannot be asserted in one replay field and absent from the durable receipt. Scan replay also requires the authorization receipt itself to be complete before delivery: a valid principal, the event-matching table-plan-scan catalog action, an affirmative allowed decision, a non-empty receipt engine, and an RFC3339 `checked_at` timestamp. Valid-but-wrong actions such as table load or commit actions are rejected before governed scan proof reaches graph or lineage sinks. This is LakeCat replay admission, not Sail planning logic; Sail

remains responsible for producing reusable table-format and scan-planning behavior, while LakeCat refuses to turn actorless or action-drifted scan evidence into graph or lineage proof. QGLake preserves the same actor and action evidence in compact handoff proof: planned and fetched scan proof carry principal subject/kind, full authorization receipt hashes, and table-plan-scan actions, and captured LakeCat replay must match those fields. That keeps archived handoffs from retaining only the restriction and task counts while dropping who was authorized to perform the governed scan. Scan replay now gets the same drain-side admission check before Grust or OpenLineage projection: planned-scan events must carry matching table identity, unsigned task counts, requested/effective projection arrays, and requested/effective stats-field arrays; fetched-task events must carry matching table identity, fetched file/delete/child-plan counts, required filters, and required/effective projection arrays. Those scan proof arrays must be non-empty, non-blank, and duplicate-free; present-but-empty projection or stats evidence is malformed, not an implicit unrestricted read. Planned and fetched governed required-filters evidence must be present and must exactly preserve the governed row predicate at service admission, so an event with missing, empty, or drifted filter proof is rejected before graph or OpenLineage projection. When there is no read-restriction row predicate, replay must keep required-filters empty; a non-empty array without that governing predicate is an unsourced filter claim, not evidence. When a governed read restriction is present, the effective projection and effective stats fields must remain inside the allowed columns, empty allowed-column arrays fail closed for both planned and fetched replay, and explicit effective projection or stats evidence cannot widen beyond the caller-requested or server-required evidence it claims to preserve. QueryGraph bootstrap replay is also checked at the drain boundary before it becomes accepted handoff material: the event must carry a valid warehouse, table/view counts matching the verified ids and artifact arrays, full SHA-256 bundle/graph/OpenLineage/import hashes, full table/view artifact hashes, view receipt and receipt-chain hashes for accepted views, the expected standards list, and full optional TypeDID or agent proof hashes when those slots are present. The nested table artifact, view artifact, and view-version receipt entries are closed over their verified fields as well: a replay sidecar cannot attach an extra semantic artifact, standards, graph, lineage, or view receipt claim beside the stable id and hash evidence that LakeCat actually checked. View receipt replay follows the same fail-closed rule at the drain boundary. A view.version-receipts-listed event is not acknowledged unless its warehouse, namespace, view, and authorization receipt principal are valid, its authorization receipt action is view-load, its receipt-count matches full SHA-256 receipt hashes, and every drop receipt hash is included in the listed receipts. The receipt-list payload is also closed over the fields current producers emit, so an archived read cannot attach unverified view-history, lineage, graph, QueryGraph, or application claims beside checked receipt hashes and authorization evidence. LakeCat also closes the wrapped receipt-read outbox envelope for view.version-receipts-listed and view.version-receipt-chains-listed, so those claims cannot ride beside an otherwise valid checked receipt-list or receipt-chain payload. A verified view.version-receipt-chains-listed event is not acknowledged unless its warehouse, namespace, authorization receipt principal, read-side view-load authorization action, chain count, receipt count, and tombstone count are valid and count-aligned, each verified chain and receipt carries full SHA-256 digest evidence, the first receipt is a version 1 upsert without previous links, and every later upsert or drop links to the previous receipt with the expected view-version transition. The receipt-chain summary payload

plus the nested receipt-chain and receipt objects are closed over the fields LakeCat verifies, so a replay sidecar cannot attach extra view-history, principal, lifecycle, graph, lineage, QueryGraph, or application claims beside a valid chain hash or receipt hash. Raw lineage-drain summary construction also decodes the archived receipt-chain objects and requires chain-verified-count to match the decoded verified chains, so compact QGLake proof cannot inflate verified view-history by changing a count beside otherwise valid structural chain evidence. That keeps malformed or drifted view-history evidence out of both graph projection and OpenLineage replay before QueryGraph ever sees a compact handoff. The verifier also requires table-commit replay to be internally consistent before delivery: `table.commit` must carry a commit object, unsigned sequence number, stable table identity, matching nested commit-table identity, a valid commit principal and a valid authorization receipt principal with matching values, an authorization receipt whose action is a known LakeCat catalog action matching the `table.commit` event, whose allowed decision is true, whose engine is non-empty, whose `checked_at` timestamp is RFC3339, an RFC3339 commit `committed_at` timestamp, and commit hash evidence that is full SHA-256 before graph or OpenLineage projection can start. Raw lineage-drain summary construction uses the same table-commit admission shape: duplicate snake-case/kebab-case aliases for the same pointer or proof field fail closed, and nested commit, top-level payload, and wrapper objects are closed over the fields LakeCat actually verifies. A compact QGLake commit summary therefore cannot smuggle an extra graph, lineage, storage, policy, or application claim beside otherwise valid request/response hashes and pointer evidence. The action, decision, engine, and timestamp fields are deliberately small but important: replay evidence must prove which catalog action was authorized, prove the catalog acted under an affirmative authorization decision, say which engine made that decision, preserve when the authorization check happened, and preserve when the catalog accepted the pointer transition. Local/default receipts identify the local allow-all compatibility engine, while real TypeSec-backed receipts identify TypeSec. That keeps replay evidence from becoming actorful but action-less, decision-less, engine-less, or timeless proof. The compact QGLake handoff now preserves the same action proof instead of collapsing it into a receipt hash. The lineage-drain response carries `authorizationReceiptAction` for the drain read itself and for each replayed event summary. The QGLake verifier requires the drain read to prove lineage-read, requires each replayed event summary to carry a non-empty action, and rejects valid-but-wrong action drift, such as `table-commit` attached to `table.commits-listed`. Captured replay agreement checks the same field in saved handoff artifacts, so an archive cannot keep valid hash-shaped proof while silently changing what catalog action was authorized. The compact handoff summary now makes the same requirement before captured-output comparison: request identity proves lineage-read, and QueryGraph bootstrap proves graph-read. The saved self-verifier sidecar carries top-level copies of both compact proof objects, and artifact verification rejects those copies if either authorization action drifts from the summary. The saved self-verifier sidecar repeats that binding for `lineageDrainArtifactSemantics`: its drain-read `authorizationReceiptAction` must still match the compact request-identity proof, so a rehashed `lakecat-handoff-verify.json` cannot describe a different lineage-read action. Commit-history replay has the same shape: `table.commits-listed` event must carry a `commit-count` that matches both full SHA-256 commit hashes and unsigned sequence numbers, plus `principal-subject` and `principal-kind` fields that match the authorization receipt principal, a known authorization

receipt action matching `table.commits-listed`, specifically the read-side `table-load` action rather than a mutation action such as `table-commit`, an affirmative authorization receipt decision, and a non-empty authorization receipt engine with an RFC3339 `checked_at` timestamp; compact QGLake proof also binds that pointer-log replay to the accepted principal subject/kind, a full authorization receipt hash, and the `table-load` action. Service replay closes the top-level `table.commits-listed` payload over the fields LakeCat producers emit, so an archived commit-history read cannot attach unverified commit, pointer, lineage, graph, `QueryGraph`, or application claims beside checked table scope, count, sequence, commit hash, principal, and authorization evidence. It also closes the wrapped `table.commits-listed` outbox envelope over the producer fields, so those unverified commit-history claims cannot be placed beside an otherwise valid checked inner pointer-log proof. Raw lineage-drain summary construction now reuses that same service replay validator before returning compact proof, so a valid-looking table history summary cannot quietly carry an extra commit-history, lineage, graph, `QueryGraph`, or application claim that full delivery replay would have rejected. The compact `tableCommitHistoryProof` object and the captured LakeCat replay `tableCommitHistory` object are closed over those compared fields, so a saved summary or captured replay sidecar cannot append an extra unverified pointer-log claim beside accepted commit count, sequence, hash, principal, authorization, graph, replay, and `OpenLineage` evidence. The raw QGLake lineage-drain verifier checks the same accepted-principal, agent kind, receipt hash, and action before compact handoff proof is generated, so malformed, denied, actor-drifted, action-drifted, action-less, decision-less, engine-less, or timeless pointer-log summaries cannot become delivered replay evidence. Individual `table.commit` replay is held to the same commit envelope before graph or `OpenLineage` delivery: it must include a positive sequence number, non-empty new metadata pointer evidence, non-blank previous pointer evidence when present, matching commit and authorization principals, the `table-commit` receipt action, positive Iceberg format-version evidence, non-negative snapshot-id evidence, and full SHA-256 request, response, and idempotency-key hashes. The store path now supplies explicit `snapshot_id: 0` proof for metadata with no current snapshot, keeping empty-table or schema-only commits compatible with the replay contract. The policy hash is the only optional hash in that envelope, because some standard commits do not pass through a policy binding. The nested commit object is also closed over the exact fields LakeCat verifies, so replay cannot smuggle an extra unverified commit, policy, storage, graph, or `QueryGraph` claim into the sidecar before QGLake proof is generated. The wrapped `table.commit` outbox envelope is closed as well, so those claims cannot be placed beside an otherwise valid checked inner commit payload. The verifier also normalizes the legacy alias boundary: `snake_case` and `kebab-case` commit fields are both accepted, but duplicate aliases for the same semantic value are rejected before acknowledgement or projection. Credential-vend replay gets the same treatment: `credentials.vend-attempted` must carry a matching credential count, full duplicate-free credential-response prefix hashes, a full redacted storage-profile location hash, a valid authorization receipt principal, a full authorization receipt hash, the `credentials-vend` authorization action that matches the outbox event type, internally consistent secret-reference presence/provider/hash fields, a top-level storage-profile id that agrees with nested storage-profile evidence, a nested storage-profile warehouse that agrees with the event table warehouse, required top-level secret-reference presence evidence that agrees with nested storage-profile evidence, and credential-response metadata

that agrees with the selected storage profile and authorization receipt before delivery. The top-level credentials.vend-attempted payload is also closed over the fields LakeCat actually emits and verifies. A replay record cannot append unverified credential, storage-scope, issuer, graph, OpenLineage, QueryGraph, or application claims beside otherwise valid read-restriction, raw-credential-exception, storage-profile, response-evidence, and authorization proof. Storage-profile upsert replay must likewise reject raw secret references and contradictory secret-reference-state evidence before delivery. Policy-binding upsert replay must carry valid catalog scope evidence before delivery, including policy id, warehouse, optional namespace/table scope, enforcement state, captured ODRL material, and a matching odr1-hash. Namespace lifecycle replay must carry a valid warehouse and namespace path or component array before create/load/drop events can be delivered. Catalog config and namespace-list read replay must likewise carry a valid warehouse, and namespace listing must preserve both an unsigned namespace count and count-aligned namespace-paths evidence. Those paths are parsed as namespace identities and must be non-empty and duplicate-free before standard catalog reads become delivered graph/OpenLineage evidence. Catalog config, namespace list, namespace lifecycle, view list, and view lifecycle replay must also carry valid authorization receipt principals, so saved replay cannot turn standard Iceberg control-plane activity into actorless QueryGraph facts. Management-list read replay applies the same rule to operational discovery: policy, project, server, storage-profile, and warehouse list events must carry unsigned counts, valid warehouse scope when warehouse-scoped, and valid optional project scope before delivery. The warehouse-list project scope is parsed as a project identifier, not accepted as an arbitrary string, so compact management proof cannot smuggle malformed project filters into QueryGraph or OpenLineage. QGLake preserves that scope as warehouseProjectId in compact managementProof, and the verifier requires it to match one of the compact projectIds. A saved handoff therefore cannot pair a project-filtered warehouse inventory with an unrelated or malformed project identity. View list and lifecycle replay must carry valid warehouse, namespace, view name, count, and receipt principal evidence before those view events become graph/OpenLineage material for QueryGraph. View-list replay also carries count-aligned view-names evidence. Each name must parse as a valid catalog view/table name and the array must be duplicate-free, so an archived view.listed event cannot inflate view discovery by repeating or forging view identities. The receipt action must be view-load, matching the compact QGLake action contract; view-manage remains mutation proof for view.upserted, while view.loaded uses view-load and view.dropped uses view-drop. View lifecycle replay with a drifted action is rejected before graph or OpenLineage projection, and top-level view lifecycle payloads are closed over the checked event type, optional interface, warehouse, namespace, view, expected-version, and authorization evidence. Top-level warehouse and namespace evidence must also match the nested view object. QueryGraph cannot accept a view mutation or read under the wrong catalog permission or scope, nor can a replay sidecar append unverified view lifecycle, lineage, graph, or application claims beside otherwise valid view evidence. LakeCat also closes the wrapped outbox envelope for view.upserted, view.loaded, and view.dropped, so those claims cannot be placed beside an otherwise valid checked inner view lifecycle payload. Table lifecycle replay applies the same identity discipline before delivery: table.created, table.loaded, table.deleted, and table.restored must carry a decodable table identity, optional payload scope hints must match it, delete replay must carry soft-delete

evidence that points at the same table with a positive unsigned version, and the authorization receipt must carry a valid principal plus the matching lifecycle action. Create, load, and restore replay must also carry the unsigned table version emitted by the catalog producer plus positive Iceberg format-version evidence. Delete replay carries the same generation and table-format evidence inside the required positive soft-delete record, accepting the durable Rust record spelling `format_version` and the Iceberg-style proof spelling `format-version`. Full table identity objects are closed over warehouse, namespace, and name, and soft-delete evidence is closed over the durable delete-proof fields LakeCat verifies. A replay sidecar cannot attach extra table-scope, delete-state, principal, storage, or application claims beside the checked lifecycle identity or soft-delete record. LakeCat also closes the top-level table lifecycle payload before delivery. That means an otherwise valid `table.created`, `table.loaded`, `table.deleted`, or `table.restored` replay cannot append unverified lifecycle, storage, lineage, graph, `QueryGraph`, or application claims beside checked table identity, version, format-version, location, soft-delete, and authorization evidence. It also closes the wrapped outbox envelope for those table lifecycle events, including delete-side soft-delete wrapper evidence, so those claims cannot be placed beside an otherwise valid checked inner lifecycle payload. The nested metadata-graph summary used for graph projection is closed over its current schema and snapshot summary fields as well; richer graph taxonomy and query behavior still belong in Grust, not in the catalog validator. When those lifecycle events carry table metadata-location, table location, or soft-delete metadata-location evidence, the values must be non-empty before the event is acknowledged or projected. The Iceberg table operation remains the standard catalog action; the stricter non-empty replay evidence is LakeCat's control-plane proof that `QueryGraph` and `OpenLineage` did not accept an empty pointer, table-format, or storage-location claim from a corrupted outbox record. Project, server, and warehouse upsert replay must carry valid tenant-root evidence too: project ids, server scopes, endpoint URLs, storage roots, identifiers, properties, and pre-redacted hash anchors are checked before delivery. Policy-binding, project, server, storage-profile, and warehouse upsert replay must also carry a valid authorization receipt principal, an event-matching catalog action, an affirmative allow decision, a non-empty receipt engine, and an RFC3339 checked-at timestamp before delivery, so compact `QueryGraph` proof can attribute management mutations to the actor and TypeSec-style action accepted by LakeCat. It also requires planned and fetched read restrictions to match before compact proof generation, requires both requested/effective projection and requested/effective stats-field evidence, requires effective projection to be a narrowed subset of the requested projection and to match the planned allowed columns, and requires effective stats fields to be both inside the planned allowed columns and a narrowed subset of the requested stats fields in source replay and compact handoff proof. Empty allowed-column evidence is rejected in planned and fetched replay instead of being interpreted as unrestricted replay. It also requires the fetched projection and filter requirements to exactly preserve the fetched allowed columns and row predicate. A fetched response that omits required-filter proof is rejected just like one that widens or changes that proof, and the compact handoff summary applies the same missing-proof check before accepting governed scan evidence. Credential replay applies the same policy-proof discipline to the two credential branches: the restricted-agent denial and trusted-human audited raw-credential exception must both carry a complete read restriction, and those restrictions must match before credential proof can feed the compact handoff summary. For

commit-history replay, the typed drain summary carries the commit count, committed sequence numbers, commit hashes, replay hashes, and OpenLineage hashes. The handoff verifier rejects compact scan proofs without those OpenLineage hashes and compact commit-history proofs whose counts, sequences, or hash arrays do not align. It also requires the compact commit, replay, and OpenLineage arrays to contain full sha256:-prefixed 64-hex digests, not prefix-shaped placeholders. Source replay validation applies the same pointer-history discipline before compact proof generation: the table commit count must match the sequence-number and commit-hash arrays, commit sequences must be positive and strictly increasing, and commit hashes must be full SHA-256-shaped before pointer-history evidence can enter the compact handoff proof. The store-level source of that proof is also row-bound: decoded commit records must match durable pointer-log table identity and principal evidence before those records can be summarized. The graph projection helper applies the same fail-closed posture to the commit-history summary it consumes: missing, count-drifted, or non-string commit-hashes fail before commit graph nodes are emitted, so graph evidence cannot silently carry a null commit hash if a future internal path bypasses the replay validator. A table with no recorded commit entries is still valid pointer-history evidence: it drains as an explicit zero-count read, emits lineage proof, and does not fabricate loaded commit graph nodes. The compact QGLake verifier accepts that zero-count proof only when the sequence-number and commit-hash arrays are also empty; once entries exist, they must satisfy the normal positive-sequence, full-hash, duplicate-free invariants. Service route coverage pins the producer side too: request hashes, response hashes, idempotency-key hashes, and commit hashes are full SHA-256 digests across the route response, pointer-log outbox payload, lineage-drain summary, and graph projection. The QGLake fixture verifier also checks the management commit-history response itself, so short readable placeholders are rejected before the later lineage-drain and compact handoff checks run. QueryGraph can therefore verify the governed Sail-planned read and pointer-history inspection without parsing the full lineage payload. The core QueryGraph bundle, graph, OpenLineage, and import anchors must be SHA-256-shaped in compact verify/import/bootstrap proof before a matching summary can pass; matching strings are not enough. The bootstrap, scan, credential, view, receipt-chain, and commit-history receipt arrays must also carry full SHA-256 digest evidence before compact proof generation can consume them. The same full-digest check applies to accepted-view receipt evidence: bootstrap view-version receipt hashes, tombstone receipt hashes, and namespace receipt-chain hashes must be full SHA-256 digest evidence before accepted-view proof feeds the compact handoff summary. Raw lineage-drain verification also rejects short sha256: placeholders for credential replay sink receipts and table commit-history replay/OpenLineage receipts before those arrays can become compact QueryGraph proof. The raw request/bootstrap side follows the same rule: lineage-read authorization, bootstrap authorization, core QueryGraph bundle/graph/OpenLineage/import hashes, agent delegation and summary signature hashes, and TypeDID envelope/proof hashes must all be full SHA-256 digest evidence before request identity or QueryGraph bootstrap proof can feed the compact handoff. Raw lineage-drain summary construction now applies the structural shape rule too: every view-version-receipt-chains entry must decode as a ViewVersionReceiptChainResponse, and a present chain-verified-count must be an unsigned count. A malformed chain object can no longer disappear from the raw summary just because its chain-hash looked valid. Dropped and active accepted-view source replay also compares the

bootstrap view-version receipt hashes with the accepted QueryGraph verification set, so a valid-looking receipt array cannot be spliced from another bootstrap proof. The compact handoff verifier repeats the same binding against `viewReceiptChainProof.views[].acceptedReceiptHash` and `viewReceiptChainProof.views[].acceptedReceiptChainHash`, including tombstoned accepted views, catching drift even when an operator validates only the saved summary. Dropped accepted-view source replay also binds the namespace receipt-chain read back to the accepted view warehouse/namespace and rejects verified-chain count or receipt-hash coverage drift before compact handoff proof is generated. The lineage-drain summary now carries the nested chain receipts as full receipt hash coverage before that check runs. Summary construction also fails closed when top-level or nested view receipt and receipt-chain hashes are malformed, or when a nested receipt-chain object or verified-count field is malformed, instead of omitting bad values from the raw replay summary. Raw lineage-drain summaries now also reuse the full view receipt-list and receipt-chain replay validators before returning compact proof, so otherwise valid view-history summaries cannot append extra lineage, graph, QueryGraph, or application claims that delivery replay would have rejected. Generated replay evidence also preserves each accepted view's `acceptedReceiptChainHash` inside the namespace `receiptChains[].chainHashes` set, even when the namespace read has its own chain hash, so the compact summary can prove the accepted chain is covered by the namespace proof it verifies. The same replay now emits catalog-facing Commit graph events for the listed sequences, leaving traversal and query semantics to Grust.

For handoff testing, LakeCat can verify a saved bootstrap bundle and a saved drain response together:

```
cargo run -p lakecat-cli -- qglake-verify-replay \
--bundle target/qglake/lakecat-bootstrap.json \
--drain target/qglake/lineage-drain.json \
--principal did:example:agent
```

That offline check replays the same boundary assertions used by the live QGLake fixture: the bundle manifest must verify, the QueryGraph import compatibility contract must match, and the lineage drain must carry matching bootstrap hashes, credential-denial receipts, management-list evidence, scan/fetch task evidence, and table commit-history receipt evidence, plus view receipt evidence when views are present. On success, the command prints the accepted bundle and QueryGraph import hashes, table/view counts, and compact control-plane lines such as:

[illegible]

aligned, matching service-side outbox admission, so a saved handoff cannot inflate the number of accepted tables or views by repeating an already accepted stable id. Raw lineage-drain replay summaries and compact handoff proof sections both keep replay, OpenLineage, credential prefix, view receipt, and view receipt-chain hash arrays duplicate-free, not only sha256: shaped. That covers the bootstrap, governed scan, management, table commit-history, view tombstone/receipt-chain, storage-profile upsert, and credential-vending proof sections, so a source replay or archived handoff cannot make an evidence set look larger by repeating an already accepted digest. The service applies the first version of that rule before a drain summary is returned at all: projection receipt hash arrays must match the lineage-event count and must not contain malformed or repeated replay or OpenLineage hashes. Commit-history summary construction now applies the same fail-closed posture to commit-count, sequence-numbers, and commit-hashes, so malformed or count-drifted commit-history entries cannot disappear from raw QGLake replay proof or inflate compact pointer-history evidence. Governed scan summary arrays for required/requested/effective projection and requested/effective stats fields also reject malformed, blank, or duplicate strings before raw replay proof is returned, so a malformed Sail-planned read summary cannot hide missing projection or stats evidence. Planned and fetched scan summary construction applies the same second-line check to required-filters: when row-predicate evidence is present, the field must remain array-shaped and exactly preserve that server-derived predicate rather than widening or omitting the mandatory filter inside raw QGLake proof. The raw summary regression suite covers both planned and fetched scan replay, so either half of the governed scan can fail closed before compact proof is returned. The verifier also compares those QueryGraph import-plan graph node and edge counts with the verified bootstrap bundle graph counts, so an import plan cannot keep the semantic hashes and table/view ids while silently dropping graph material. It also parses the saved lineage-drain response, reruns the typed QGLake replay verifier, and regenerates the LakeCat replay evidence that proves request identity, QueryGraph bootstrap replay, governed scan replay, pointer history, view receipt chains, storage-profile replay, and credential-vending decisions. It then compares that regenerated replay evidence to the compact lakecatReplayVerification proof. The governed scan proof includes both the requested and effective stats-field arrays from the scan-planned replay, and the verifier rejects handoffs where the effective fields no longer match the allowed columns or no longer prove policy narrowing. Credential-vending proof is not just a count: each restricted-agent and trusted-human branch carries the redacted storageProfile graph anchor and maxCredentialTtlSeconds, including profile id, provider, issuance mode, secret-reference presence, and the graph event count emitted by replay. The storage-profile upsert proof also carries its own positive graphEvents count, and captured LakeCat replay must match it. The verifier also rejects a handoff when the credential branches do not bind back to the same storage-profile upsert proof: profile id, provider, issuance mode, location-prefix hash, and secret-reference state must all match the replayed management event. A saved handoff is rejected if the archived lineage drain proves lineage receipt hashes but omits that credential TTL cap or credential-root graph projection. It also recomputes the captured LakeCat replay and QueryGraph verify/import output hashes, so terminal captures cannot drift from the compact summary. It compares the legacy string path aliases for the LakeCat replay, QueryGraph verify, and QueryGraph import captures with the hashed capturedOutputs paths they duplicate. It also hashes the service log through a full-digest serviceLogHash, so

archived operational logs cannot drift behind a stable path or a short placeholder hash. The final local summary also binds the first LakeCat handoff-verifier capture with a required full-digest `lakecatHandoffVerifyOutputHash`. A saved handoff that names `lakecat-handoff-verify.json` but omits that hash, sets it to null, or uses a short placeholder is rejected. Because that output can only exist after a successful verifier run, the harness performs a second sidecar self-check: first it writes `target/qglake-handoff/lakecat-handoff-verify.json`, then it records the file's hash in the summary, then it verifies the summary again without overwriting the declared artifact. The verifier checks that saved JSON is a `lakecat.qglake.handoff-verification.v1` success for the same principal, catalog URL, warehouse, namespace, and table, and that its table/view counts, stable ids, standards, request-identity proof, and QueryGraph bootstrap proof still match the compact handoff summary. It also checks the saved self-verifier output's bundle, lineage-drain, QueryGraph import-plan, captured-output, and service-log hashes against the summary's artifact manifest. The compact summary's own `artifacts.lineageDrain` and `artifacts.querygraphImportPlan` objects are also closed over a single full-digest sha256 field before the saved self-verifier layer is trusted, so short placeholders and extra unverified hash mirrors fail at the summary boundary. The saved sidecar must keep the core `artifactFiles.bundle`, `artifactFiles.lineageDrain`, and `artifactFiles.querygraphImportPlan` hash objects present, and it must keep the nested `artifactFiles.capturedOutputs` manifest present before any captured LakeCat or QueryGraph output hash is trusted. The lineage-drain and QueryGraph import-plan artifact hashes must still match the compact handoff summary; an archived sidecar cannot rewrite either nested artifact digest and then recompute only the outer verifier-output hash. Their saved hash objects are closed over a single full-digest sha256 field as well, so short placeholders or extra unverified fields fail before those artifacts are trusted. The saved sidecar must also keep the `artifactFiles.capturedOutputs.lakecatReplay`, `artifactFiles.capturedOutputs.querygraphVerify`, and `artifactFiles.capturedOutputs.querygraphImport` hash objects present before the captured LakeCat replay or QueryGraph transcripts are trusted. Each capture hash object is closed over a single full-digest sha256 field; short placeholder hashes or extra unverified hash fields fail before the saved QueryGraph verify/import transcripts can be trusted. Those saved QueryGraph verify/import hashes must also match the compact handoff summary, so an archived sidecar cannot rewrite the nested transcript digest and then recompute only the outer verifier-output hash. The regression suite covers the core bundle, lineage-drain, QueryGraph import-plan, and captured-output hash objects as required saved-sidecar entries. The saved output cannot keep a valid verifier-output hash while rewriting `artifactFiles.serviceLogHash`; that operational log digest must still be present, non-null, full-length, and match the compact handoff summary. The regression suite covers each saved-sidecar shape explicitly: missing, null, short placeholder, and drifted service-log hashes all fail before the archived handoff proof is accepted. Artifact paths are checked before hashing as well: every path must resolve under the handoff summary directory, so an archived summary cannot splice in an absolute path or .. traversal to matching bytes outside the bundle. The semantic readers use the same bundle-local resolver, so parsing the captured outputs, bootstrap bundle, QueryGraph import plan, and lineage drain cannot bypass the containment rule that hash verification already enforced. It also checks the saved self-verifier output's own semantic sections: captured replay semantics must match the compact LakeCat and

QueryGraph proof, bundle artifact semantics must match QueryGraph verification, import-plan semantics must match QueryGraph import verification, lineage-drain semantics must match the accepted replay proof, and saved import-plan graph counts must still match the saved bundle graph counts. Then it parses the archived lineage-drain artifact and requires the saved lineage-drain semantics' delivered count, event type list, graph event count, lineage event count, and drain authorization action to match before accepting the verifier-output hash. The archived drain itself must also reconcile those same top-level counts with its replay summary array, including repeated event-type multiplicity and the exact eventTypes to replay-summary order. Then it parses those captured JSON files and checks that the replay schema/status, table/view counts, semantic hashes, standards, request-identity proof, QueryGraph bootstrap proof, governed scan proof, storage-profile upsert proof, and credential-vending proof inside the captures still match the summary. It also rejects malformed TypeDID hash slots in the request-identity and QueryGraph bootstrap proofs before a consumer has to interpret those slots. The local handoff harness runs it automatically and writes the captured verifier output to `target/qqlake-handoff/lakecat-handoff-verify.json`. Before the harness writes compact proof, it checks the replay and OpenLineage hash arrays it lifts from LakeCat replay evidence as full SHA-256-shaped and duplicate-free, so malformed compact proof is rejected before the archived handoff summary is treated as accepted.

The end-to-end result is a chain:

```
catalog write
-> audit event
-> outbox event
-> graph projection
-> OpenLineage projection
-> QueryGraph import evidence
```

If graph or lineage sinks are down, catalog state should not be lost or rolled back accidentally. The outbox lets LakeCat retry projection from committed state. A drain acknowledges delivery only after every projection in the batch succeeds. If OpenLineage fails after a graph event has already been emitted, the drain fails and the catalog event remains pending, so recovery starts from the committed outbox rather than from guesswork. If the graph sink fails first, LakeCat fails the drain before emitting lineage and still leaves the outbox event pending, so graph and lineage consumers recover from the same committed catalog fact instead of diverging.

Replay order is part of that contract. LakeCat selects undelivered outbox events by `created_at, event_id` in both embedded memory tests and the durable Turso store, so a QGLake replay does not depend on writer interleaving or database row-return quirks. Delivery acknowledgement is duplicate-safe as well: if a drainer accidentally reports the same event id twice, LakeCat marks the event once and the receipt count remains tied to committed catalog facts. Pending batch validation happens before projection: if a store returns duplicate event IDs in the same drain batch, LakeCat fails the drain with only the duplicate event-id hash and does not emit graph, OpenLineage, or acknowledgement side effects for that batch. The same redaction rule applies to malformed pending records. If a custom or corrupted store hands the drain an event whose payload cannot be projected, LakeCat reports only the outbox event-id hash and

stops before graph emission, lineage emission, or delivery acknowledgement. Malformed table and principal identity JSON decode failures, as well as unsupported event types, follow that same pattern: they carry event-hash evidence for correlation without echoing the raw event identifier into diagnostics.

1.9.9 An Agentic QGLake Flow

The agentic path is the reason LakeCat has to be more than a passive catalog. Imagine a resilience supervisor agent investigating incidents:

1. The supervisor delegates table triage to a specialist agent.
2. The specialist asks LakeCat to plan a scan over `local.default.events`.
3. LakeCat resolves the agent identity and TypeDID context.
4. TypeSec authorizes the table scan and returns a restricted capability.
5. LakeCat narrows the projection and appends the required row predicate.
6. Sail plans against the current Iceberg metadata and delete manifests.
7. LakeCat returns governed plan and fetch-task responses.
8. The specialist summarizes only the allowed result shape.
9. LakeCat records scan and credential decisions into audit/outbox.
10. QueryGraph imports graph, policy, lineage, and bootstrap evidence.

The key point is the absence of raw storage reach. The specialist agent does not need broad cloud credentials to do its job. It needs a governed plan, a bounded task set, and a receipt trail.

The local fixture compresses this story into a short artifact-producing sequence:

```
cargo run -p lakecat-cli --features qglake-fixture -- qglake-fixture \
  --output target/qglake/lakecat-bootstrap.json \
  --drain-output target/qglake/lineage-drain.json \
  --principal did:example:agent
cargo run -p lakecat-cli -- qglake-verify-replay \
  --bundle target/qglake/lakecat-bootstrap.json \
  --drain target/qglake/lineage-drain.json \
  --principal did:example:agent
```

The fixture generator opts into `lakecat-cli`'s `qglake-fixture` feature because it writes local Iceberg metadata and manifest files through Sail. Verification commands stay in the default CLI surface, which keeps ordinary catalog inspection, replay, and handoff checks available without pulling in the local fixture writer.

The one-command handoff wraps the same evidence in a live local service run and then asks QueryGraph to verify and import it:

```
scripts/qglake-handoff-local.sh
cargo run -p lakecat-cli -- qglake-verify-handoff \
  --summary target/qglake-handoff/handoff-summary.json \
  --json
cat target/qglake-handoff/handoff-summary.json
```

That fixture creates the sample table shape, installs a restricted policy, verifies governed scan planning, verifies fetch-scan-task reapplication, checks requested/effective stats-field narrowing in replay and handoff proof, exercises delete manifest handling, probes credential-vend behavior for agents and trusted humans, verifies compact table commit-history evidence, exports QueryGraph bootstrap artifacts, drains the outbox, and proves the resulting bundle through QueryGraph's Rust verifier/importer. It then asks LakeCat to verify its own compact handoff summary and recompute the raw artifact file hashes, making the summary a first-class acceptance artifact rather than an unchecked convenience file. The compact handoff summary root is closed over its known fields before any artifact is accepted, so an archived summary cannot add an unverified top-level proof claim beside a valid LakeCat replay and QueryGraph import result. The saved self-verifier output is checked too: its internal `artifactFiles` entries for the bootstrap bundle, lineage drain, QueryGraph import plan, captured LakeCat and QueryGraph outputs, and service log must carry full SHA-256 digests before they are compared with the compact summary. The saved sidecar also has to stay closed over that known manifest: extra top-level artifact claims and extra nested captured-output claims are rejected instead of being carried as unverified evidence beside the accepted hashes. Each nested artifact and captured-output hash object is also closed over `sha256` only, so an otherwise accepted hash cannot carry an alternate unverified hash claim beside it. The sidecar root and `capturedOutputSemantics` object are closed over their known schema keys for the same reason: a saved verifier output should not append proof sections that no verifier compares. The individual LakeCat, QueryGraph, bundle, import-plan, and lineage-drain semantic sections are closed the same way; a saved `querygraphImportPlanSemantics` block cannot append an extra import-plan proof beside matched graph counts and hashes. The import-plan artifact itself now follows the same rule: extra root fields, verification fields, table-entry fields, or view-entry fields fail before the artifact's semantic hashes, standards, stable ids, and graph counts can be accepted. It is small, but it is not decorative. It is the acceptance story for a catalog that participates in the user workflow from notebook to agent. The summary file gives automation a single stable place to find the accepted table/view counts, semantic hashes, bundle, lineage drain, import plan, captured verifier outputs, and raw artifact hashes without scraping terminal text.

1.10 Operating The Book's Example System

The local development posture is intentionally small:

```
cargo run -p lakecat-cli -- config
cargo run -p lakecat-cli -- storage-profile-list
cargo run -p lakecat-cli -- policy-list
cargo run -p lakecat-cli --features qglake-fixture -- qglake-fixture \
  --output target/qglake/lakecat-bootstrap.json \
  --drain-output target/qglake/lineage-drain.json \
  --principal did:example:agent
cargo run -p lakecat-cli -- qglake-verify-replay \
  --bundle target/qglake/lakecat-bootstrap.json \
  --drain target/qglake/lineage-drain.json \
  --principal did:example:agent
scripts/qglake-handoff-local.sh
cargo run -p lakecat-cli -- qglake-verify-handoff \
```

```
--summary target/qglake-handoff/handoff-summary.json \  
--json  
cargo run -p lakecat-cli -- bootstrap-export \  
--output lakecat-bootstrap.json
```

The important thing is what these commands exercise. They are not a separate product surface. They touch the same catalog, policy, bootstrap, and QueryGraph export contracts that the service uses.

1.11 Standards And Engine Boundary Decision Record

This chapter is the release decision record for the current catalog concepts. It answers three questions that come up whenever LakeCat vocabulary gets close to Iceberg vocabulary:

1. Which terms are standard Iceberg parlance?
2. Which terms are LakeCat, QueryGraph, TypeSec, Grust, or Sail extensions?
3. Which ideas are plausible future Iceberg-adjacent optional profiles?

The distinction matters because LakeCat should be ambitious without making ordinary Iceberg clients pay for that ambition. A PySpark job should be able to load and commit a table through normal REST catalog behavior. A governed agent should be able to receive a TypeSec-backed, Sail-planned, replayable proof of bounded work. QueryGraph should be able to import QGLake evidence, OpenLineage anchors, view receipt chains, credential posture, management proof, and commit proof. Those are three different audiences using one catalog foundation.

The implementation can be pictured as four concentric contracts:

standard Iceberg client contract
namespace, table, metadata location, optimistic commit

LakeCat catalog-control contract
identity, tenancy, Turso rows, CAS, idempotency, audit, outbox, replay

TypeSec and Sail governed-work contract
capability decision, restriction, receipt, projection, predicate, scan plan

QueryGraph semantic contract
QGLake handoff, OpenLineage, Croissant, CDIF, OSI, ODRL, graph, agents

The outer layers are additive. They can observe, prove, and compose catalog state, but they must not redefine the inner standard table contract.

1.11.1 Catalog Concepts In Plain Terms

This section names the catalog concepts directly, because the words can sound similar even when they live at different layers. The simplest way to read the architecture is this:

- Iceberg defines the portable table contract.
- LakeCat defines the durable catalog-control contract around that table.
- TypeSec defines the policy and capability evidence used by governed work.
- Sail defines the engine-grade interpretation of Iceberg table state.

- QueryGraph defines the semantic workflow built from accepted catalog proof.

The Rust service/catalog spine is a LakeCat implementation choice, not an Iceberg concept. Iceberg does not care whether a catalog is written in Rust, Java, Go, or Python. Iceberg cares that a client can ask for catalog config, list namespaces, load a table, and commit a new metadata pointer under ordinary REST catalog rules. LakeCat chooses Rust because the proof path should be short, typed, and close to the engine. In one process, LakeCat can parse the request, resolve identity, check tenancy, call TypeSec-style authorization, ask Sail for table interpretation, update the CatalogStore, append audit and outbox rows, and return the standard response. The portable idea is not “Rust catalogs are standard.” The portable idea is that a catalog state transition can be deterministic, auditable, replayable, and bound to engine facts without extra indirection.

The Turso-backed local store is also a LakeCat implementation choice, not an Iceberg term. Iceberg needs durable catalog state and atomic metadata-pointer movement; it does not name Turso, SQLite, PostgreSQL, FoundationDB, or any other storage engine. LakeCat uses the Rust turso crate for the local durable spine because it gives the project a fast embeddable store while keeping the store contract explicit. The important abstraction is CatalogStore: tables, views, namespaces, policies, storage profiles, idempotency records, pointer logs, audit rows, outbox rows, and replay markers must be read and written in ways that bind decoded data back to the requested tenant, warehouse, namespace, and table. Turso is the current local vehicle. The future portable concept is stronger than a database choice: exact catalog retry, atomic pointer movement, row-binding checks, transactional event identity, and replayable durable proof.

The Iceberg REST namespace and table paths are the compatibility floor. These are standard Iceberg parlance: catalog config, namespace, table identifier, metadata location, table metadata, requirements, optimistic commit, snapshot, manifest list, manifest, data file, delete file, partition spec, sort order, and schema. A PySpark client should be able to use LakeCat as an Iceberg REST catalog without knowing what QGLake, TypeSec, Grust, or QueryGraph are. That is the line LakeCat must not cross. If normal table load or commit requires a non-standard QueryGraph endpoint, then LakeCat has stopped being an Iceberg-compatible catalog and has become a private application protocol. The LakeCat proof work should happen behind the route or beside the route, not in place of the route.

Commit CAS straddles standard language and LakeCat hardening. In Iceberg, a commit is optimistic: the client presents requirements that describe the table state it believes it is changing, and the catalog accepts the new metadata location only if the requirements still hold. LakeCat implements that as metadata-pointer compare-and-swap, which is standard in spirit: move from the previous metadata location to the new one only if the current state matches the expected state. LakeCat then adds proof around it: the idempotency record, metadata-object create-only posture, pointer log, audit row, outbox event, redacted conflict evidence, and replay admission verdict. CAS itself belongs to Iceberg-compatible commit correctness. The proof envelope is LakeCat’s catalog reliability layer, and a stripped-down form of it could become a future optional catalog reliability profile.

Idempotency, pointer logs, audit/outbox, and replay validation are not the normal vocabulary a PySpark user brings to Iceberg. They are LakeCat’s control-plane vocabulary. Idempotency means a retried commit with the same key must return the same accepted response or fail if the request drifted. Pointer logs mean operators and QueryGraph can reconstruct which metadata location was current after each accepted transition. Audit rows mean the catalog can explain who did what, under which principal and tenant, without exposing credentials or raw secrets. Outbox rows mean graph and lineage side effects are durable catalog facts that can be drained later rather than fragile request-path side effects. Replay validation means the drained event is checked again against a closed schema before Grust, OpenLineage, QGLake, or QueryGraph can inherit it. Those are extensions today. They are also among the best future proposal candidates because they can be described without adopting LakeCat’s product names: “catalog event admission,” “exact retry,” “pointer history,” and “transactional lineage outbox” are useful neutral ideas.

Governed scan and credential paths are LakeCat plus TypeSec plus Sail. Iceberg itself says an engine can use table metadata to plan a scan. It does not say that a catalog must return a TypeSec receipt, a purpose string, a policy hash, an allowed-column list, a mandatory predicate, a credential TTL cap, or a redacted proof of storage scope. LakeCat introduces those as additive governed surfaces because agents and untrusted principals should not default to broad object-store credentials. In the governed scan path, TypeSec-style policy logic decides the capability and restriction; LakeCat binds that receipt to the current pointer, request identity, tenant, and audit context; Sail turns the restriction into engine work. In the governed credential path, raw credential vending is intentionally narrower: it is an audited exception for a principal that is allowed to read directly, and its response must leave redacted prefix hashes, storage-profile hashes, TTL evidence, and secret-ref posture rather than leaking credential material. This is an extension today. The future proposal candidate is not “TypeSec in Iceberg”; it is a policy-engine-neutral governed access profile with proof-carrying scan and credential posture.

QueryGraph and QGLake handoff surfaces are product integration surfaces, not standard Iceberg semantics. QueryGraph needs bootstrap evidence, management inventory, table and view manifests, OpenLineage event hashes, graph import hashes, view receipt chains, credential posture, storage-profile posture, commit-history proof, replay verdicts, and artifact hashes. LakeCat should emit those anchors because QueryGraph needs a trustworthy substrate. But ordinary Iceberg clients should not need to produce or consume them. The standardization candidate is narrower than the QueryGraph feature set: event identity, lineage binding, replay-admissible payloads, view lifecycle proof, and commit proof could be phrased neutrally. Croissant, CDIF, OSI, ODRL, TypeDID workflow composition, QGLake artifact layout, Grust graph import shape, and agent workflow reasoning should remain QueryGraph, TypeSec, and Grust architecture unless and until a narrower neutral profile emerges.

Typed Iceberg v4 support is different again. If Iceberg v4 introduces or solidifies table-format semantics, those semantics are Iceberg work and engine work, not LakeCat product invention. LakeCat can preserve unknown JSON fields as a compatibility bridge, but passthrough is not interpretation. The durable direction is for Sail to own typed support: metadata structures, manifest and delete semantics, scan planning, metadata-as-data, commit validation, and view/

table behavior. LakeCat can then advertise an honest state: standard REST compatibility now, JSON preservation where necessary, and typed Sail-backed proof as those engine APIs mature.

That is the extension answer. LakeCat has extensions in the product sense, because it extends what a catalog can prove. Most of those extensions should not be proposed as mandatory Iceberg behavior. The ones worth considering for Iceberg-adjacent discussion are the parts that can be stated without proper nouns: exact retry, pointer history, redacted conflict proof, transactional catalog events, replay-admissible lineage, governed scan proof, governed credential posture, and view lifecycle proof. The proper nouns are where the implementation is being proven. The neutral concepts are what might someday be shared.

1.11.2 Decision Matrix

Concept	Iceberg meaning	LakeCat or Query-Graph meaning	Decision
Rust service/catalog spine	Iceberg does not specify an implementation language.	LakeCat keeps route handling, identity, tenancy, store transactions, Sail calls, TypeSec receipts, CAS, idempotency, audit, outbox, and replay close together in Rust.	Project implementation. The portable behavior is deterministic and replay-safe catalog state transition proof.
Turso-backed local store	Iceberg needs durable catalog state and atomic metadata-pointer movement, not a named database.	LakeCat uses the Rust turso crate as the durable local CatalogStore spine for tables, views, policies, storage profiles, idempotency, pointer logs, audit, outbox, and replay markers.	Project implementation. Generalize CAS, exact retry, pointer history, row binding, and transactional event identity, not Turso.
Iceberg REST namespace and table paths	Standard catalog compatibility: config, namespace operations, table load/create/drop, metadata locations, requirements, and optimistic commit.	LakeCat serves these paths while attaching server-side audit, outbox, and replay evidence.	Standard floor. Do not make normal clients depend on QueryGraph, TypeSec, Grust, QGLake, or LakeCat proof schemas.
Commit CAS	Optimistic metadata-pointer movement under commit requirements.	LakeCat binds the accepted transition to idempotency, pointer logs, audit, outbox, redacted conflict proof, and replay admission.	CAS is standard. The proof envelope is a future optional reliability-profile candidate.
Idempotency, pointer logs, audit/outbox, replay validation	Not generally standard table semantics, except insofar as they protect commit correctness.	LakeCat makes retries exact, pointer history inspectable, side effects recoverable, and down-serial provenance.	LakeCat extension today. Strong candidate for neutral catalog reliability and event-admission proof extension today.
QueryGraph CatalogStore lineage/bootstrap/management/view/commit proof	Not generally standard table semantics, except insofar as they protect commit correctness.	LakeCat makes retries exact, pointer history inspectable, side effects recoverable, and down-serial provenance.	LakeCat extension today. Strong candidate for neutral catalog reliability and event-admission proof extension today.

The standards answer is therefore not a blanket yes or no. LakeCat implements many extensions in the practical sense, because it extends what a catalog can prove. Only the parts that survive without LakeCat-specific names should be considered future Iceberg-adjacent proposals.

1.11.3 The Proper-Noun Test

The proper-noun test is the simplest review tool:

- If the concept requires the words LakeCat, QueryGraph, QGLake, TypeSec, Grust, Sail, or Turso to be meaningful, keep it as project architecture.
- If the concept can be described as behavior any Iceberg-compatible catalog or engine could implement, consider it a future optional-profile candidate.
- If the concept changes what a standard client must send or understand for an ordinary table load or commit, treat it as suspect until compatibility is proven.

That test keeps the proposal set small. Rust and Turso are excellent LakeCat choices, but they are not standards work. TypeSec receipts and QueryGraph import plans are essential product interfaces, but they should not become Iceberg requirements. Exact retry, pointer-history proof, redacted conflict proof, transactional event identity, replay-admissible evidence, governed scan proof, credential posture proof, view receipt chains, and lineage receipt binding are more plausible because they can be expressed without adopting the LakeCat stack.

1.11.4 Why The Engine Owns The Truth

The strongest technical reason to push work into Sail is that Iceberg table truth is not a set of easy strings. A governed proof that says “only these columns” is not reliable until it is mapped through Iceberg field ids, nested schema evolution, aliases, and projection rules. A proof that says “only these rows” is not reliable until the predicate has been interpreted by the same expression system that will plan the scan. A proof that says “these files” is not reliable until manifest metrics, residual predicates, partition transforms, delete files, sequence numbers, and snapshot context have been interpreted by the engine.

LakeCat can prove authority:

```
principal -> tenant -> warehouse -> namespace -> table
request -> TypeSec receipt -> current pointer -> CAS/idempotency
accepted state -> audit row -> outbox row -> replay admission
```

Sail should prove table work:

```
metadata -> schema field ids -> projection
metadata -> snapshots/manifests/statistics -> pruning
metadata -> equality/position deletes -> row visibility posture
metadata -> scan tasks / metadata-as-data / commit validation
```

When those two proofs are bound together, QueryGraph gets evidence that is both authorized and data-close. When LakeCat tries to do both jobs itself, it becomes a shadow engine: slower, less correct, and less reusable. Manifest metric decoding, delete planning, metadata-as-data, scan tasks, and typed v4 support should benefit Sail users, LakeCat governed reads, QueryGraph import, and the wider Rust lakehouse stack at the same time.

Sail is a particularly good engine boundary because it keeps the proof-heavy path Rust-to-Rust. LakeCat can receive a REST request, bind identity and tenancy, call TypeSec-style governance, ask Sail for typed table interpretation or a plan, commit through the Turso-backed CatalogStore, and persist replay proof without crossing a JVM sidecar, Python shim, or remote plugin boundary in the hot path. The public compatibility boundary remains Iceberg REST. The internal correctness boundary stays close to the Rust engine.

1.11.5 Workflow Consequences

For a PySpark user, the catalog remains ordinary. The user configures an Iceberg REST catalog, lists namespaces, loads a table, reads metadata, and commits with optimistic requirements. LakeCat’s proof work stays server-side. The PySpark job does not need to understand TypeSec receipts, QGLake artifacts, Grust graph imports, or QueryGraph bootstrap.

For an operator, the catalog becomes inspectable. The operator can ask which principal moved a pointer, whether an idempotent retry drifted, which durable row was accepted, which outbox event is pending, which replay validator admitted the evidence, and whether downstream graph or OpenLineage projection came from accepted catalog state.

For a governed service, the catalog becomes enforceable. TypeSec narrows the request by purpose, capability, policy hash, allowed columns, mandatory predicate, TTL cap, and credential posture. LakeCat binds that decision to the current pointer and request identity. Sail turns the restriction into field-id projection, predicate binding, manifest/file pruning, delete posture, scan tasks, and plan evidence.

For an agent, the safest default becomes bounded work. Raw credentials are still possible for deliberately authorized principals, but they are audited exceptions. A restricted agent should usually receive Sail-planned tasks or a proof-backed fetch path whose scope, TTL, projection, predicate, and credential posture are already bound.

For QueryGraph, the catalog becomes a foundation rather than an application. LakeCat emits stable anchors: bootstrap hashes, management inventory hashes, policy-binding hashes, storage-profile hashes, commit-history hashes, view receipt-chain hashes, credential posture hashes, OpenLineage event hashes, graph/import hashes, artifact hashes, and replay verdicts. QueryGraph composes those anchors with Croissant, CDIF, OSI, ODRL, TypeDID, Grust graph state, and agent workflow state. LakeCat does not import QueryGraph. QueryGraph consumes LakeCat proof.

That separation is the release discipline. Standard clients get standard Iceberg behavior. Operators get durable Rust and Turso proof. Governed callers get TypeSec decisions bound to Sail engine work. QueryGraph gets a semantic handoff that is broad enough to build on and narrow enough not to fork Iceberg.

1.12 First Release Readiness

The first LakeCat release should not try to finish every idea in this book. It should release the catalog substrate that can already be proven locally. The right question is not “does LakeCat

contain the whole future QueryGraph stack?” It is “does LakeCat provide a compatible, durable, governed catalog foundation that QueryGraph can trust?”

For the first release, the release-blocking behavior is the catalog spine: standard Iceberg REST config, namespace, table-load, table-create, and table-commit paths; warehouse-prefixed routing; Rust service identity handling; the CatalogStore seam; the Turso-backed local store; memory-store parity for embedded tests; metadata-pointer CAS; idempotent commit replay; pointer logs; audit rows; outbox rows; and replay admission that rejects malformed durable evidence before graph or OpenLineage projection.

The governed path is also release-blocking because it is the reason LakeCat exists for agents. A restricted agent should be able to ask for a governed read, receive a TypeSec-style receipt, get Sail-planned work instead of broad storage authority, and leave behind replayable scan, fetch, credential, management, view, and commit-history evidence. Trusted raw credential vending can exist only as an audited exception with redacted storage-scope proof. Fetched scan replay treats the returned plan-task as evidence rather than arbitrary text: if it is present, it must be a non-empty LakeCat-issued token and it must not contain decorated location, query/fragment, or credential material before graph, OpenLineage, QGLake, or QueryGraph import can inherit the fetch proof. That keeps governed plan/fetch receipts about Sail-planned work, not a carrier for raw path or token claims. The same fetched replay path treats stats-fields as checked evidence when it is present: the array must be non-empty, duplicate-free, and bound to the effective stats fields before downstream proof can use it. The compact QGLake handoff proof now preserves that fetched side as its own fetchedRequestedStatsFields and fetchedEffectiveStatsFields evidence, so a handoff cannot prove only the planned stats narrowing while silently dropping what the fetch path actually returned. The local handoff script applies the same nonblank, duplicate-free array rule to projection, stats-field, and read-restriction allowed-column evidence before it writes that compact proof into the summary.

The QueryGraph handoff is release-blocking as an acceptance proof, not as a requirement for ordinary Iceberg clients. The local QGLake workflow must keep creating a bootstrap bundle, draining LakeCat lineage/outbox evidence, verifying replay, running QueryGraph verification/import, and saving a handoff summary whose artifact hashes, table/view counts, standards, OpenLineage receipts, graph hashes, policy anchors, credential proof, view receipt chains, and commit history agree. If that handoff cannot be reproduced locally, LakeCat may still serve an Iceberg endpoint, but it has not proven the QueryGraph foundation. The saved handoff verifier output must also repeat the lineage-drain catalogConfigProof; omission is treated as proof failure, not as an unspecified optional field. Extra fields inside that repeated proof are also rejected, so an archived verifier sidecar cannot append a new endpoint or compatibility claim beside the validated catalog configuration evidence. Archived handoff file paths are part of that proof. LakeCat resolves artifact paths under the handoff summary directory before hashing or parsing them, and the verifier rejects relative traversal outside that bundle for both hash verification and captured-output semantic reads. A matching hash from another directory is not accepted as QGLake evidence. The artifact manifest itself is also closed: the primary artifacts object, nested capturedOutputs object, and individual bundle, lineage-drain, QueryGraph import-plan, and

captured output artifact objects may carry only the path and SHA-256 evidence LakeCat checks. A saved summary cannot add an alternate hash, mirror artifact, or unverified capture beside an otherwise valid handoff bundle. The captured files themselves are closed at the root as well. Saved LakeCat replay output and QueryGraph verify/import output may carry only the fields LakeCat compares to the compact summary; a matching hash does not make an extra replay, QueryGraph, import, or application claim part of the proof. The same release gate treats raw view-lineage proof hashes as real digests, not placeholders: view replay receipts, tombstone view receipts, namespace receipt-chain hashes, and receipt-chain replay/OpenLineage hashes must be full SHA-256-shaped values before QGLake can archive them as accepted view proof.

The first release should explicitly defer work that belongs elsewhere or is not yet ready to claim. Typed Iceberg v4 semantics belong in Sail; LakeCat should advertise only the current JSON passthrough bridge with `typed-sail=unavailable` until Sail exposes stable typed support. Reusable graph taxonomy, traversal, Cypher, graph stores, and algorithms belong in Grust. Capability composition, TypeDID envelopes, secure-agent proof, and richer policy semantics belong in TypeSec. Croissant, CDIF, OSI, ODRL application composition, and agent-facing reasoning belong in QueryGraph. Cloud SDK secret managers beyond the current Vault and file-backed provider roots are future credential backends, not blockers for the catalog substrate.

As a working estimate, the first-release LakeCat catalog substrate is roughly 90 percent complete. That number is not a promise about the whole future QueryGraph architecture. It means the release-blocking LakeCat pieces are mostly present and locally proven: standard Iceberg REST namespace and table paths, the Rust service spine, the Turso-backed local store direction, metadata-pointer CAS, idempotency, pointer logs, audit and outbox rows, replay admission, governed scan and fetch proof, credential-vending receipt proof, management proof, view receipt chains, QueryGraph bootstrap, OpenLineage replay, and QGLake handoff/import evidence.

The remaining 5-10 percent is concentrated in release engineering and dependency-boundary cleanup rather than a new conceptual layer. The broad local gate has already been recorded from the current handoff path, including QueryGraph verification and import under `--locked`; the release task is to keep that gate green after each dependency-boundary change and rerun it from the final release commit. The temporary Sail helper bridge is release-explicit: LakeCat depends on local Sail paths plus checked-in helper patches until the required Sail APIs are published. The Grust contract is likewise explicit: LakeCat and QueryGraph follow the active local Grust 0.10 path checkout, and LakeCat binds Turso-backed graph projection to the dedicated `grust-turso` crate. `grust-local` keeps fast memory-backed projection tests, while `grust-turso-local` proves durable graph projection through a bootstrapped `grust_turso::TursoGraphStore`. QueryGraph's `qg-rust` checkout uses the same local Grust path for `lakecat-verify` and `lakecat-import`, keeping the end-to-end graph import path aligned with the catalog graph sink. LakeCat's own graph responsibility stops at the catalog-facing envelope: graph sinks reject blank projection identity, non-object properties, and table-scoped labels without table identity before handing anything to Grust. Persistence, traversal, Cypher, and graph mutation semantics stay in Grust. The service also redacts Grust Turso graph-sink connect/bootstrap failures to graph-

store-path-hash and backend-error-hash evidence so release logs do not capture raw graph database paths.

README, status, changelog, book artifacts, and version notes must be refreshed from the same clean proof commit. The already-published v0.1.0 tag should not move; current post-tag hardening stays under Unreleased while the workspace version remains 0.1.0. For a future version-bump release, tag only after the broad local gate, QGLake handoff, QueryGraph locked verify/import, dependency-contract check, and book validation all pass together. Tracked book artifacts are refreshed deliberately with docs/book/build.sh; the release-candidate gate validates a fresh EPUB/PDF/MOBI build in a temporary LAKECAT_BOOK_DIST_DIR and gives Calibre a temporary CALIBRE_CONFIG_DIRECTORY, so neither binary artifact metadata nor converter preference state can dirty the candidate commit or operator profile.

Manual GitHub Actions is deliberately narrower than the local release proof. When an operator triggers it, the workflow runs the local dependency contract, the workflow-trigger self-test, and the release-version contract before the Rust matrix. It does not own release-proof freshness. That check belongs to the clean local release-candidate gate, because ordinary post-proof code hardening should report “the old proof is stale” locally rather than create a surprising failing cloud run. In practice: use manual CI as a second environment after the local gates are green, not as the source of release truth.

That leaves important work after the first release, but it should stay out of the first-release blocker list unless the scope changes. Typed Iceberg v4 semantics belong in Sail. Cloud SDK-backed secret resolvers belong behind the existing TypeSec-gated provider seam. Reusable graph taxonomy, traversal, stores, and Cypher behavior belong in Grust. Full Croissant, CDIF, OSI, ODRL application composition and agentic workflow semantics belong in QueryGraph and TypeSec above LakeCat. The first release should prove the catalog foundation, not absorb every future system.

The release evidence is concrete:

```
scripts/check-release-readiness.sh --release-candidate
scripts/qglake-handoff-local.sh
docs/book/build.sh
scripts/check-book-artifact-contract.sh docs/book/dist
scripts/check-local-dependency-contract.sh
```

The current full local release-candidate proof was refreshed on June 24, 2026 from clean head 72df4eed. It passed with tracked book artifact validation, the checked-in release-proof contract in clean candidate mode, the strengthened post-tag release-posture contract for the published v0.1.0 baseline, stable git patch-id verification for the Sail helper patch bridge, temporary book build, executable book artifact contract, QueryGraph locked verify/import, Grust Turso graph projection proof, bundle hash sha256:ee1f996eeac976cc07edb468f41039bc24275adba11b8113c38b85a9b4d5a66, graph hash sha256:2c32eaec43a9043c4a764e749afb851f68a59efcb471790ff9126fef5b8010ed, OpenLineage hash sha256:2b280c4fe4815dbc74c3a43a91c2ec2d7edae0c26b661cc1318f699afe7b778b, QueryGraph import hash

sha256:3f86e0f8b7948b8cc97b7f8dda9af4c26f672c4fc8862541eee2110b45b56333, and the final clean-tree check.

LakeCat also carries a smaller proof-freshness contract for the release docs themselves:

```
scripts/check-release-proof-contract.sh
```

That command verifies the active docs agree on the latest full release-candidate proof commit, that the proof commit is an ancestor of the current tree, and that any commits after it are limited to documentation and checked-in book artifact refresh. This avoids an infinite proof-update loop: the heavy gate proves the candidate, then a narrow documentation commit records the proof. If Rust code, manifests, workflows, dependency bridges, release scripts, or other executable behavior changes after the cited proof commit, the old proof is no longer enough and the full release-candidate gate must run again. The quick and ordinary full release-readiness gates surface that distinction without becoming release proof themselves. They print a non-failing freshness note when executable changes exist after the recorded proof commit, so a narrow development slice can still finish while the operator sees that final release proof requires a new `scripts/check-release-readiness.sh --release-candidate` run.

The contract requires a clean tree by default. While editing the proof contract or release docs, maintainers can run:

```
LAKECAT_RELEASE_PROOF_ALLOW_DIRTY=1 scripts/check-release-proof-contract.sh
```

That is only a local self-test mode. It still checks unstaged, staged, and untracked paths against the same post-proof allowlist, so a dirty Rust source file or manifest does not masquerade as documentation-only release evidence.

The full release-candidate gate runs the same contract in candidate mode:

```
LAKECAT_RELEASE_PROOF_CANDIDATE=1 scripts/check-release-proof-contract.sh
```

That mode is intentionally narrow. It still requires a clean tree and coherent active proof references, but it allows the current clean HEAD to become the next proof commit. After the heavy gate passes, the follow-up documentation and book artifact refresh records that new proof commit.

The already-published `v0.1.0` tag is a baseline, not something to move. While the workspace version remains `0.1.0` and HEAD is past `v0.1.0`, post-tag hardening stays under Unreleased. The release version contract checks that shape directly so a follow-up proof commit cannot accidentally look like a second same-version release. It also requires the release checklist to scope tagging chores to a future version-bump release, not the already-published baseline. Finally, it derives the expected versioned changelog heading date from the existing tag, not the current day, so the published `0.1.0 - 2026-06-23` release heading remains stable while hardening continues under Unreleased.

The quick check is acceptable while landing a narrow slice:

```
scripts/check-release-readiness.sh --quick
```

The quick check is not a release claim, but it does validate the tracked docs/book/dist artifact contract so stale EPUB/PDF/MOBI deliverables are caught before a full release-candidate build regenerates temporary artifacts. The full gate is the first-release claim. It runs local dependency-contract checks, workflow-trigger checks, formatting, default workspace tests, feature matrix tests, Turso rows, Sail/TypeSec/Grust integration rows, all-features workspace tests, an out-of-tree book build with artifact validators, and the QGLake handoff proof. Cloud CI remains manual-only until that local proof is boring. A release should be cut from the state those commands prove, not from a hope that a remote runner will discover the truth later.

The handoff portion must stay locked in QueryGraph as well as in LakeCat. The local gate intentionally runs QueryGraph verification and import with Cargo's `--locked` mode so a stale QueryGraph dependency graph is a local failure, not a cloud surprise. When Grust or TypeSec publish new crates, the right sequence is to refresh the QueryGraph lockfile locally, prove `lakecat-verify`, `lakecat-import`, and QueryGraph tests under `--locked`, then rerun LakeCat's full release gate. That makes the QueryGraph import contract part of the release artifact instead of a best-effort downstream hope.

The current gate also proves a subtle QGLake contract that is easy to miss. The handoff readiness probe reads `GET /catalog/v1/config`, and that read is a catalog event. It must therefore carry the same QGLake agent identity as the rest of the handoff, otherwise a harmless-looking readiness curl becomes an anonymous `catalog.config-read` proof in the drained evidence. The handoff script binds that config-read proof to the agent, writes a canonical handoff-verifier output artifact, hashes it into the summary before strict verification, and keeps management list proof, policy-upsert proof, and storage-profile upsert proof in the distinct sections the verifier compares. That is the local substitute for cloud hope: the artifact graph is checked before the summary is accepted.

For this release, replay admission is part of that local proof rather than a best-effort projection filter. `catalog.config-read` replay is closed over the checked warehouse, defaults, overrides, advertised endpoints, and authorization receipt, with defaults and overrides closed again as key/value entries and tenant-root records closed over their known record fields. Optional raw server endpoint and warehouse storage-root values in those config tenant records are also bound back to matching full hash evidence before graph, OpenLineage, QGLake, or QueryGraph import can accept them. `table.commit` replay is closed over the checked table identity or scope hints, authorization receipt, and nested commit evidence, and the nested commit object is closed over the pointer transition, principal, hashes, format, snapshot, and timestamp fields LakeCat verifies. The new and previous metadata-location evidence in that commit object must also remain undecorated and free of credential-like material before graph, OpenLineage, QGLake, or QueryGraph proof can inherit it. `querygraph.bootstrap` replay is closed over the checked warehouse, counts, verified table/view manifests, artifact hashes, view-version receipts, standards, bundle hash, graph hash, OpenLineage hash, QueryGraph import hash, and authorization receipt. Its request-identity object is also closed over the known envelope fields for principal, source, TypeDID hashes, delegation hashes, token hash, attestation state, and raw-secret posture. That means an old or corrupted outbox row cannot add an unverified compatibility claim, endpoint claim, graph claim, standards claim, OpenLineage claim, QueryGraph claim, or application claim and have it flow into Grust,

OpenLineage, QGLake, or QueryGraph import proof. The raw lineage-drain summary follows the same posture for bootstrap standards: if `standards` is present, it must be a string-array-shaped claim without blank, non-string, duplicate, or unsupported entries. Compact QGLake handoff verification applies the same exact expected standards set, so an archived handoff cannot append a future-looking or local-only standards claim beside the accepted Iceberg, Croissant, CDIF, OSI, Grust, OpenLineage, and ODRL evidence. Malformed standards evidence now rejects the summary instead of disappearing from the QGLake proof that QueryGraph later compares. The same summary path treats `table-artifacts` and `view-artifacts` as evidence arrays: malformed non-array artifact fields are rejected rather than summarized as zero artifact counts.

1.13 Future Work

LakeCat is a direction more than a single release. The next stage should keep the architecture more true without pushing engine, graph, or security semantics back into the catalog.

1. Replace temporary Sail patch bridges only after their helpers are published upstream, then remove the corresponding LakeCat bridge contract.
2. Keep pushing reusable table-status, planning, metadata-as-data, manifest, delete, and commit helpers into Sail.
3. Treat typed Iceberg v4 as a Sail project after formal specification adoption. The active v4 draft includes relative metadata locations, but LakeCat should not turn a moving draft into catalog behavior; its bridge remains explicit and JSON-only until Sail can model and plan the adopted semantics.
4. Make the transactional outbox the only path for graph and lineage side effects, while keeping graph persistence, traversal, and Cypher in Grust.
5. Expand TypeSec-backed capability checks until every privileged path is unbypassable, with raw credentials remaining an explicit audited exception.
6. Keep OSI, Croissant, CDIF, ODRL composition, and agent workflows in QueryGraph. Before the next public release, refresh QueryGraph's older dependency-guide examples to match its live Grust 0.10.0 and TypeSec 0.8.0 runtime manifest; its LakeCat importer already validates the current receipt-chain and Grust graph handoff contract.
7. Prove the bootstrap bundle through QueryGraph import on every meaningful public-surface change, and keep local release evidence ahead of cloud CI.

The payoff is a catalog foundation that feels ordinary to Iceberg clients and rich to QueryGraph. Tables remain portable. Policies become enforceable. Graph and lineage become replayable. Sail plans close to the data. QueryGraph gets a trustworthy substrate for the next version of its lakehouse intelligence.